



Storage Fundamentals Training



Senthilkumar Muthusamy

Agenda

- ▶ **Storage Foundation - 10 Days**
 - ▶ Storage Fundamentals
 - ▶ FC Fundamentals
 - ▶ FC Switch Configuration & Management
- ▶ **NetApp ONTAP9 Cluster Administration – 7 Days**
- ▶ **HPE 3par StorServ Administration - 8 Days**



Storage Foundation Training

- ▶ Module1: Introduction to Data-Storage
- ▶ Module2: Storage Hardware & Ecosystem
- ▶ Module3: RAID
- ▶ Module4: Component of Storage Array
- ▶ Module5: iSCSI
- ▶ Module6: Storage & Data Management
- ▶ Module7: Introduction to NAS
- ▶ Module8: Storage Virtualization
- ▶ Module9: Monitoring / Reporting / Capacity Management
- ▶ Module10: Introduction to Fibre Channel



Agenda

- ▶ **Module I I: Initial Setup**
 - ▶ Brocade
 - ▶ Cisco
- ▶ **Module I 2: Fabric Zoning (Brocade and Cisco)**
- ▶ **Module I 3: Advanced Concepts**
- ▶ **Module I 4: Troubleshooting**

- ▶ **Lab – Cisco MDS DM & FM**



Introduction to Data-Storage

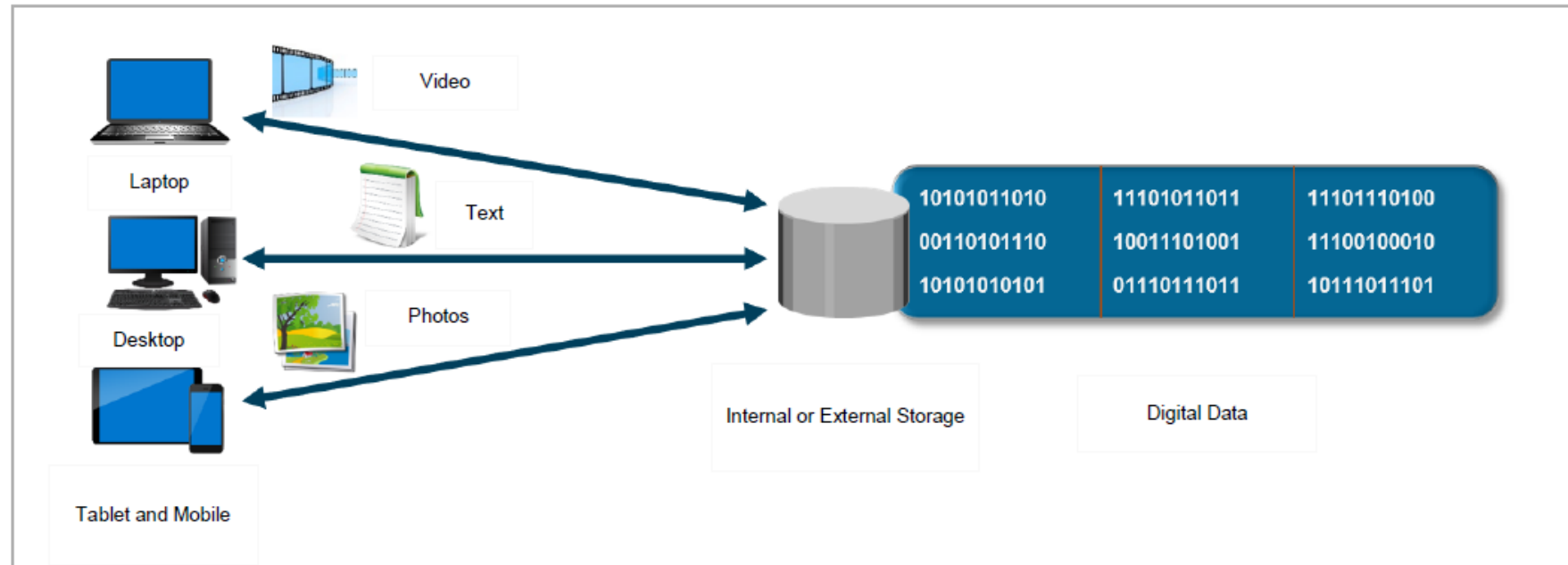
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Digital Data



Definition: Digital Data

A collection of facts that is transmitted and stored in electronic form, and processed through software.



Information Storage

- Information is stored on storage devices on non-volatile media
 - Magnetic storage devices: Hard disk drive and magnetic tape drive
 - Optical storage devices: Blu-ray, DVD, and CD
 - Flash-based storage devices: Solid-state drive (SSD), memory card, and USB thumb drive
- Storage devices are assembled within a storage system or “array”
 - Provides high capacity, scalability, performance, reliability, and security
- Storage systems along with other IT infrastructure are housed in a data center



Storage challenges in business

- Data explosion
 - Management and analysis
- Business continuity
 - High availability and disaster recovery solutions
- Consolidation
 - Simplify architecture and maximize performance
- Ease of use
 - Simplified management for complex storage infrastructure



Digital Transformation

Digital transformation puts technology at the heart of an organization's products, services, and operations.



Why Information Storage and Management

- Organizations are dependent on continuous and reliable access to information
- Organizations seek to store, protect, process, manage, and use information
- Organizations are increasingly implementing intelligent storage solutions:
 - To efficiently store and manage information
 - To gain competitive advantage
 - To derive new business opportunities



Demands on Data

The demand on data is very high given its critical nature

- ▶ Data must be highly available
- ▶ Data must be secure
- ▶ Data must be reliable
- ▶ Speed of data access
- ▶ Data Protection



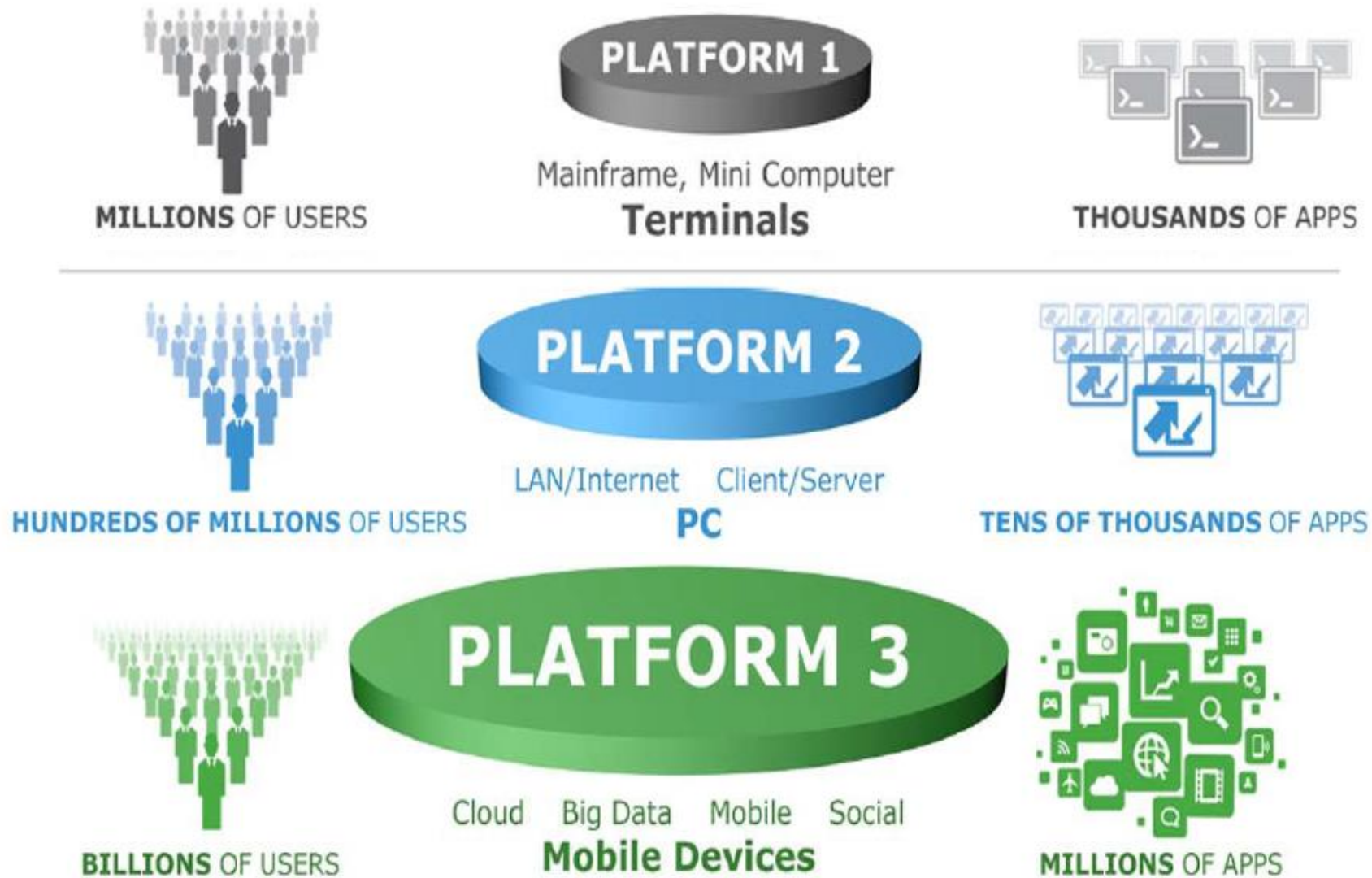


Storage Architectures



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Evolution of Computing Platforms



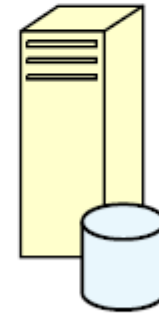
Storage architectures

- Direct attached storage
 - Dedicated storage
- Network attached storage
 - Share files over TCP/IP
- Storage area networks
 - Centrally managed, shared storage
 - Dedicated storage network

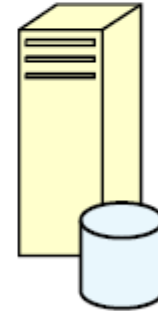


Direct attached storage

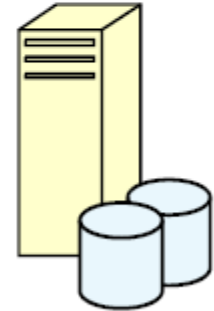
- Storage is accessed directly from the server host
 - Internal or external from server
- Data access is at block level
- The server communicates directly to the DAS storage device
- Storage device is addressed using SCSI ID



Block
level



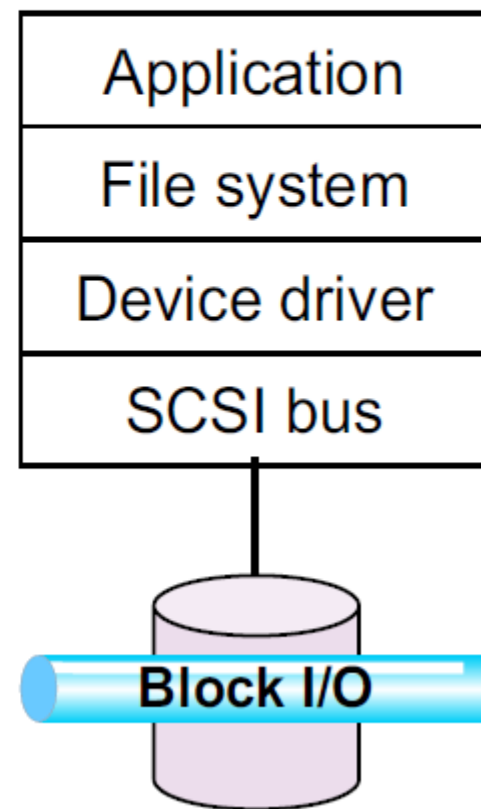
Block
level



Block
level

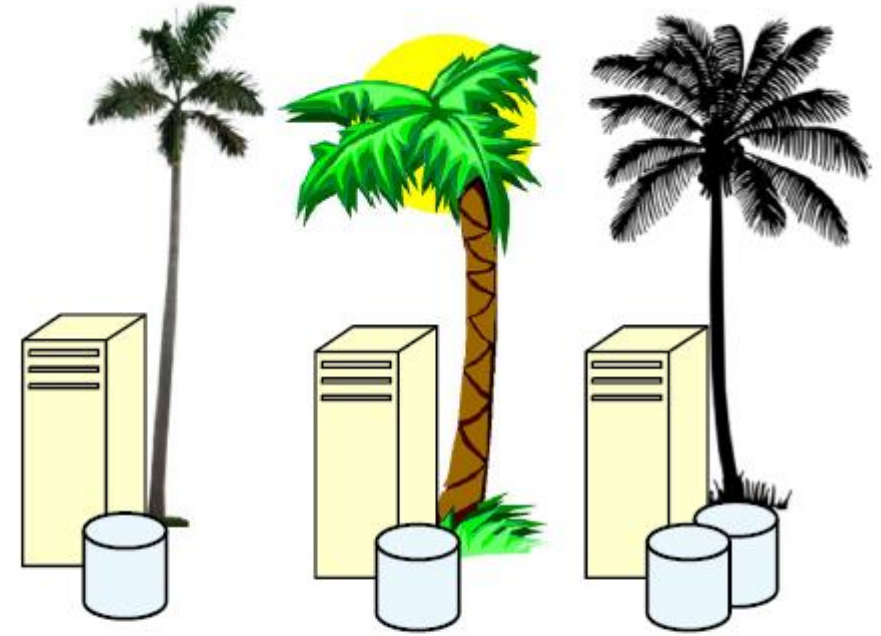
Direct attached storage data access

- Application requests file I/O
- File system software translates request
- Commands are sent to the physical storage device to retrieve the data
 - SCSI architecture and commands are employed in most server based systems
- Data blocks are retrieved from storage device
- Communications is through direct connection to the storage device
 - Internal or external



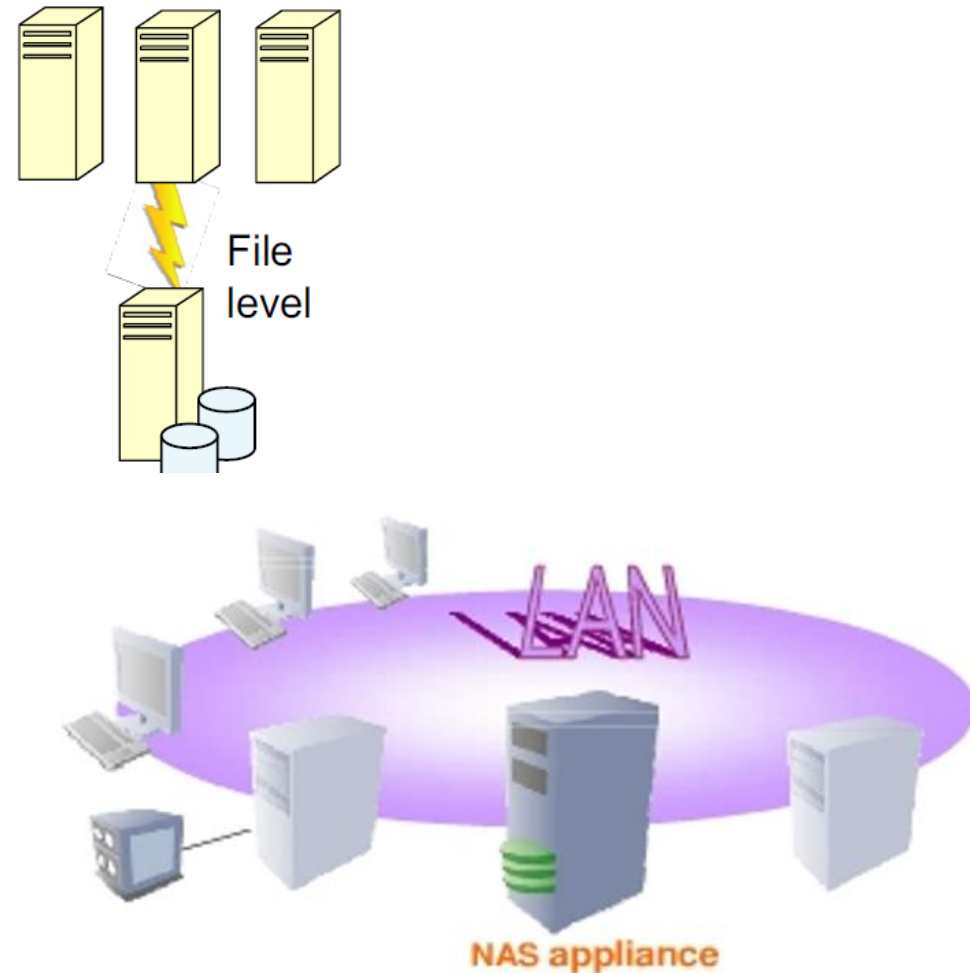
Direct attached storage issues

- Islands of storage
 - Inefficient resource usage
 - Difficult to scale capacity
- SCSI limitations
 - Cable, device addressing, performance
- Difficult to manage overall storage
 - Different file systems, backup/recovery, data sharing
- Inconsistent copies of information



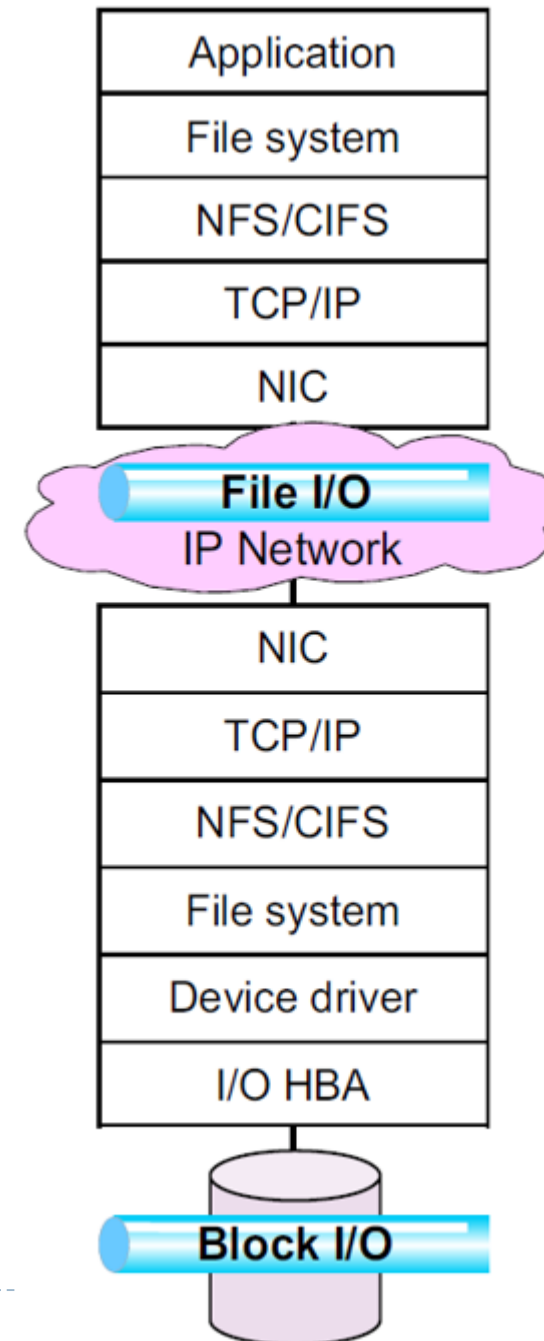
Network attached storage

- Storage is accessed using specialized appliances over a TCP/IP network
- Client/server model is employed
 - Examples include NFS and CIFS
- Supports multiple clients
- Data access is at file level
- The server communicates using TCP/IP to the NAS storage device
- NAS appliance accesses storage local to the system
- NAS gateway accesses storage located external to the system
- Storage device is addressed using TCP/IP socket



Network attached storage data access

- Application requests file I/O
- File system software translates request
 - File system redirects through NFS/CIFS
- TCP/IP packet is delivered to NAS server
- NFS/CIFS redirects to file system software
- Commands are sent to the physical storage device to retrieve the data
 - SCSI architecture and commands are employed in most server based systems
- Data blocks are retrieved from the storage device
- Communications is through a TCP/IP connection to the file server



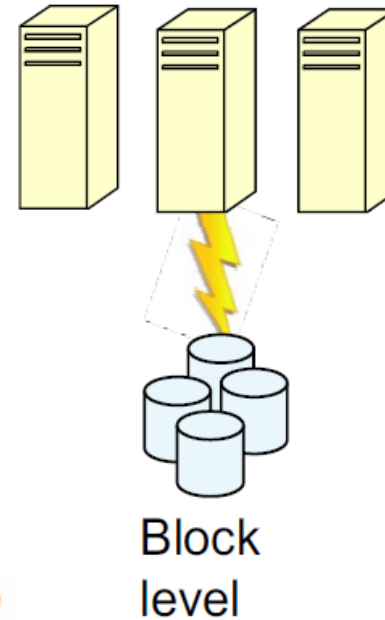
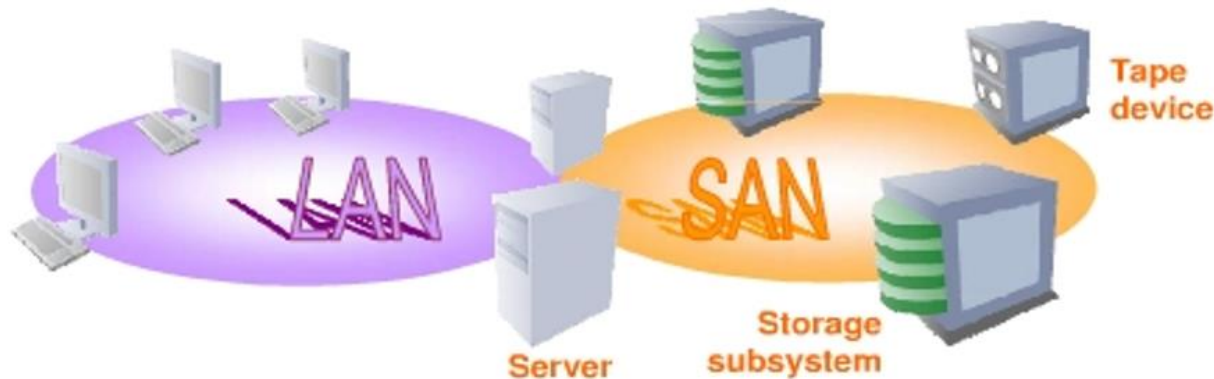
NAS Limitations

- ▶ Performance depends on the protocol
- ▶ Slow down for video application or multiple large files
- ▶ It is file oriented
- ▶ Increased LAN traffic
- ▶ The file transfer speed is not as fast as DAS
- ▶ Limited scalability



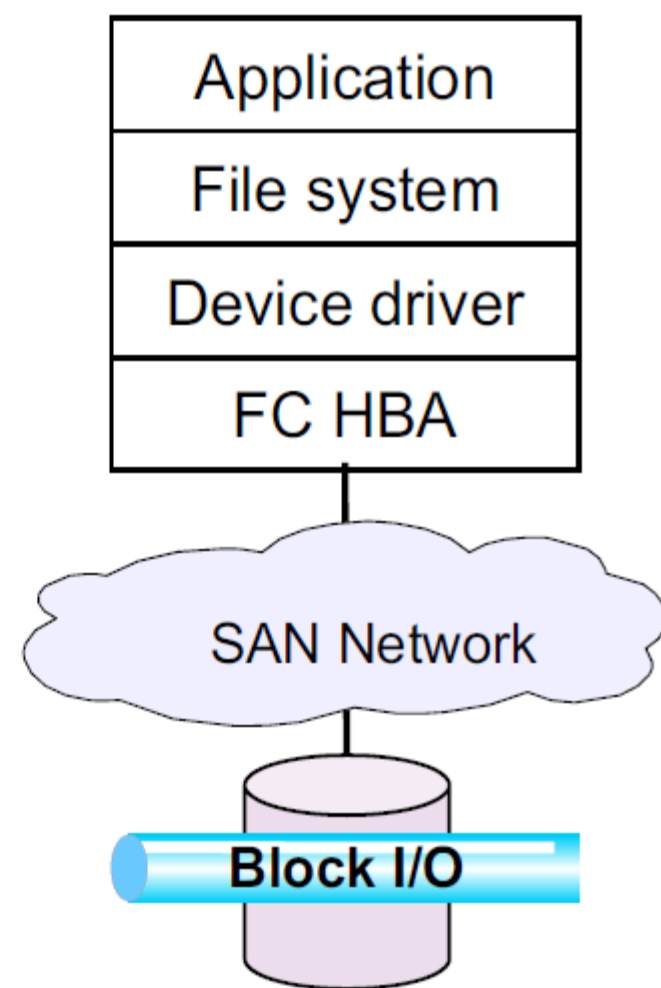
Storage area network

- Storage is accessed directly through a specialized, high speed, highly available network from the host
- Data access is at block level
- The server communicates using a SAN protocol to the storage device
 - Generally, Fibre Channel protocol
- Supports any to any connection
- Storage device is addressed using worldwide name
 - Fibre Channel based



Storage area network data access

- Application requests file I/O
- File system software translates request
- Commands are sent to the SAN storage controller to retrieve the data
 - SCSI architecture and commands are encapsulated in Fibre Channel protocol
- Storage controller retrieves data blocks from storage device
 - Data returned to server using Fibre Channel protocol
- Communications is through SAN network connection to the storage device



SAN benefits

- Improvements to application availability
 - Storage is independent of applications and accessible through multiple data paths for better reliability, availability, and serviceability
- Higher application performance
 - Storage processing is offloaded from servers and moved onto a separate network
- Centralized and consolidated storage
 - Simpler management, scalability, flexibility, and availability
- Data transfer and vaulting to remote sites
 - Remote copy of data that is enabled for disaster protection and against malicious attacks
- Simplified centralized management
 - Single image of storage media simplifies management

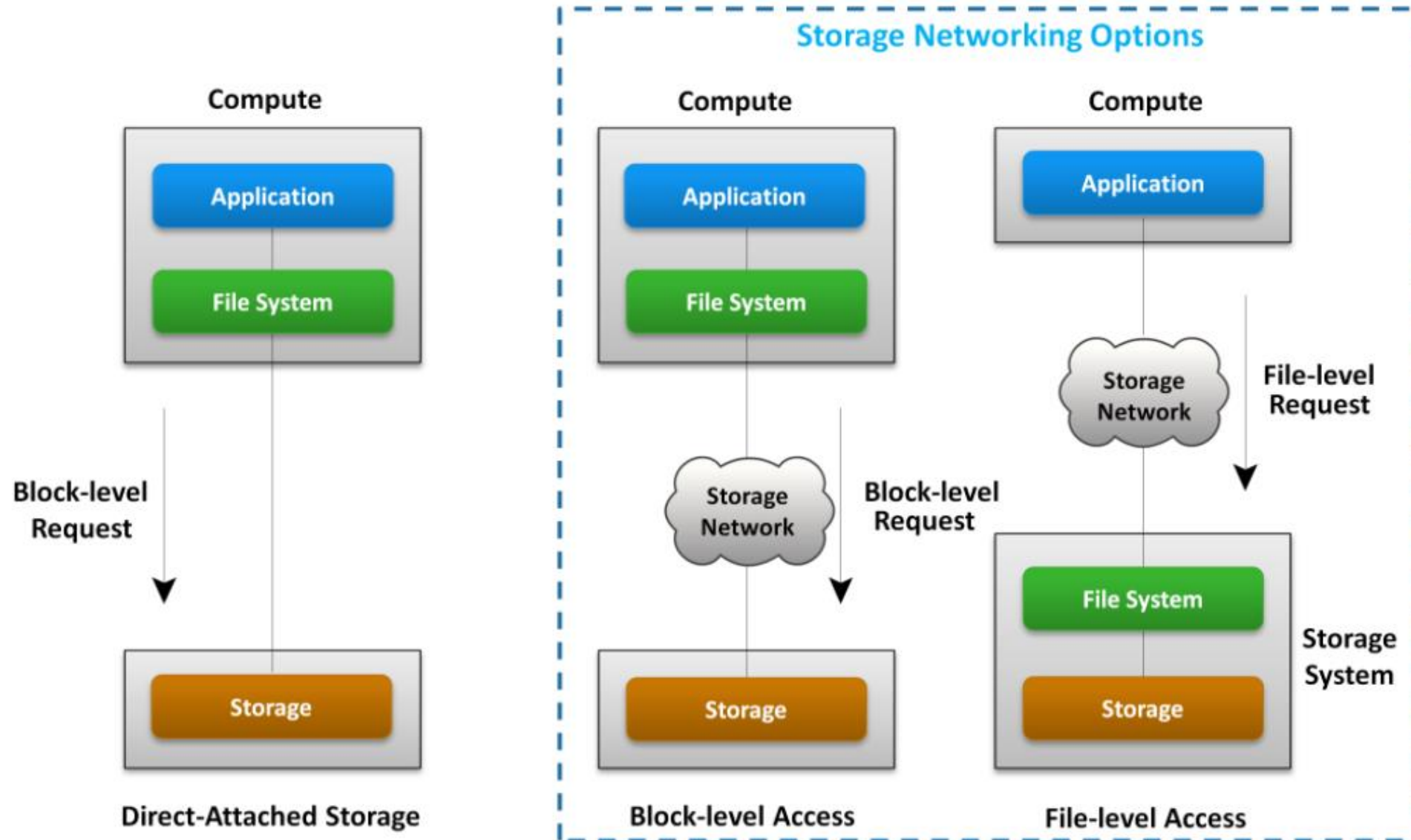


Storage architecture components

	Server	Network	Storage
DAS	• Server HW • Device driver	• Direct cable	• Controller + storage devices
NAS	• Server HW • NAS client SW • NIC	• TCP/IP net • ETH cables • Switch/Rtr	• Controller + storage devices • NAS server SW • NIC
SAN	• Server HW • HBA • Multipath SW	• SAN switch • SAN cables	• Controller + storage devices • HBA



Host Access to Storage

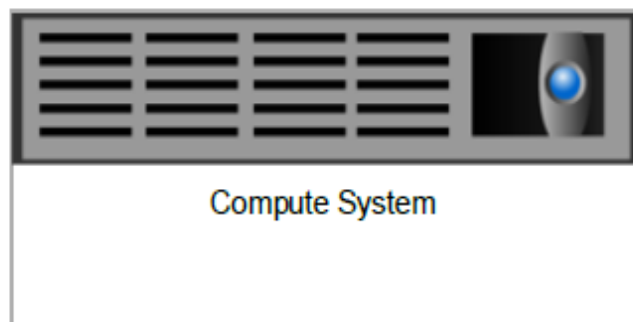


Storage Hardware Eco System

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What is a Compute System?

- A computing platform (hardware and system software) that runs applications
 - Physical components include processor, memory, internal storage, and I/O devices
 - Logical components include OS, device drivers, file system, and logical volume manager



Examples of Hosts



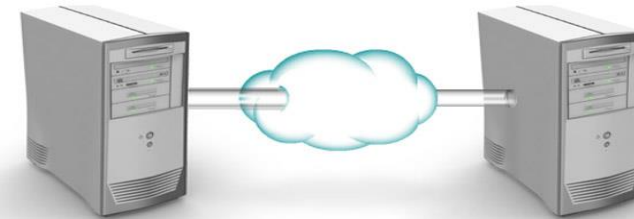
Laptop



Server



Mainframe



HA - Cluster

Types of Compute Systems



Tower Compute System



Rack-mounted Compute System



Blade Compute System



DC Rack





Introduction to Connectivity



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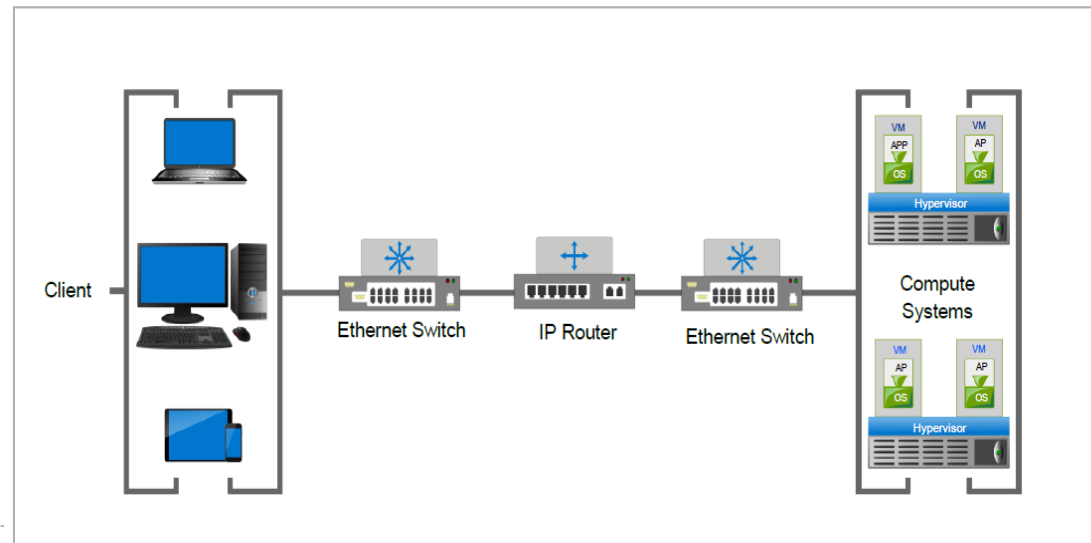
Introduction to Connectivity

- Communication paths between IT infrastructure components for information exchange and resource sharing
- Types of connectivity
 - Compute-to-compute connectivity
 - Compute-to-storage connectivity



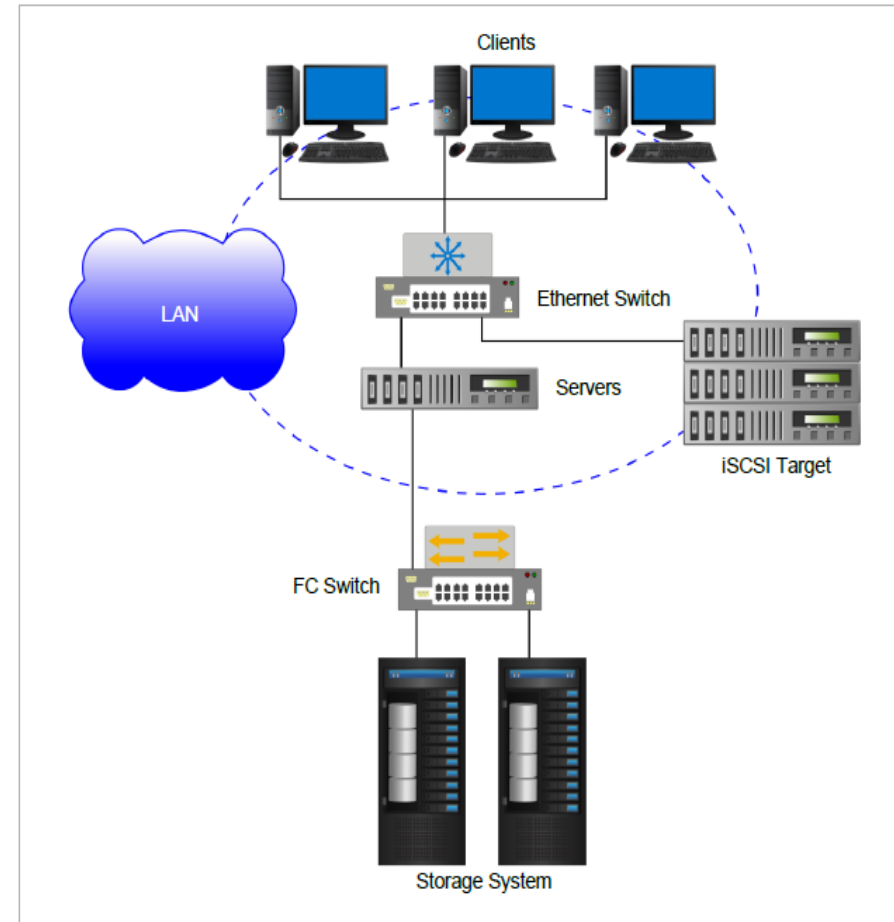
Compute-to-Compute Connectivity

- Compute-to-compute connectivity typically uses protocols based on the Internet Protocol (IP)
 - Each physical compute system is connected to a network through one or more host interface devices, called a network interface controller (NIC)
 - Physical switches and routers are the commonly used interconnecting devices
 - A switch enables different compute systems in the network to communicate with each other



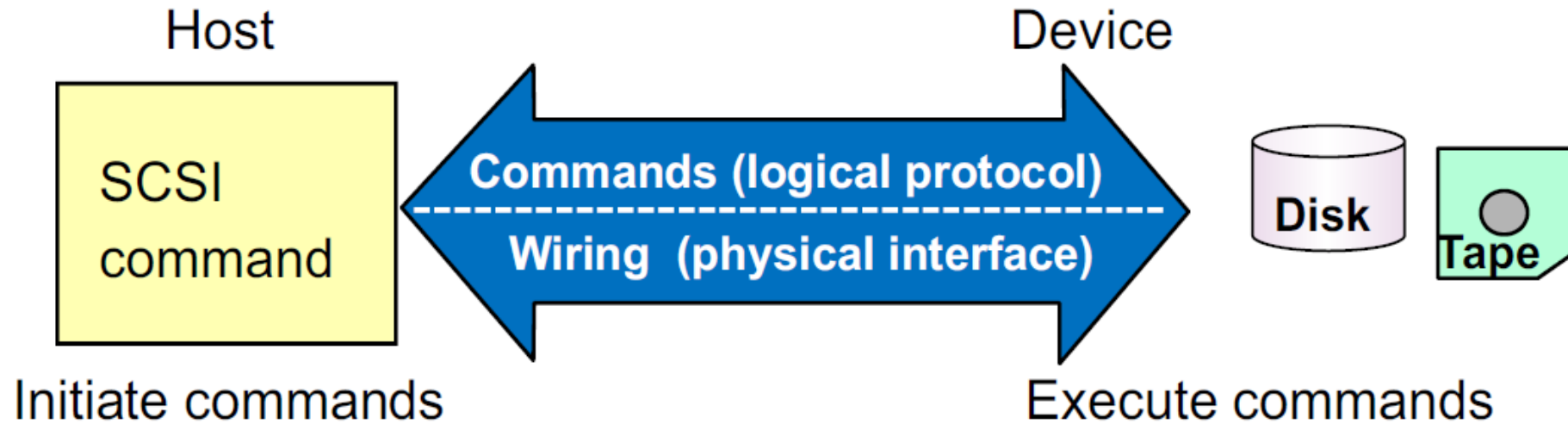
Compute-to-Storage Connectivity

- ▶ Enabled through physical components and interface protocols
- ▶ Physical connectivity components
- ▶ Host bus adapter, port, and cable
- ▶ Storage may be connected directly or over a SAN

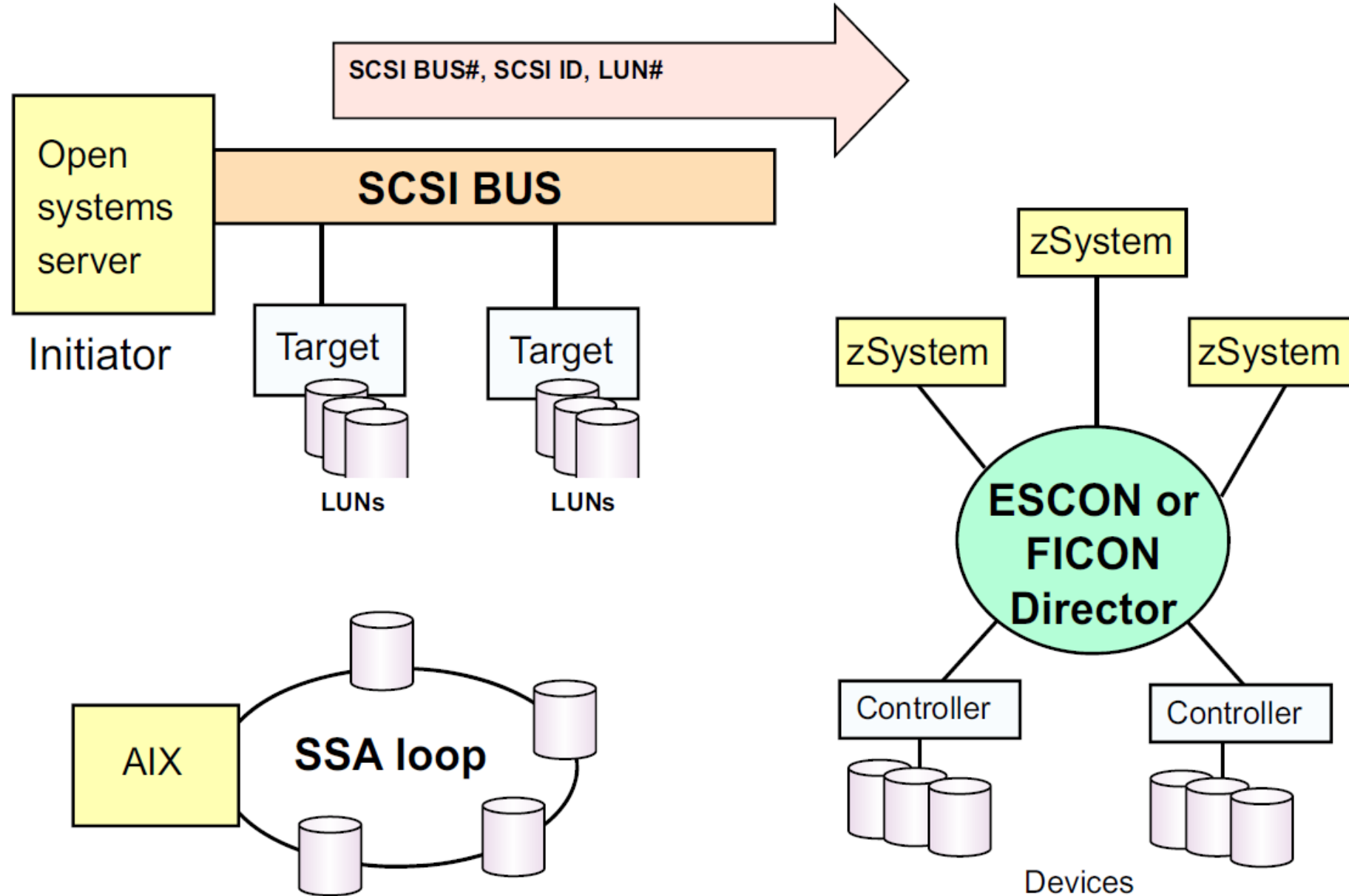


Logical protocol and physical interface

- Host initiates I/O commands
- Storage device executes I/O commands
- Data result is returned to the host
- Interface between host and storage device consists of:
 - Data transfer protocol commands
 - Physical transport protocol



Physical interfaces and topologies



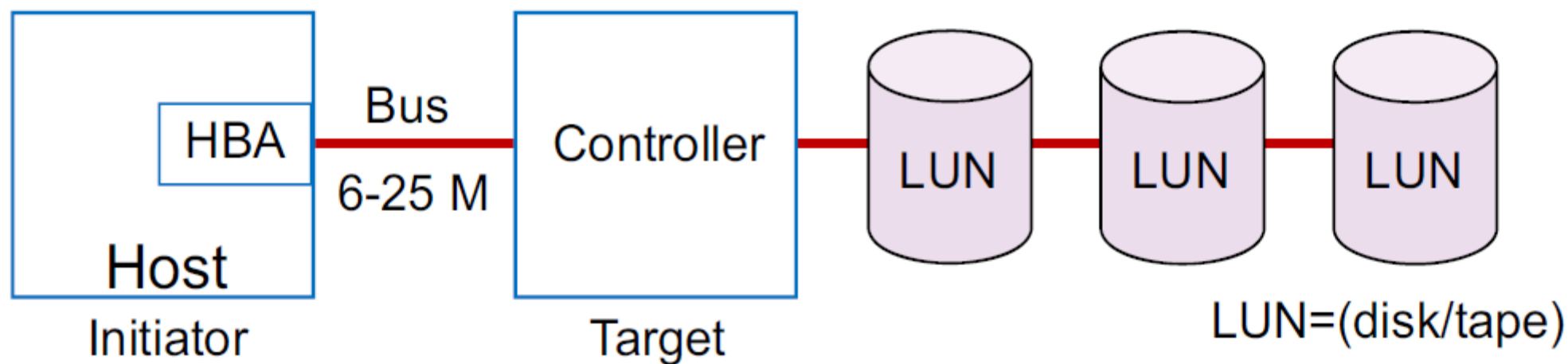
Storage Connectivity Protocols

Protocol	Description
IDE/ATA	• Popular interface used to connect hard disks and optical drives • The Ultra DMA/133 version of ATA supports a throughput of 133 MB/s
Serial ATA	• Serial version of the IDE/ATA specification typically used for internal connectivity • Provides data transfer rate up to 16 Gb/s (standard 3.2)
SCSI	• Popular standard for compute-to-storage connectivity • Supports up to 16 devices on a single bus • Ultra-640 version provides data transfer speed up to 640 MB/s
SAS	• Point-to-point serial protocol replacing parallel SCSI • Supports data transfer rate up to 12 Gb/s (SAS 3.0)
FC	• Widely-used protocol for high speed compute-to-storage communication • Provides a serial data transmission that operates over copper wire and/or optical fiber • Latest version of the FC interface '16FC' allows transmission of data up to 16 Gb/s
IP	• Existing IP-based network leveraged for storage communication • Examples: iSCSI and FCIP protocols

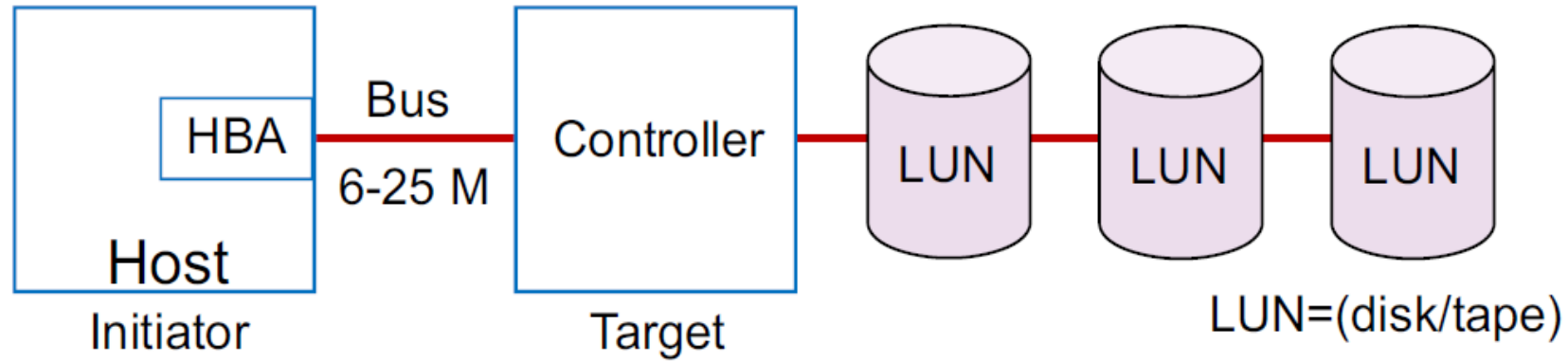


Small Computer System Interface

- ANSI standards for physical transfer of data between computers and peripheral devices
- Defines commands, protocols, and electrical / optical interfaces
- Addressing includes SCSI ID and LUN
- Bus architecture maximum limits are 8 or 16 devices
- SCSI initiator sends commands to SCSI target



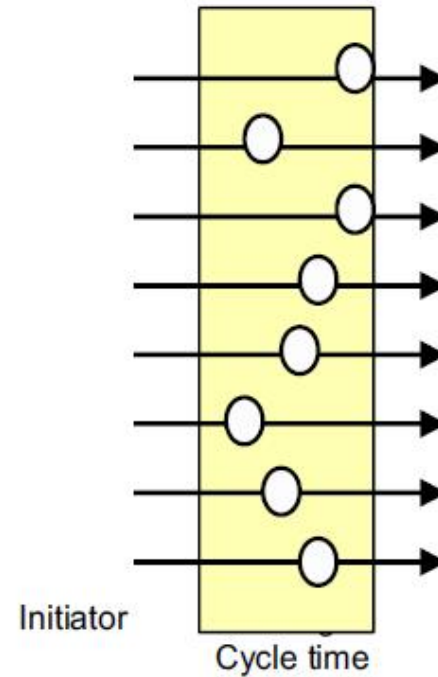
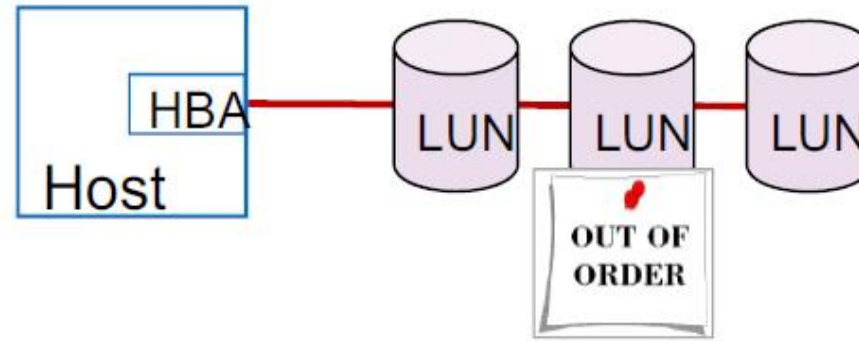
Parallel SCSI



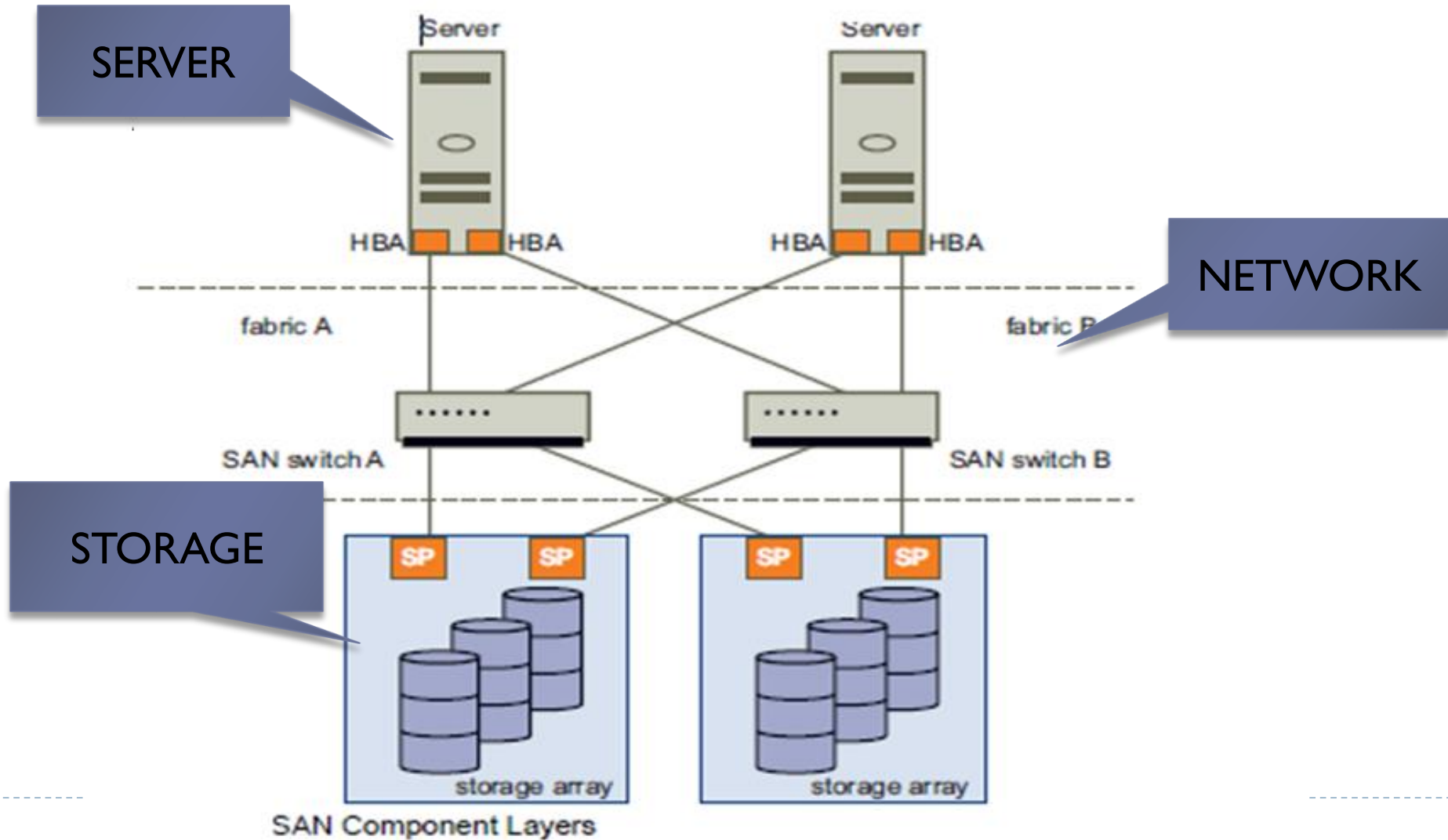
SCSI bus type	8-bit width (narrow)	16-bit width (wide)
SCSI-1	5 MB/s	
SCSI-2 Fast	10 MB/s	20 MB/s
SCSI-3 Ultra	20 MB/s	40 MB/s
SCSI-3 Ultra2	40 MB/s	80 MB/s
SCSI-3 Ultra3 (Ultra160)		160 MB/s
SCSI-3 Ultra320		320 MB/s
SCSI-3 Ultra640		640 MB/s

Parallel SCSI limitations

- Scalability
- Reliability and availability
- Speed and latency
 - Cycle time
 - Skew
- Distance
- Device sharing



Storage Components



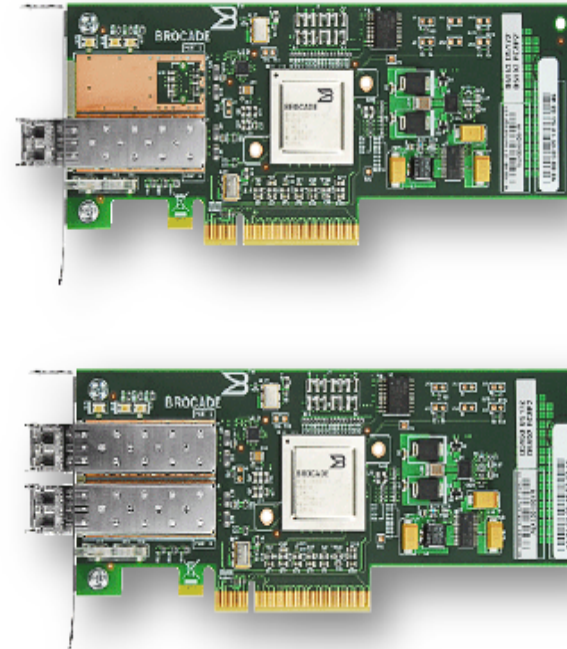
Host Adapters

- ▶ NIC (Network Interface Card) – LAN and iSCSI
- ▶ HBA (Host Bus Adapter) – SAN FC
- ▶ CNA (Convergence Network Adapter) – SAN FCoE



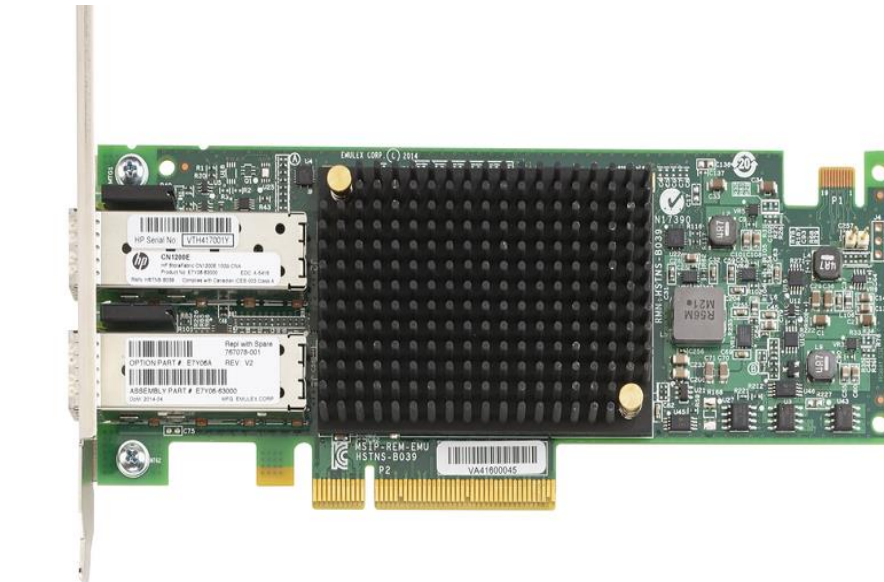
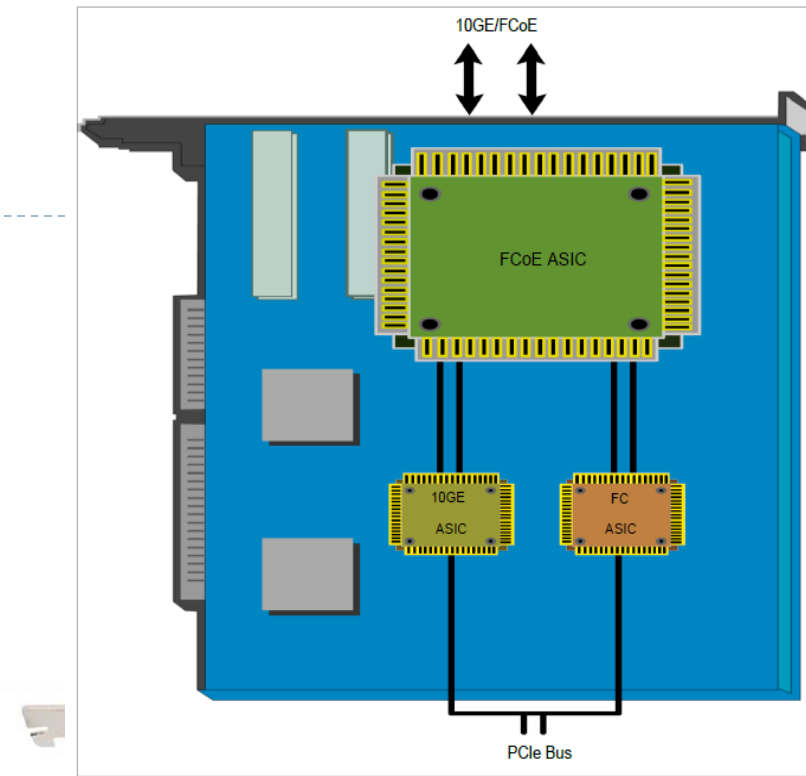
Fibre Channel Host Bus Adapters

- Provides an interface between the server or workstation internal bus (e.g. PCIe) and the Fibre Channel network
- HBA software driver provides the storage information required by the operating system
 - Handles I/O and control requests
 - Copper/Optical media support (may be dual port cards)
- Looks like a SCSI adapter to the host OS

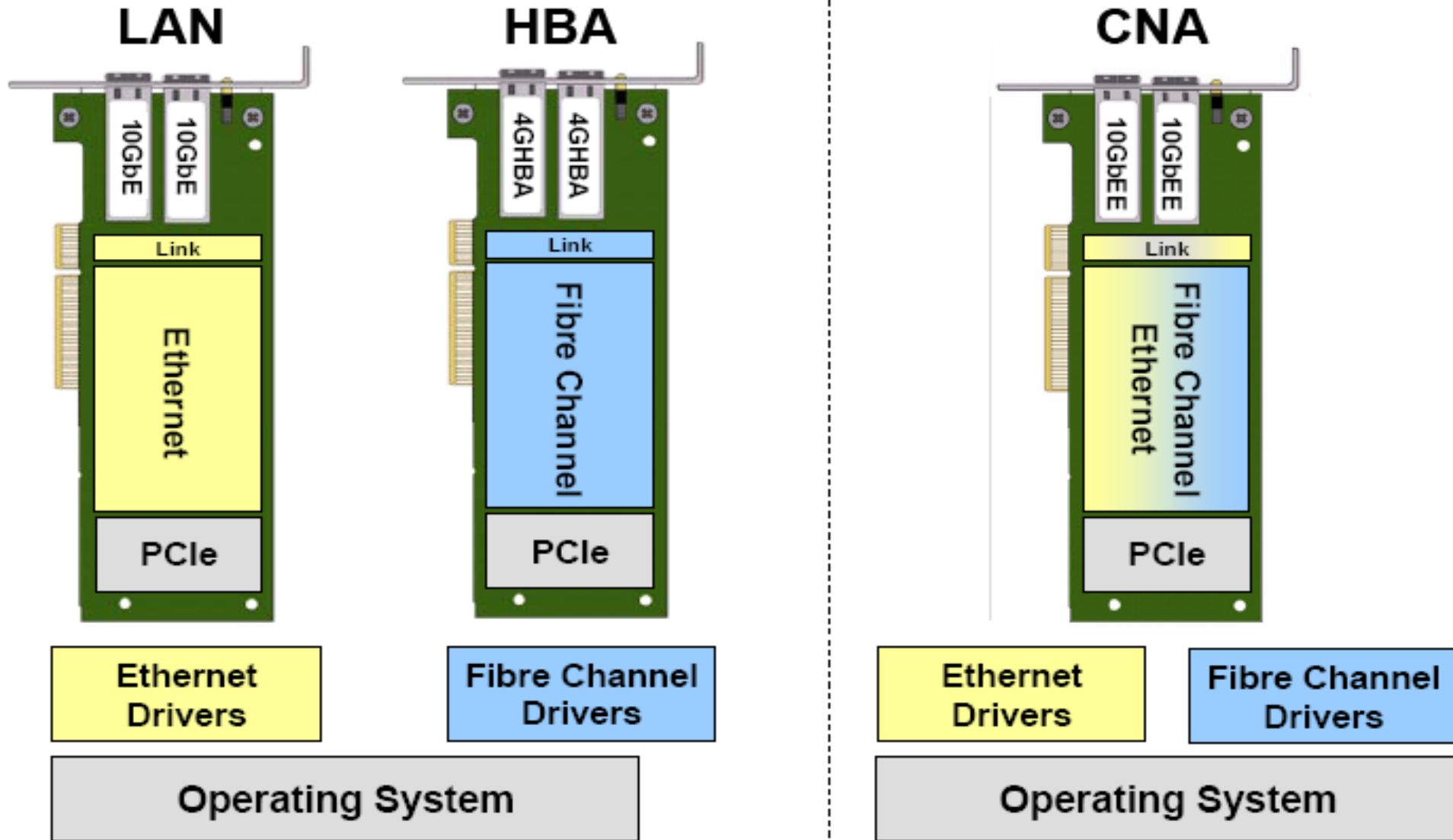


What is CNA?

- ▶ A physical adapter that provides functionality of both NIC and FC HBA
- ▶ Encapsulates FC frames into Ethernet frames and forwards them over CEE links
- ▶ Contains separate modules for 10 GE, FC, and FCoE ASICs



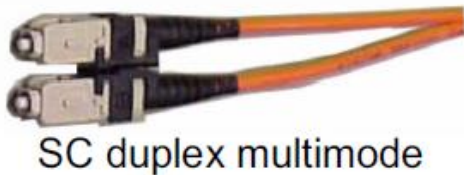
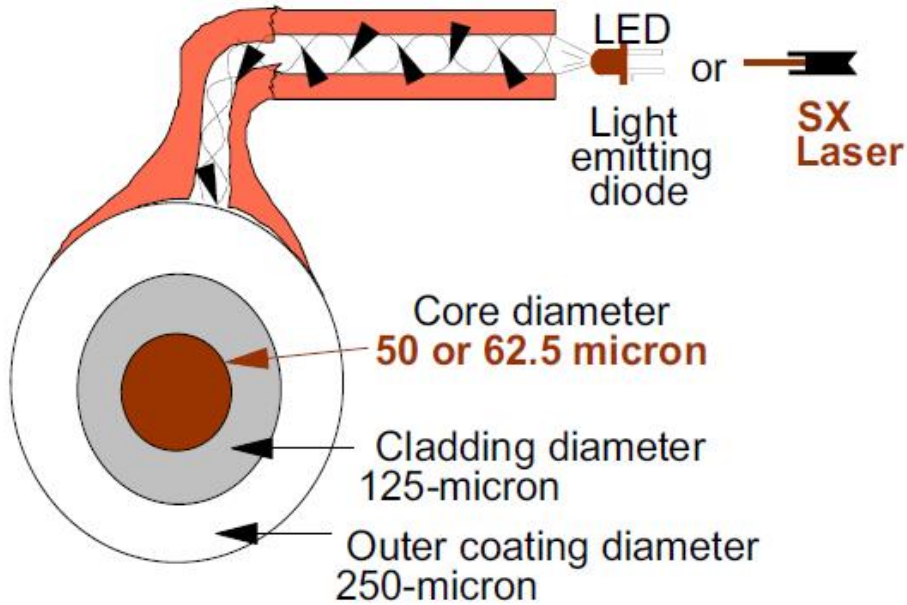
Adapter Evolution: Consolidation Network Adapter



Fiber optic technology

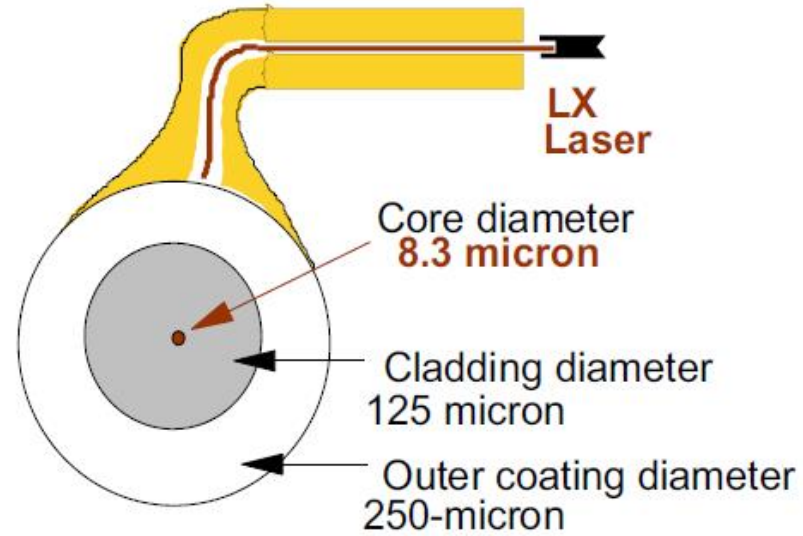
Multimode fiber

Multiple paths for the light to travel



Single mode fiber

Single path for the light to travel



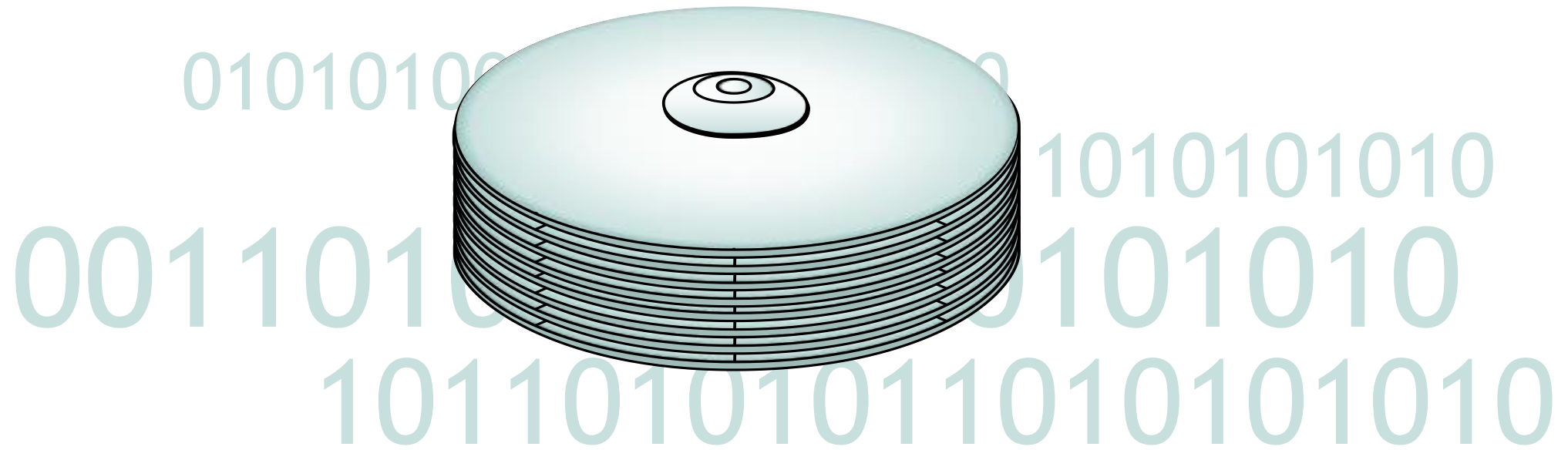


Disk Drives

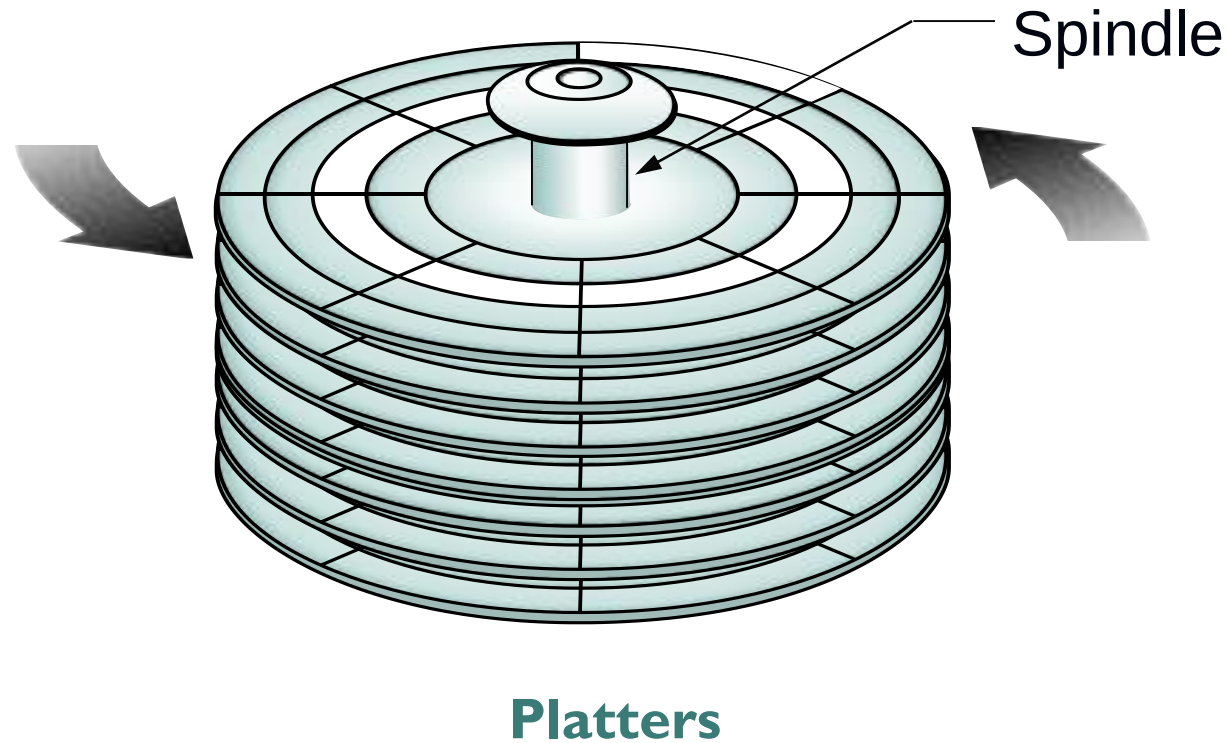


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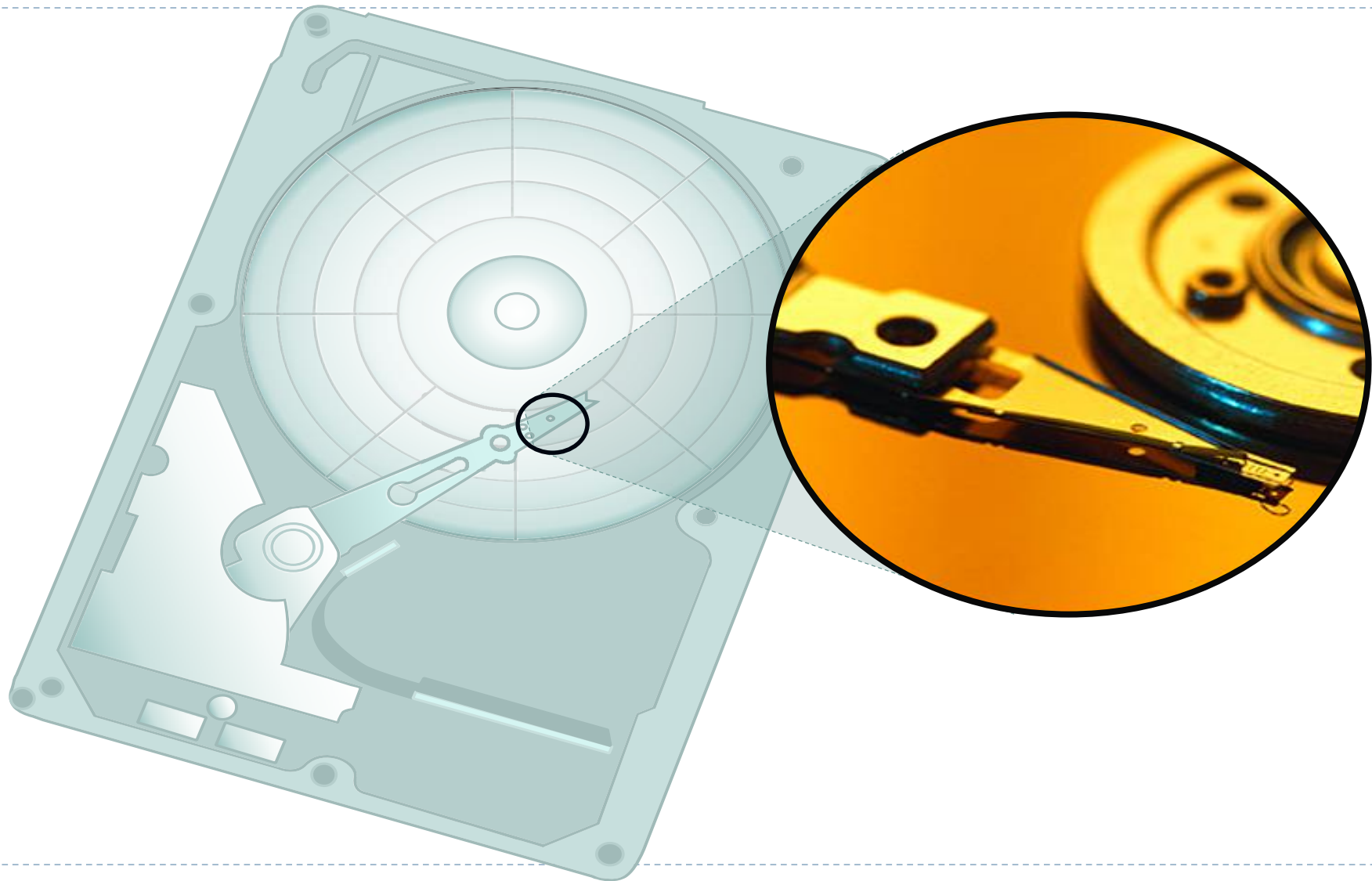
Disk Drive Components: Platters



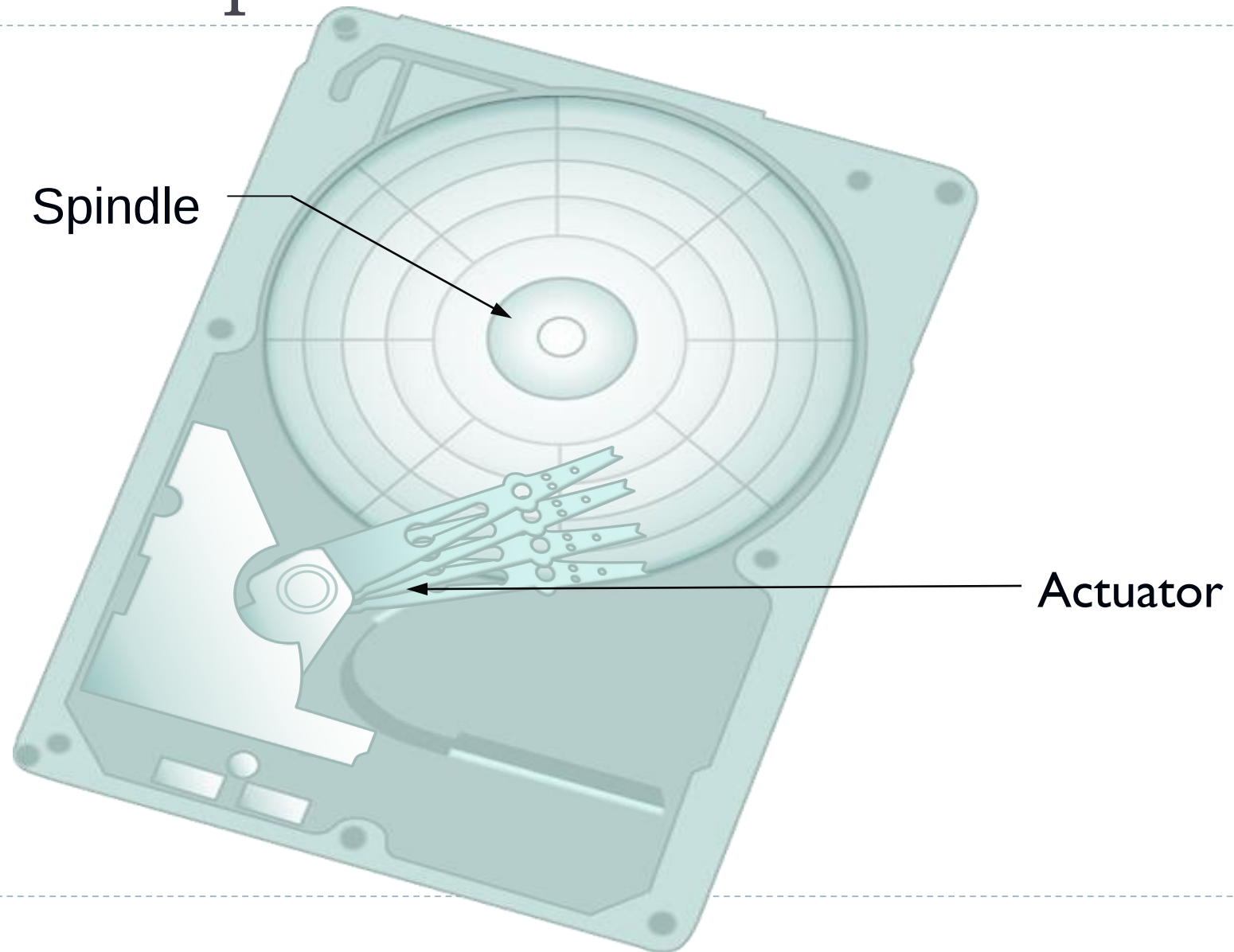
Disk Drive Components: Spindle



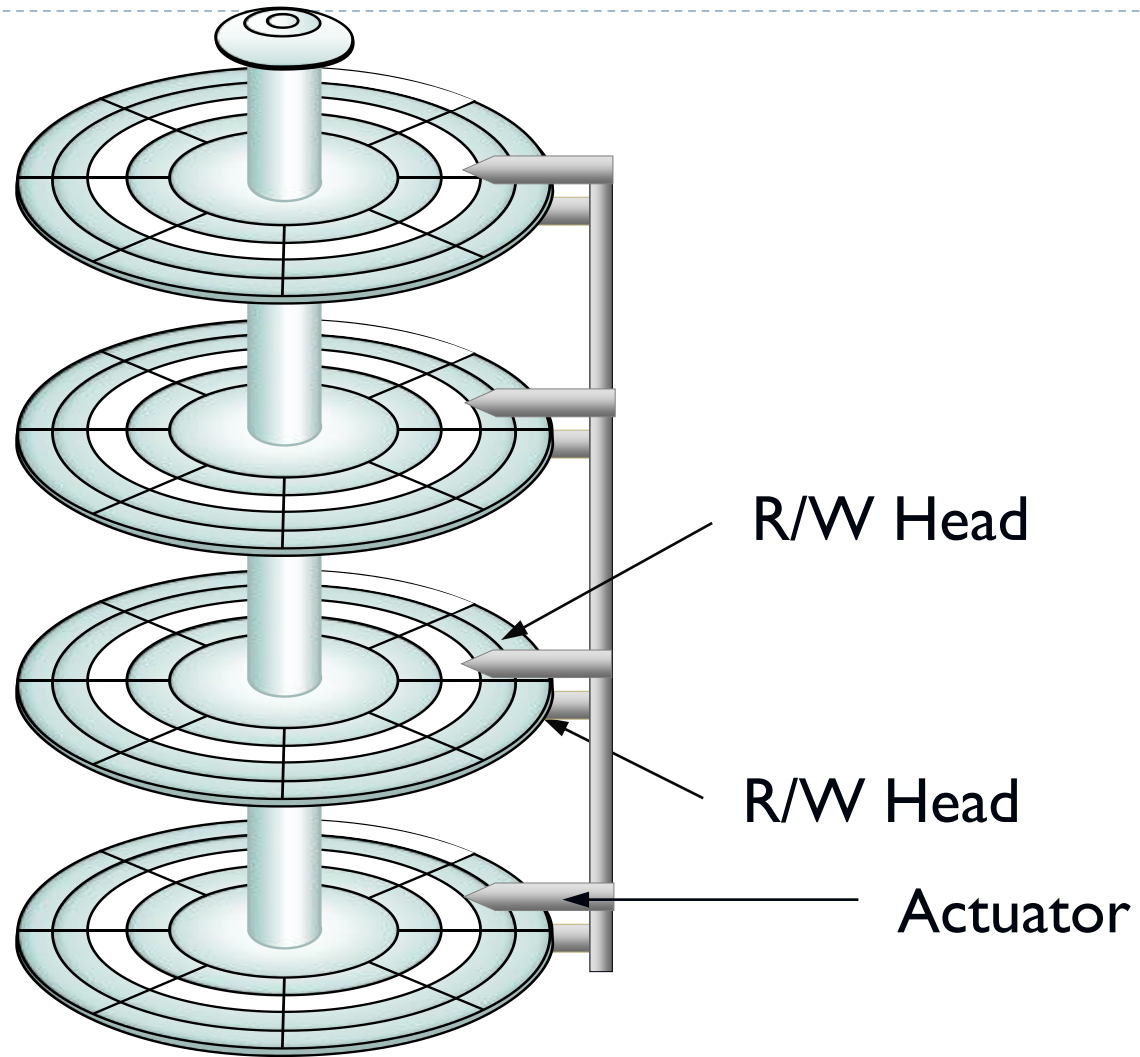
Disk Drive Components: Read/Write Heads



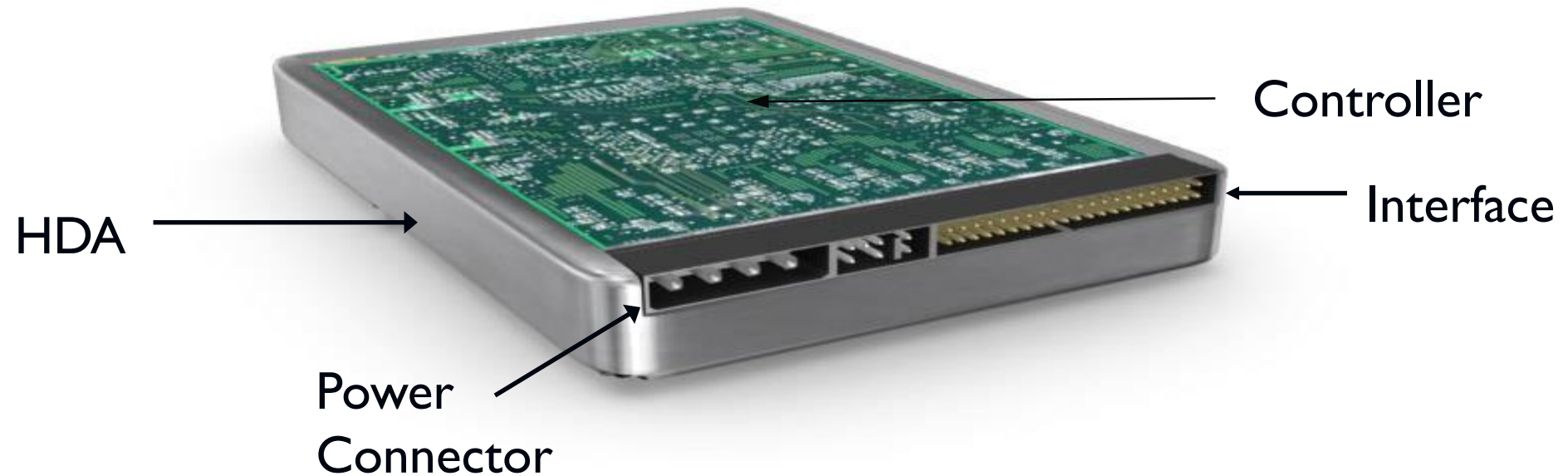
Disk Drive Components: Actuator



Physical Disk Structures: Actuator Arm Assembly



Disk Drive Components: Controller



Bottom View of Disk Drive



Solid State Drive Components

- Solid state drives (SSDs) are:
 - Storage devices that contain non-volatile flash memory
- Internally, a solid state drive's hardware architecture consists of the following components:
 - I/O interface
 - Controller
 - Mass storage
- The I/O interface enables connecting the power and data connectors to the solid state drives.
- Solid state drives typically support standard connectors such as SATA, SAS, or FC.

SDD Performance

- Access type
 - SSD performs random reads the best
 - SSDs use all internal I/O channels in parallel for multithreaded large block I/Os
- Drive state
 - New SSD or SSD with substantial unused capacity offers best performance
- Workload duration
 - SSDs are ideal for most workloads

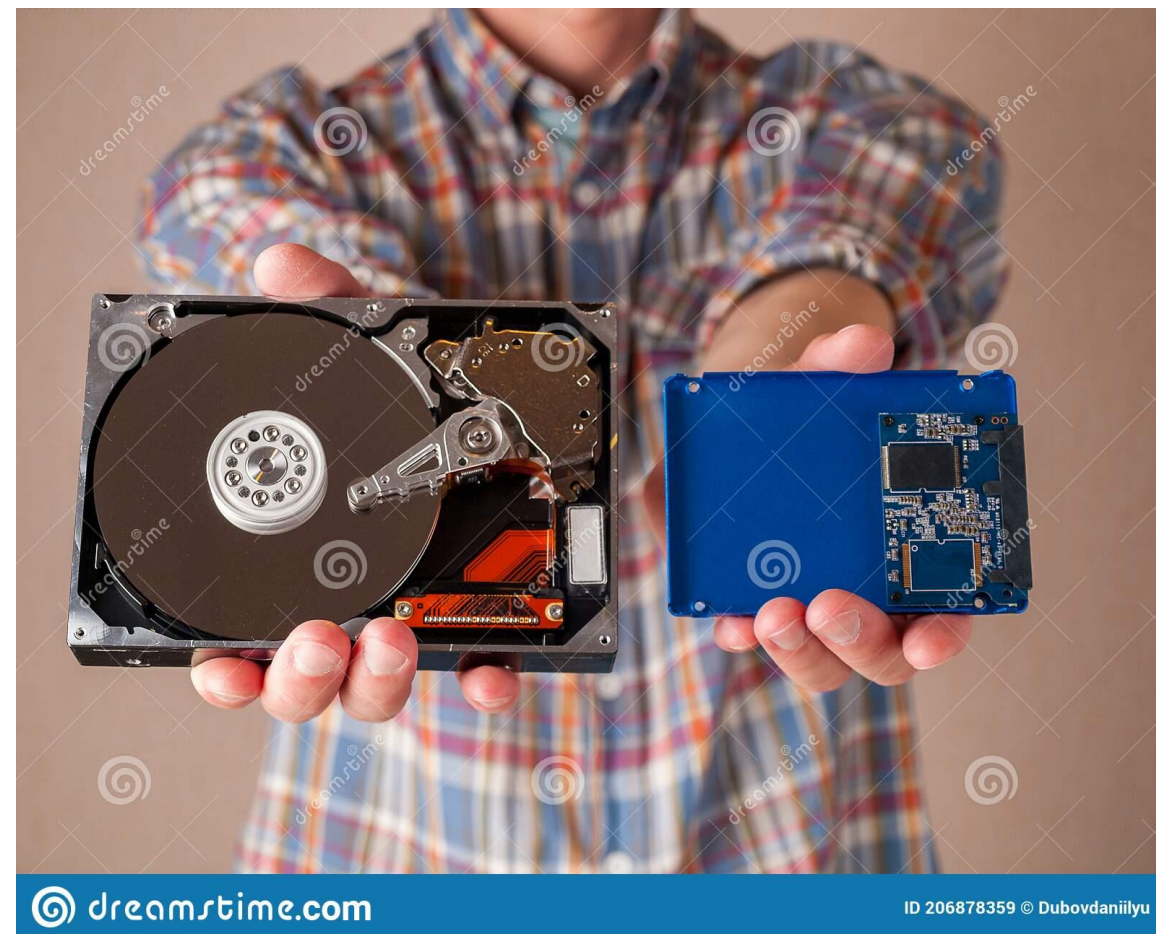




Traditional hard disk drive



Solid state hard drive





RAID



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Why RAID?

RAID

A technique that combines multiple disk drives into a logical unit (RAID set) and provides protection, performance, or both.

- Provides data protection against drive failures
- Improves storage system performance by serving I/Os from multiple drives simultaneously
- Two implementation methods
 - Software RAID implementation
 - Hardware RAID implementation

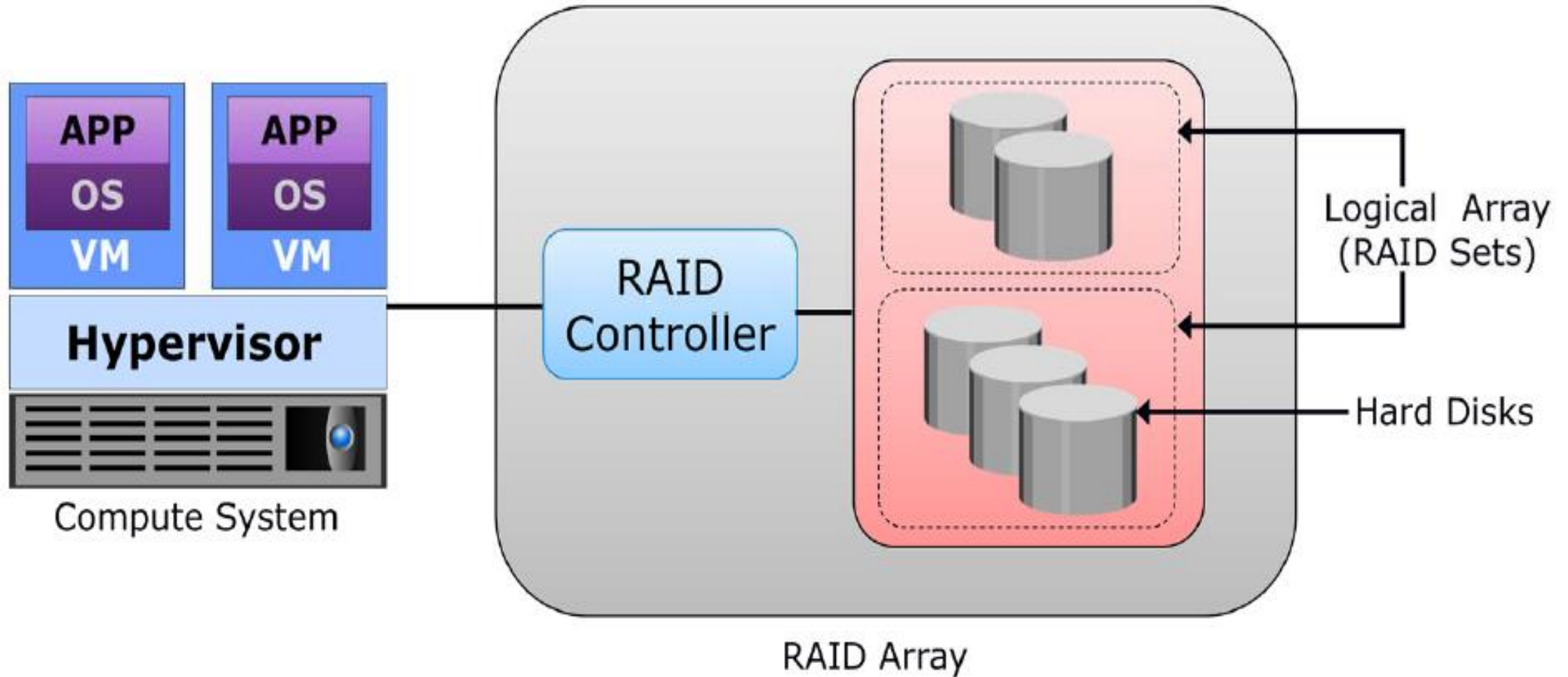


RAID Implementation Methods

- Software RAID implementation
 - ▶ Uses host-based software to provide RAID functionality
 - ▶ Limitations
 - ▶▶ Use host CPU cycles to perform RAID calculations, hence impact overall system performance
 - ▶▶ Support limited RAID levels
 - ▶▶ RAID software and OS can be upgraded only if they are compatible
- Hardware RAID Implementation
 - ▶ Uses a specialized hardware controller installed either on a host or on an array



RAID Array Components



RAID Techniques

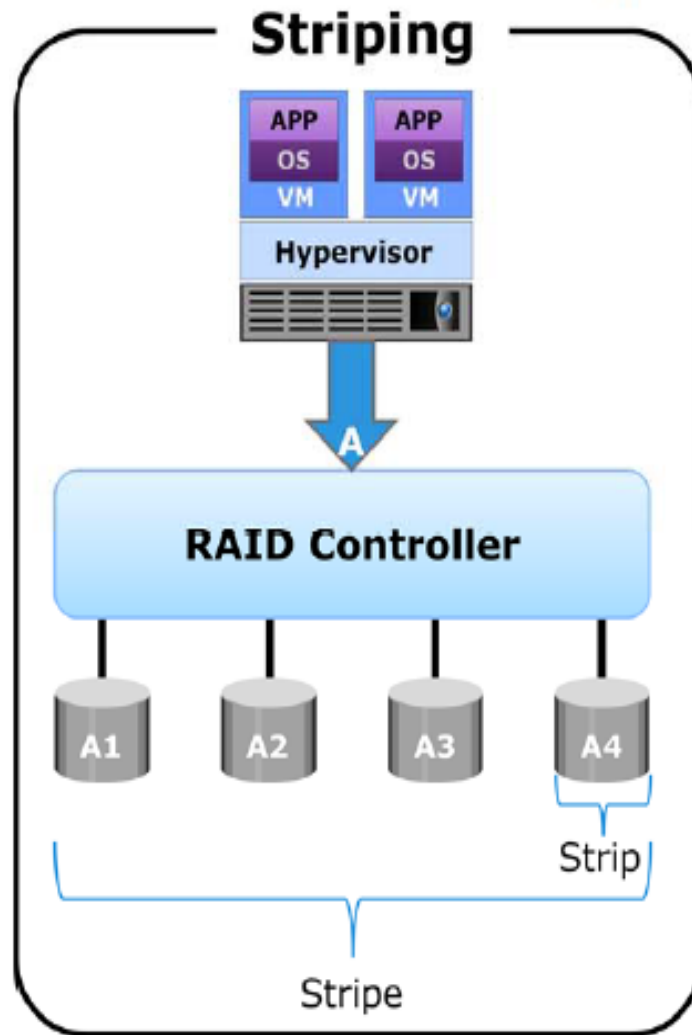


Figure 1

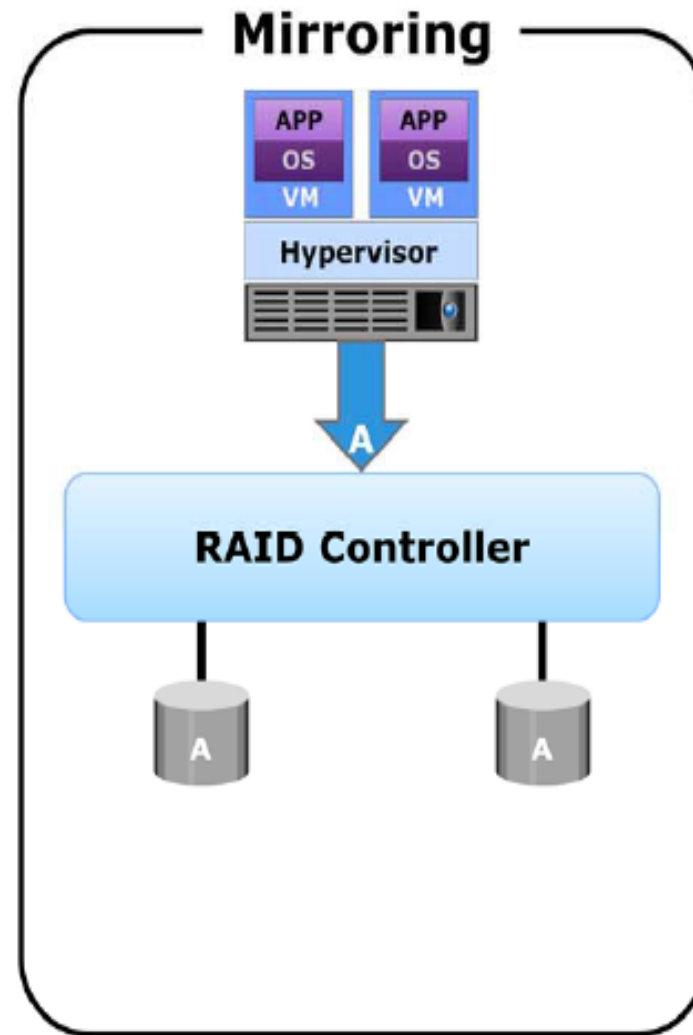


Figure 2

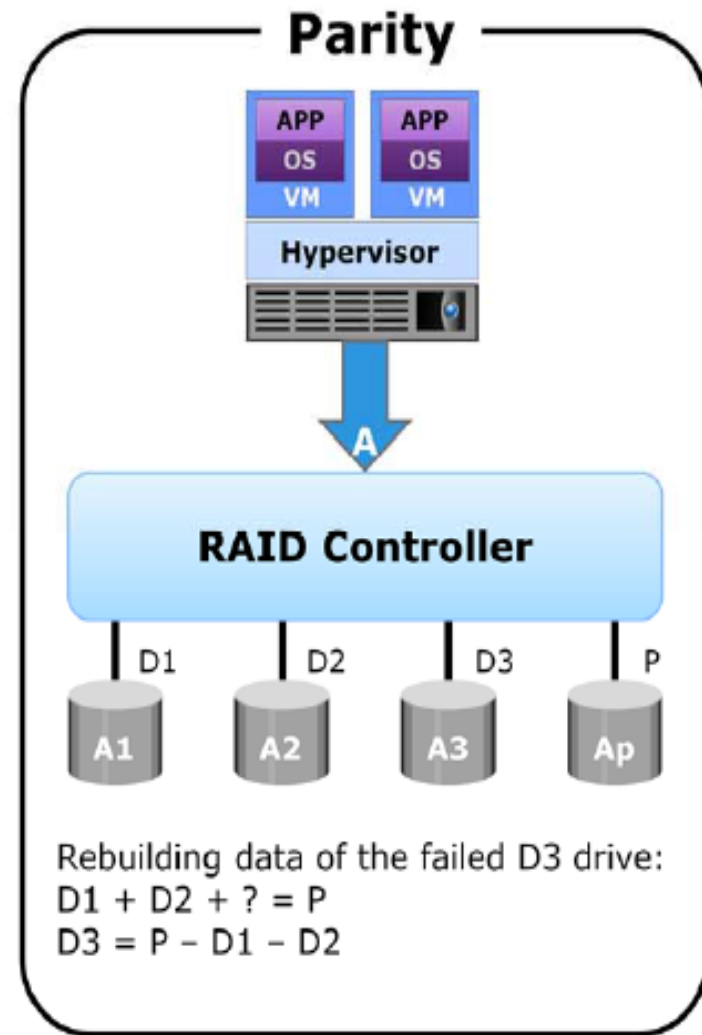


Figure 3



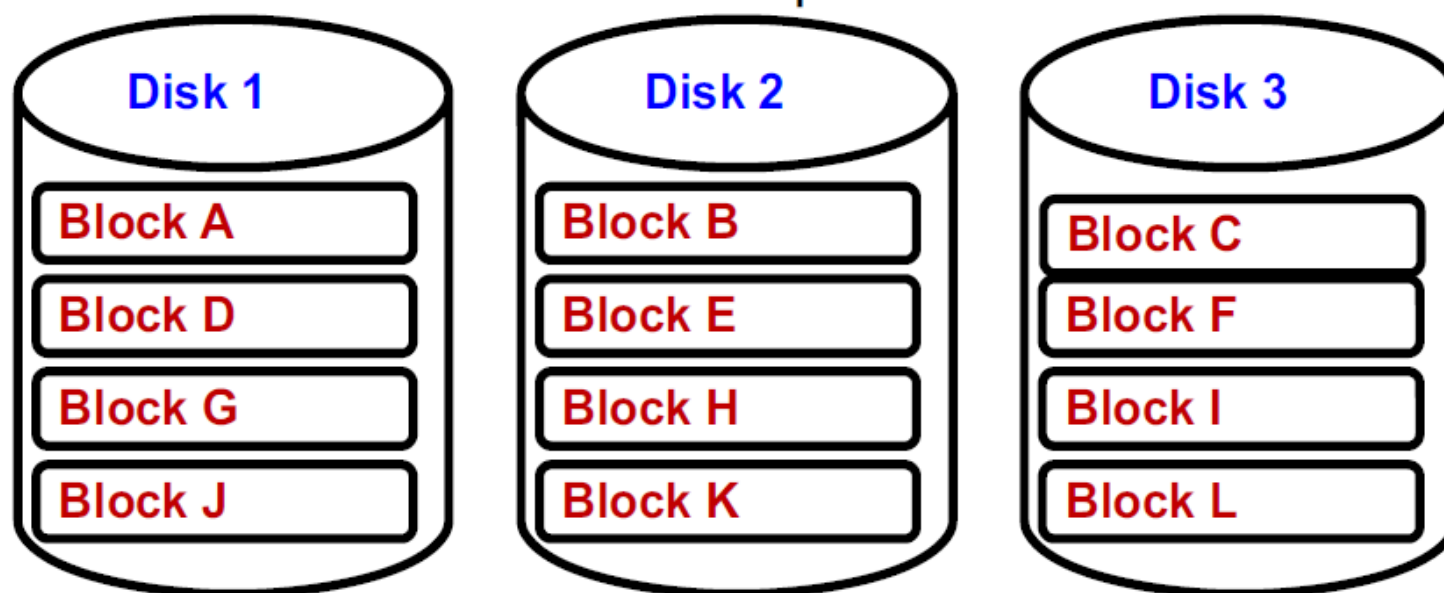
RAID Levels

- RAID 0 Block striping without parity or protection
- RAID 1 Mirrored disk
- RAID 2 Byte striping with ECC
- RAID 3 Byte striping with dedicated **parity** disk
- RAID 4 Block striping with dedicated **parity** disk
- RAID 5 Block striping with distributed **parity**
- RAID 6 Block striping with two sets of distributed **parity**
- RAID 0+1 Striped mirroring array
- RAID 1+0 Mirrored striping array



RAID 0 striped array

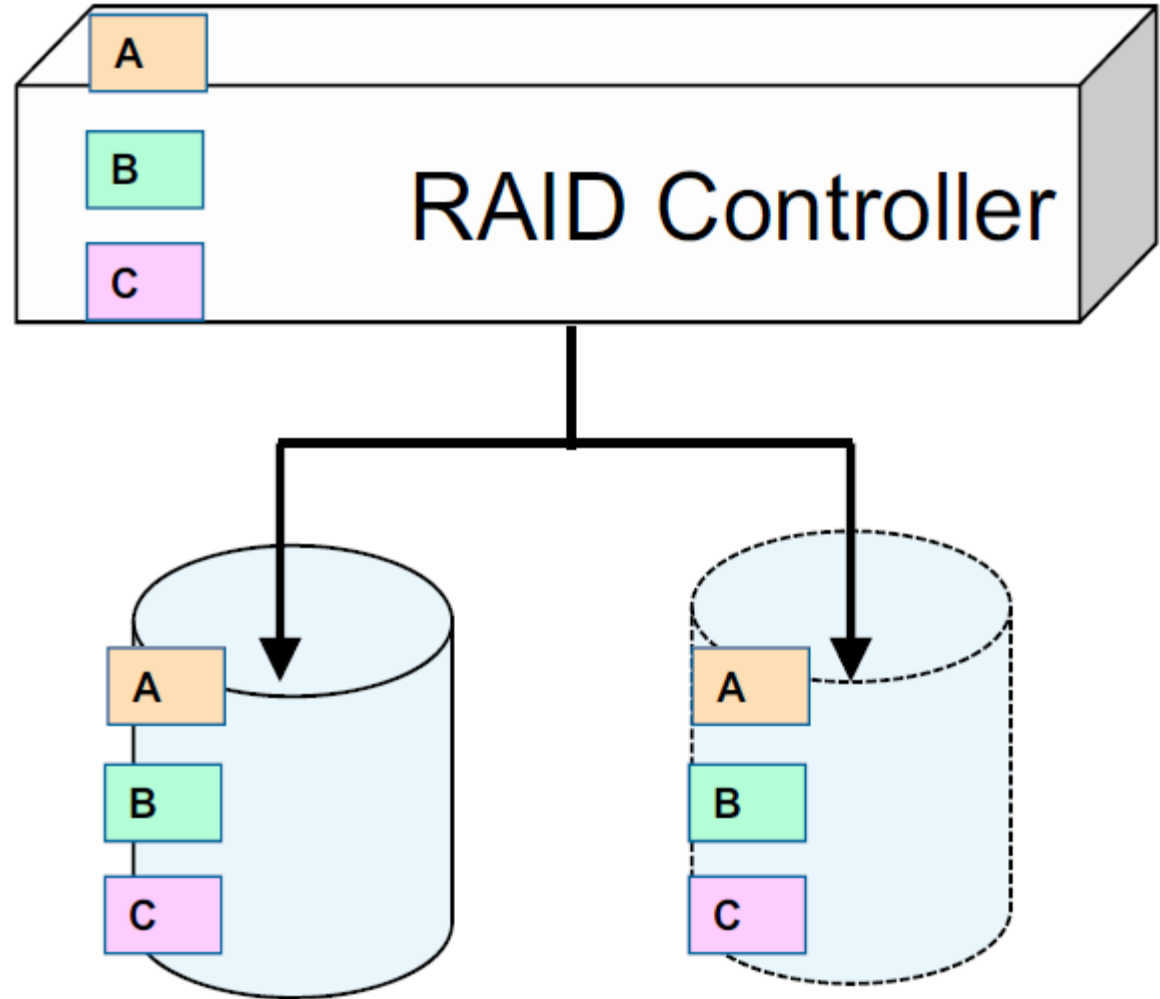
- RAID 0 implements a striped disk array.
- Each block of data is written to a separate disk drive.



- Same size required on all disk drives:
 - 40 GB + 40 GB + 40 GB = Logical disk of 120 GB
 - 40 GB + 20 GB + 40 GB = Logical disk of 60 GB
- One disk failure results in loss of all data in the array.

RAID 1 mirrored array

- RAID 1 technology provides automatic data redundancy
- RAID 1 writes data to each of two HDDs
- RAID 1 array also known as mirrored disk
- RAID 1 improves read performance by using both HDDs



RAID 3 – Byte Striping



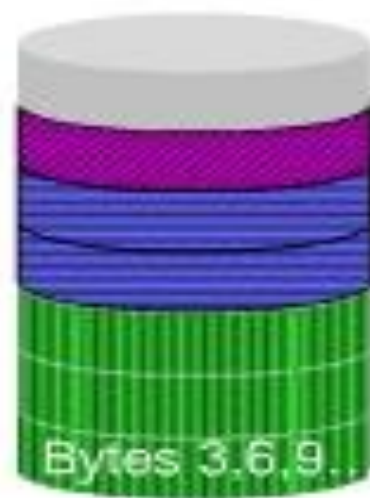
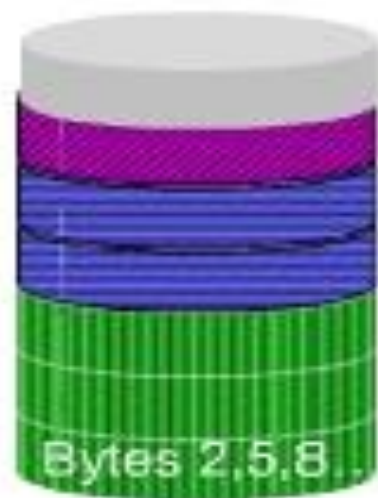
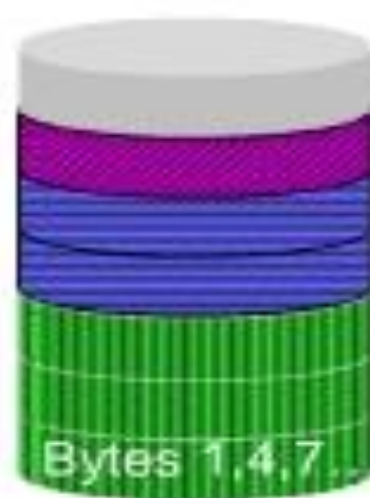
Reliability



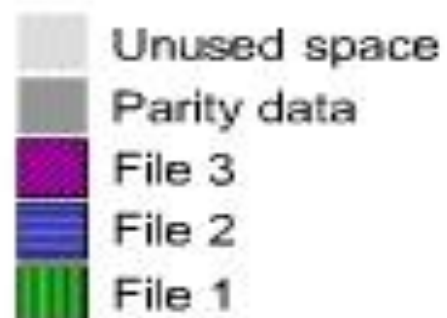
Read Performance



Write Performance



RAID 4 – Block Striping



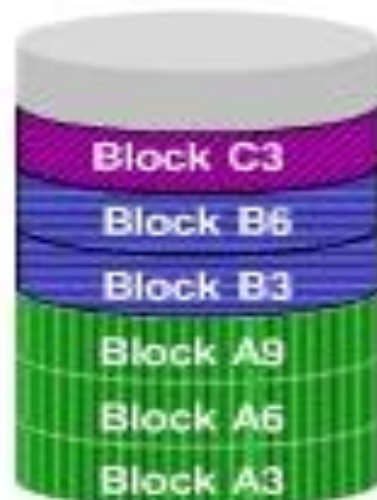
Reliability



Read Performance

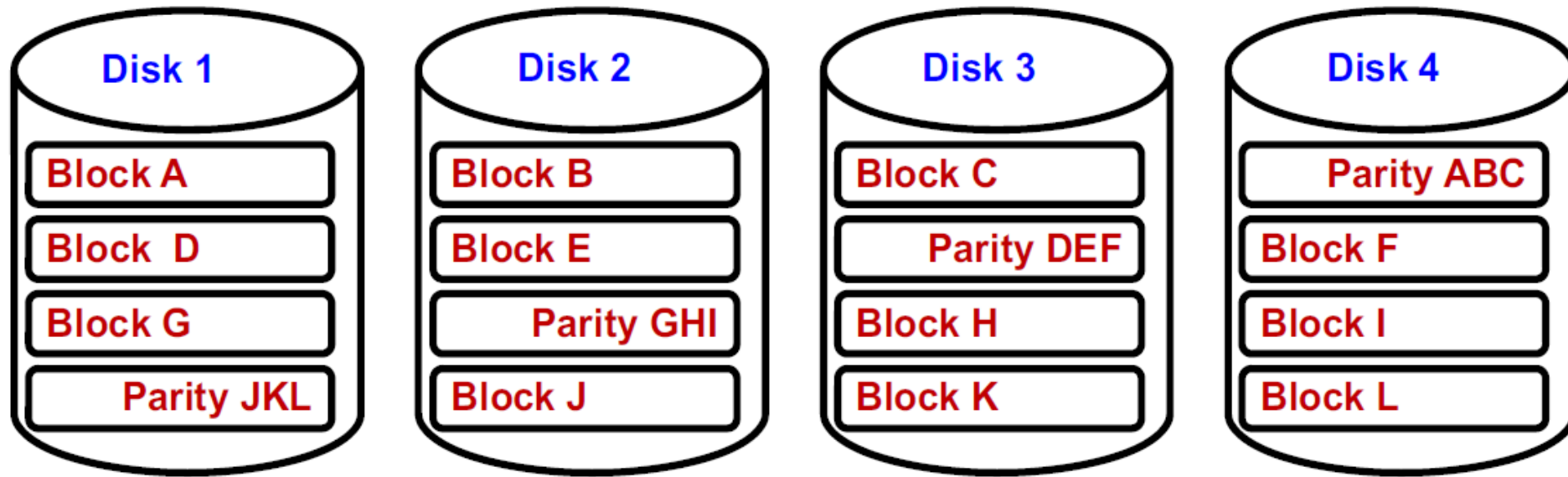


Write Performance



RAID 5 block-level stripe with distributed parity

- Most common secure RAID level.
- Read data transactions are faster.



- Minimum of three disks for a RAID 5 array
 - Three drives of 100 GB = 200 GB available
 - 10 drives of 100 GB = 900 GB available.
- Write data transactions are slower due to parity calculation

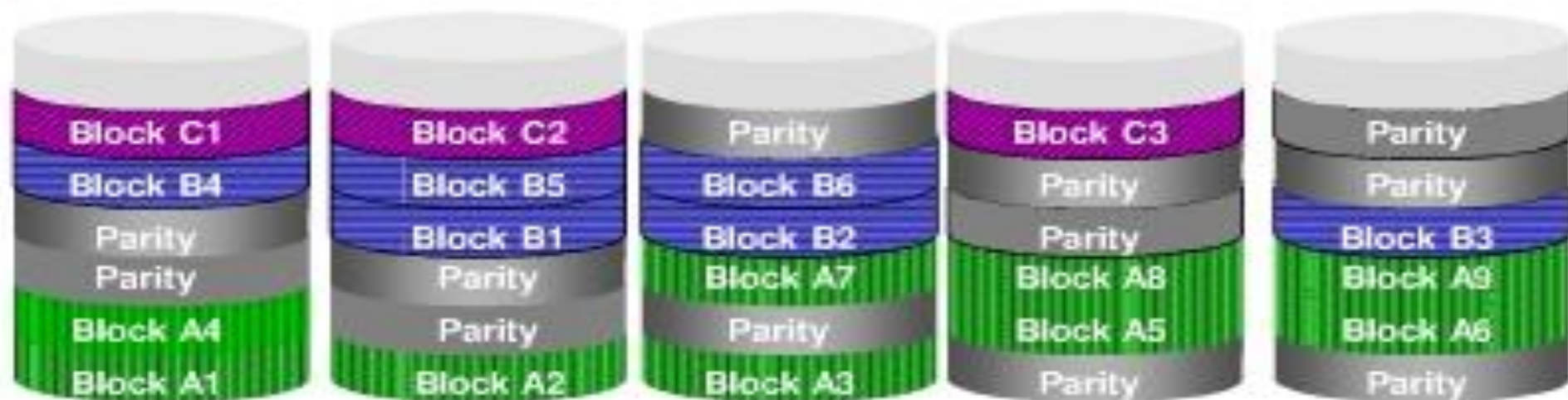
RAID 6 – Block Striping



Reliability ★★★★★

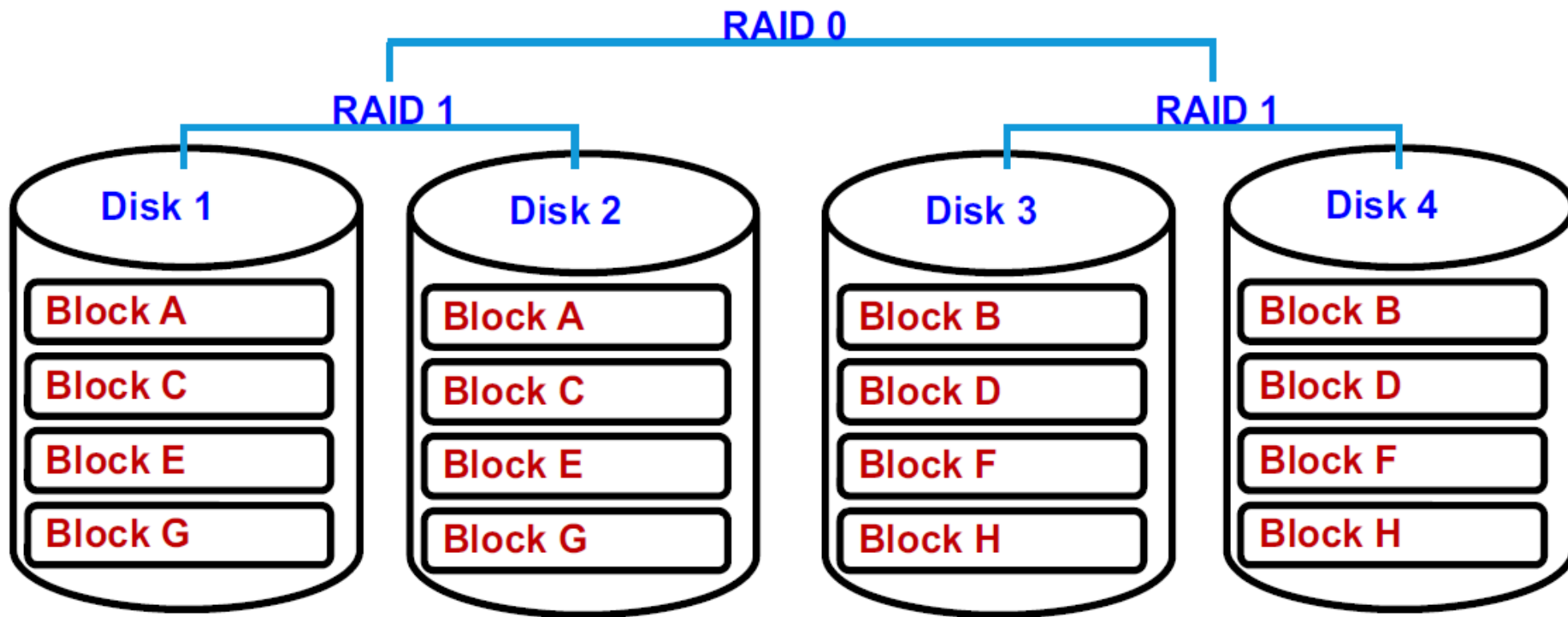
Read Performance ★★★★★

Write Performance ★★★



RAID 10 mirror with stripe

- A stripe of mirrors
- Each RAID 1 cluster has a minimum of two drives



- Minimum of four drives to implement
- Fault-tolerance improved

BREAKDOWN OF COMMON RAID LEVELS

Enterprise

RAID LEVEL	METHOD	HARDWARE / SOFTWARE	MINIMUM # OF DISKS	COMMON USAGE	PROS	CONS
JBOD	SPANNING	 	2	INCREASE CAPACITY	COST-EFFECTIVE STORAGE	NO PERFORMANCE OR SECURITY BENEFITS
0	STRIPING	 	2	HEAVY READ OPERATIONS	HIGH PERFORMANCE (SPEED)	DATA IS LOST IF ONE DISK FAILS
1	MIRRORING	 	2	STANDARD APP SERVERS	FAULT TOLERANCE, HIGH READ PERFORMANCE	LAG FOR WRITE OPS, REDUCED STORAGE (BY 1/2)
5	STRIPING & PARITY		3	NORMAL FILE STORAGE & APP SERVERS	SPEED + FAULT TOLERANCE	LAG FOR WRITE OPS, REDUCED STORAGE (BY 1/3)
6	STRIPING & DOUBLE PARITY		4	LARGE FILE STORAGE & APP SERVERS	EXTRA LEVEL OF REDUNDANCY, HIGH READ PERFORMANCE	LOW WRITE PERFORMANCE, REDUCED STORAGE (BY 2/5)
10 (1+0)	STRIPING & MIRRORING		4	HIGHLY UTILIZED DATABASE SERVERS	WRITE PERFORMANCE + STRONG FAULT TOLERANCE	REDUCED STORAGE (1/2), LIMITED SCALABILITY

What Happened to 2-4 and 6-9?

The RAID levels described above are the most common levels used in enterprise scenarios. The levels in between are highly specialized and only make sense in very specific scenarios.

Components of Storage Array

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What is an Intelligent Storage System?

Intelligent Storage System

A feature-rich RAID array that provides highly optimized I/O processing capabilities.

- Has a purpose-built operating environment
 - Provides intelligent resource management capability
- Provides large amount of cache
- Provides multiple I/O paths

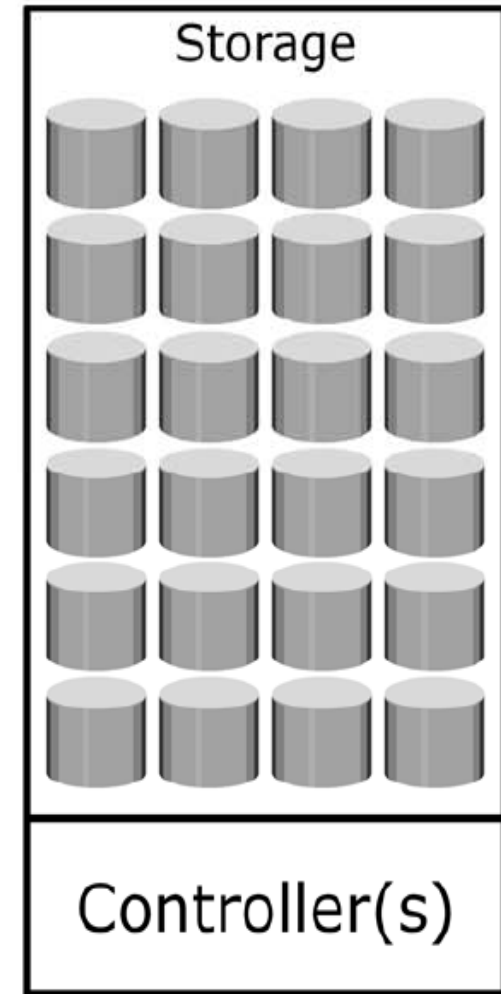
Features

- Supports combination of HDD and SSD
- Service massive amount of IOPS
- Scale-out architecture
- Deduplication, compression, and encryption
- Automated storage tiering
- Virtual storage provisioning
- Multi-tenancy
- Supports APIs to integrate with SDDC and cloud
- Data protection



Components of Intelligent Storage System

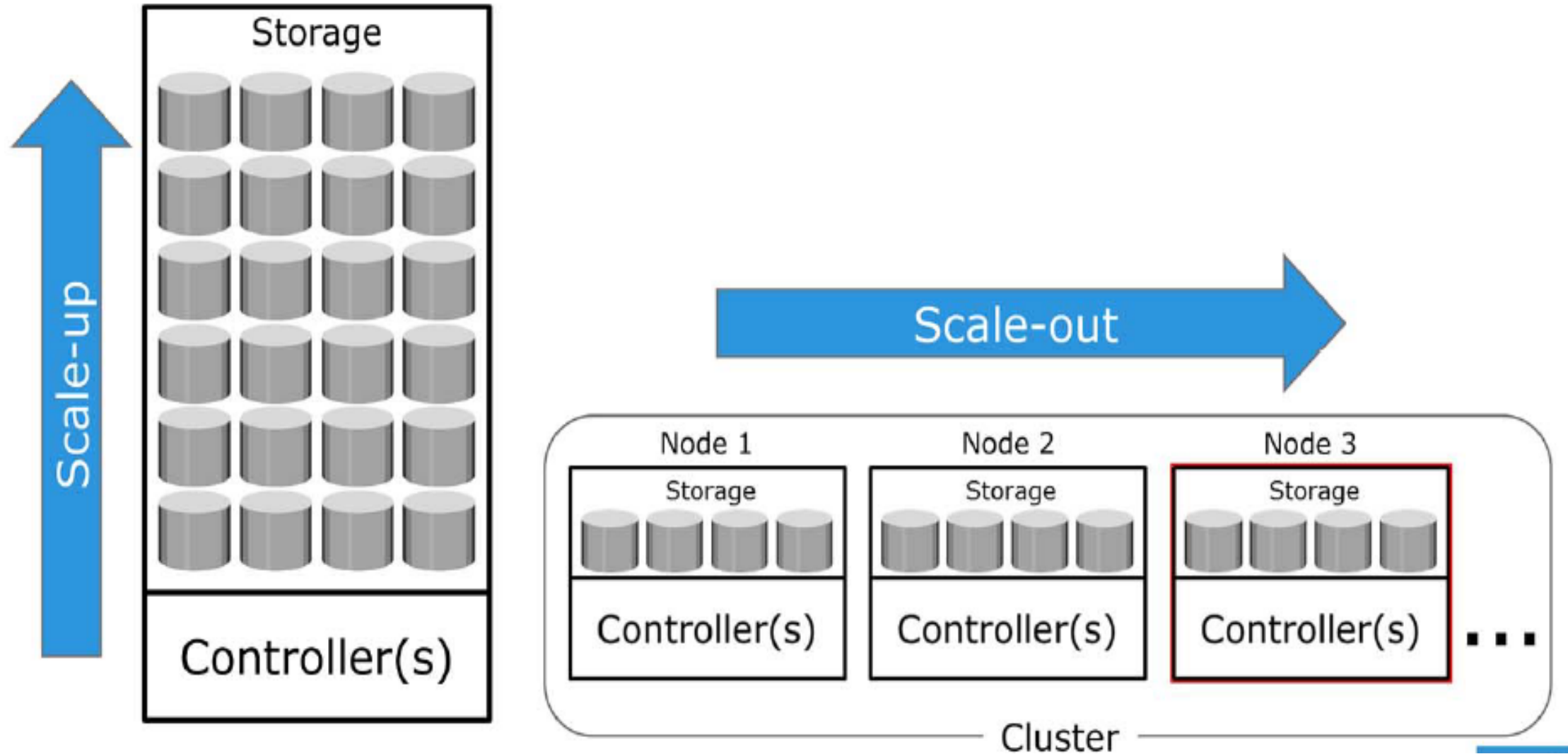
- Two key components of an ISS
 - Controller
 - Block-based
 - File-based
 - Object-based
 - Unified
 - Storage
 - All HDDs
 - All SSDs
 - Combination of both



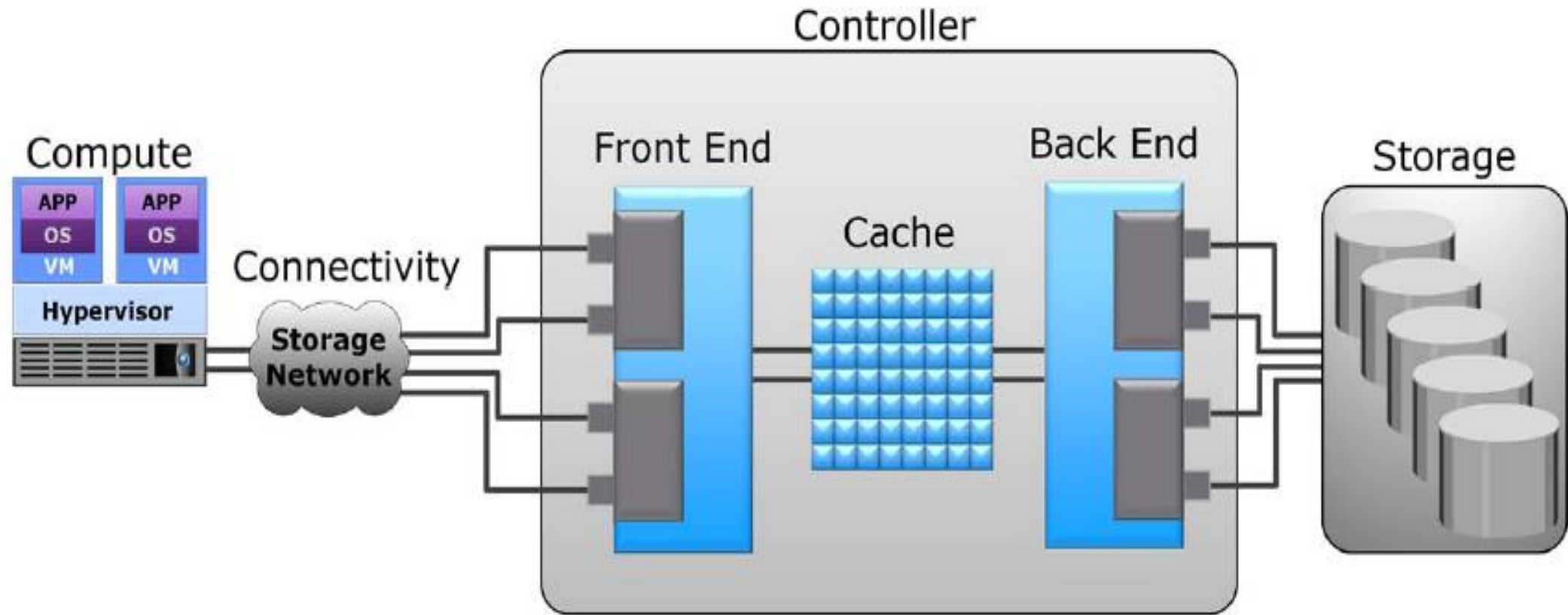
Intelligent Storage System



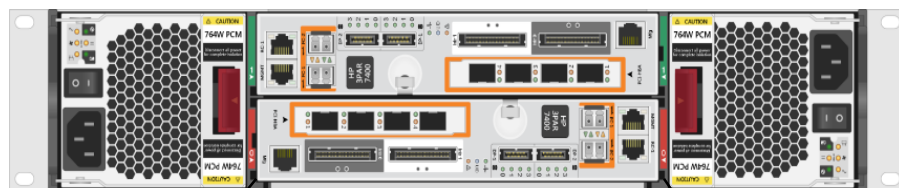
Scale-up Vs. Scale-out Architecture



Components of a Controller



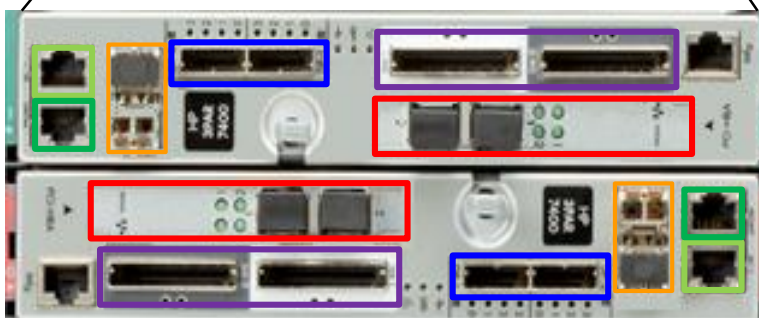
Controller enclosure



Rear view



Front view

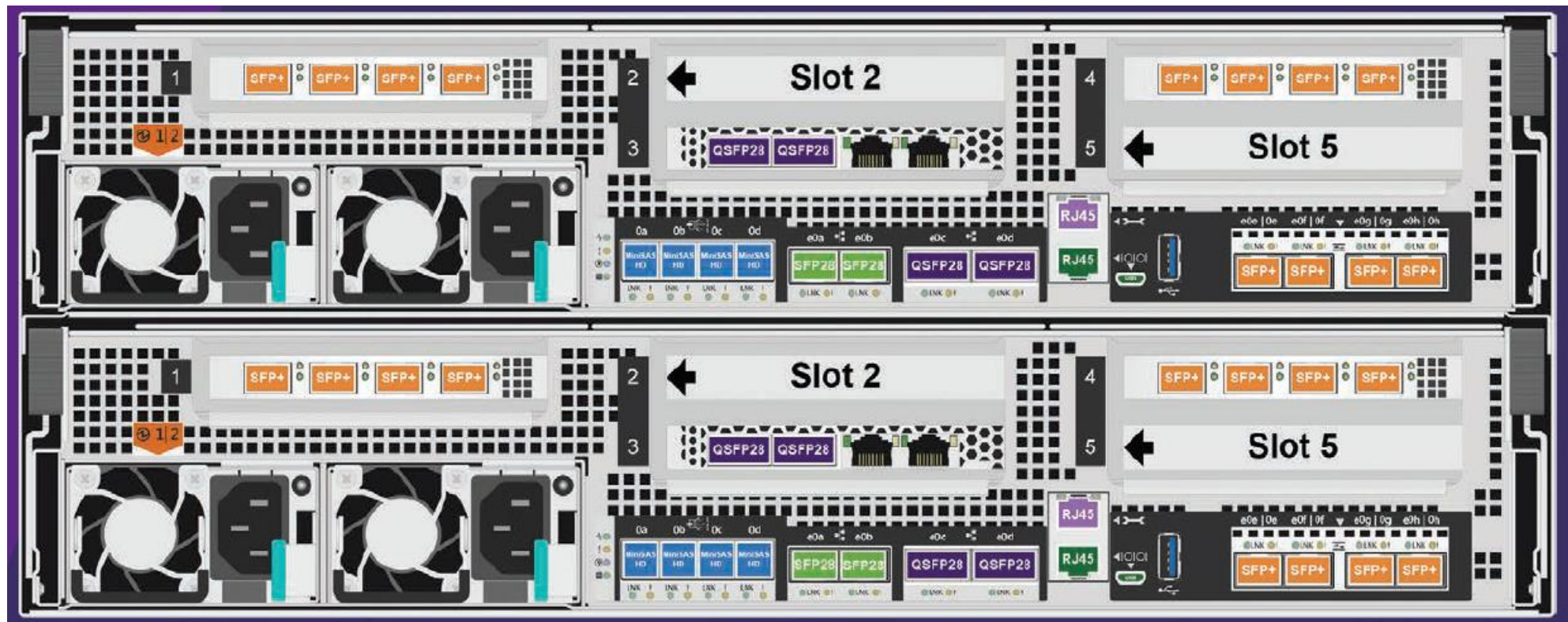


Node 1

Node 0

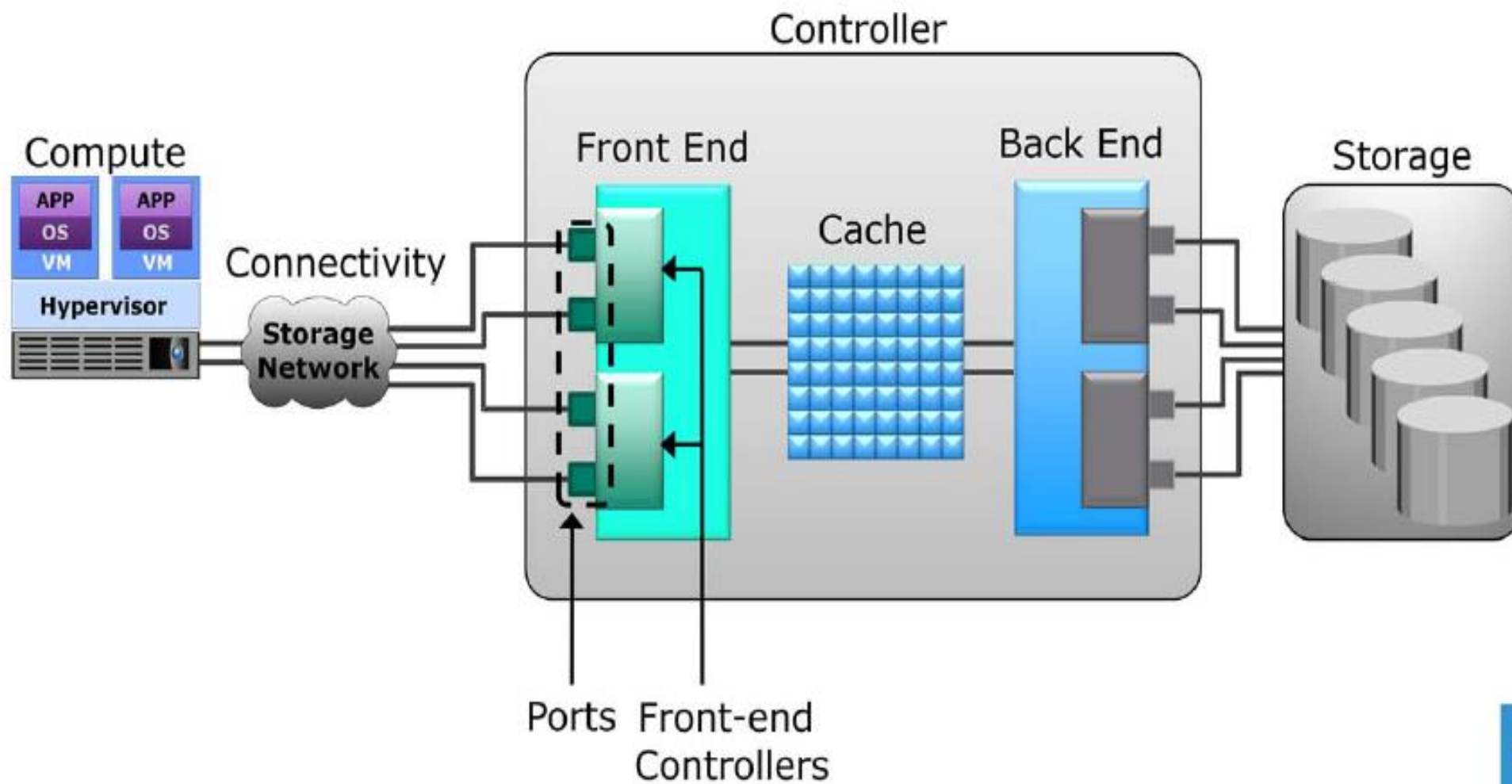
- ☐ Built-in Eth Remote Copy Port
- ☐ Eth Mgmt. Port
- ☐ 2x built-in 8Gbit/s FC
- ☐ 2x 4-lane 6Gbit/s SAS for drive chassis connection
- ☐ Optional PCIe card
 - 4x 8GB/s HBA or
 - 2x 10Gbit/s CNA
- ☐ Controller interconnect

Controller Rear View



Components of a Controller

Front End



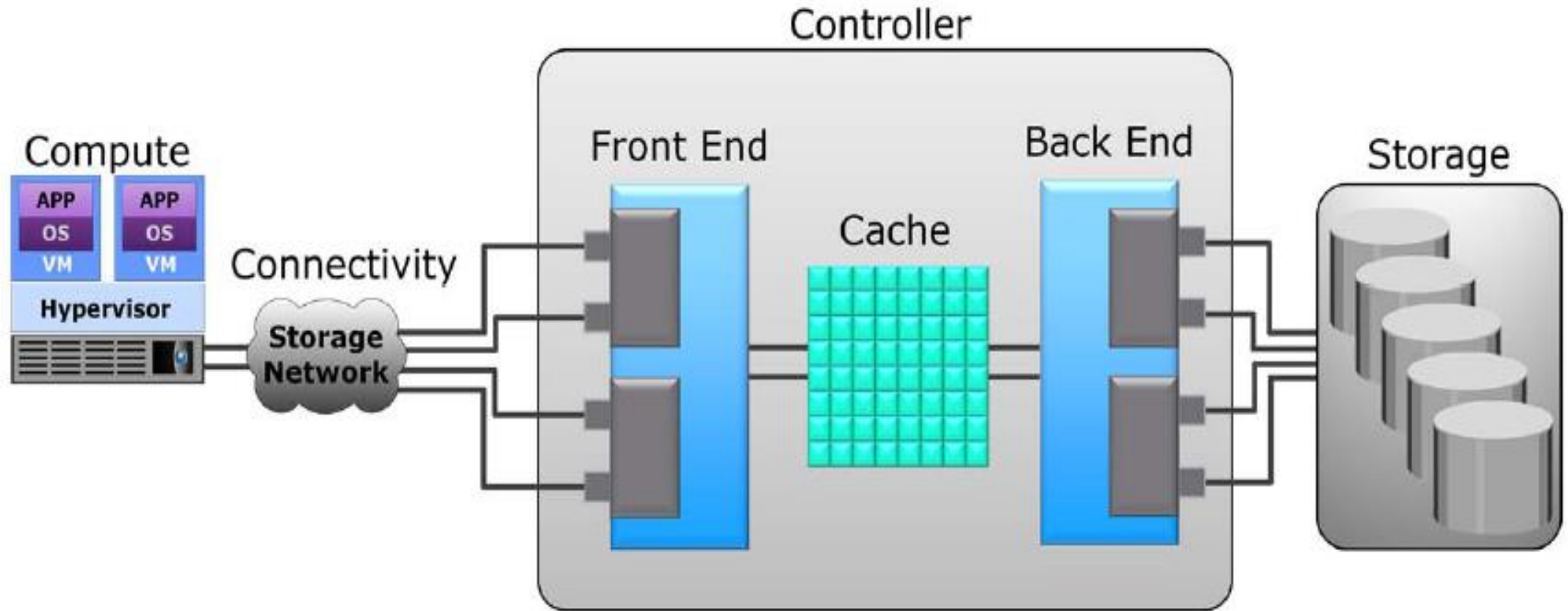
Host Ports or Front End Ports

- ▶ **Ports are used to connect to Hosts.**
 - ▶ FC Ports
 - ▶ FCoE Ports (UTA – Universal Target Adapter)
 - ▶ iSCSI Ports
 - ▶ Ethernet Ports



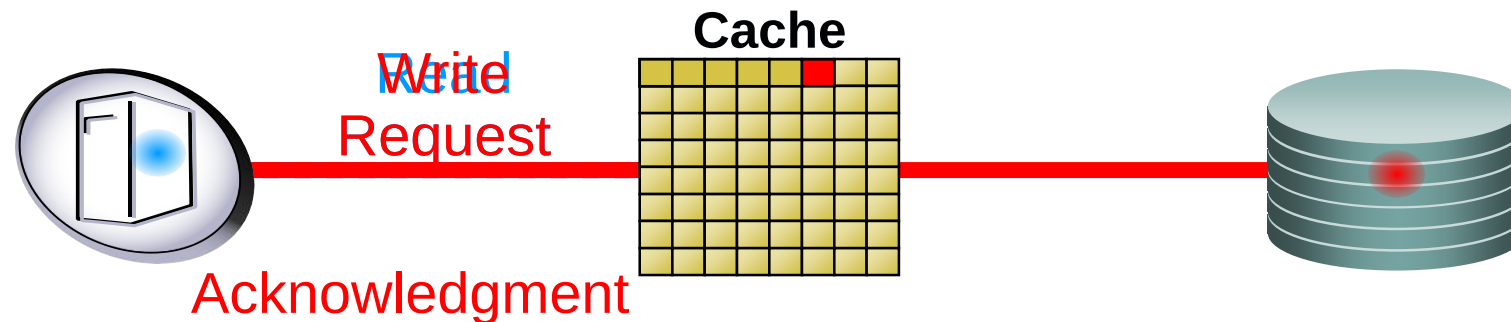
Components of a Controller

Cache

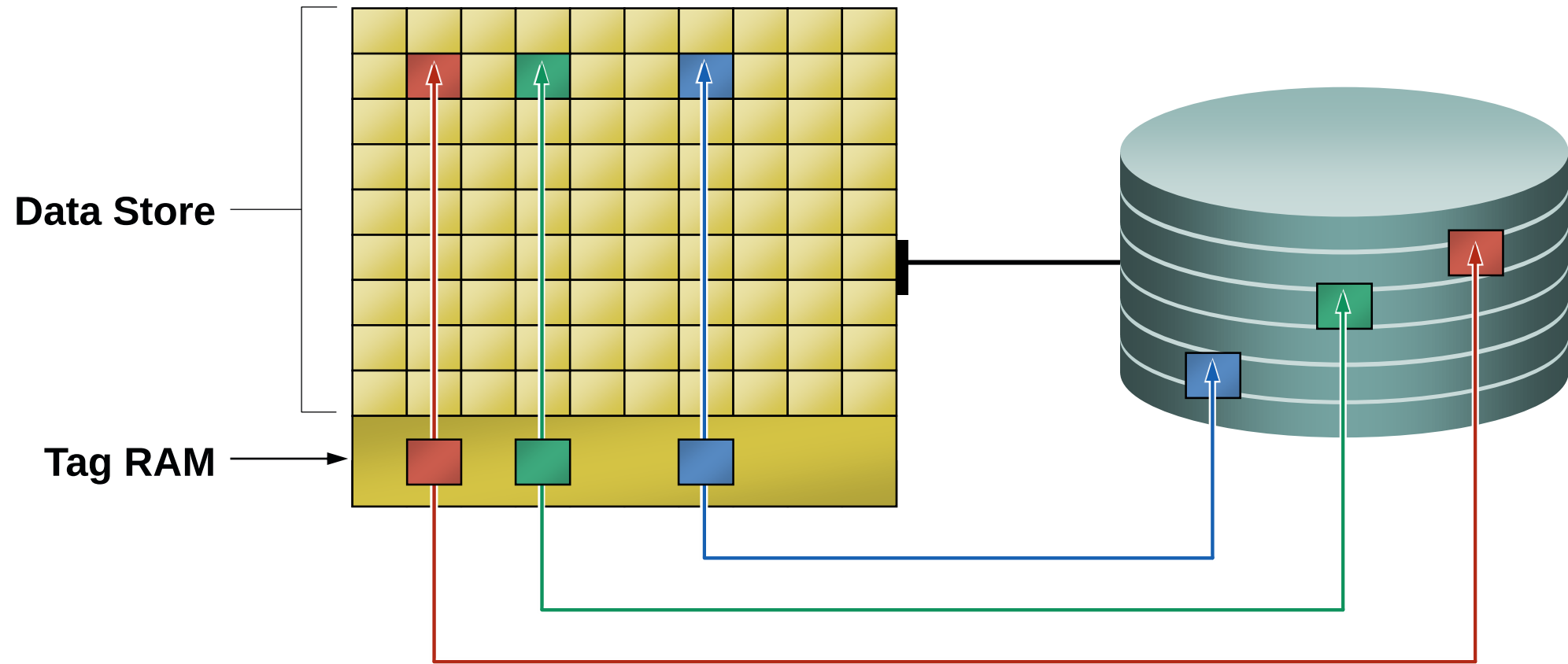


What is Cache in a Storage System

A memory space used by an intelligent storage system to reduce the time required to service I/O requests from the host

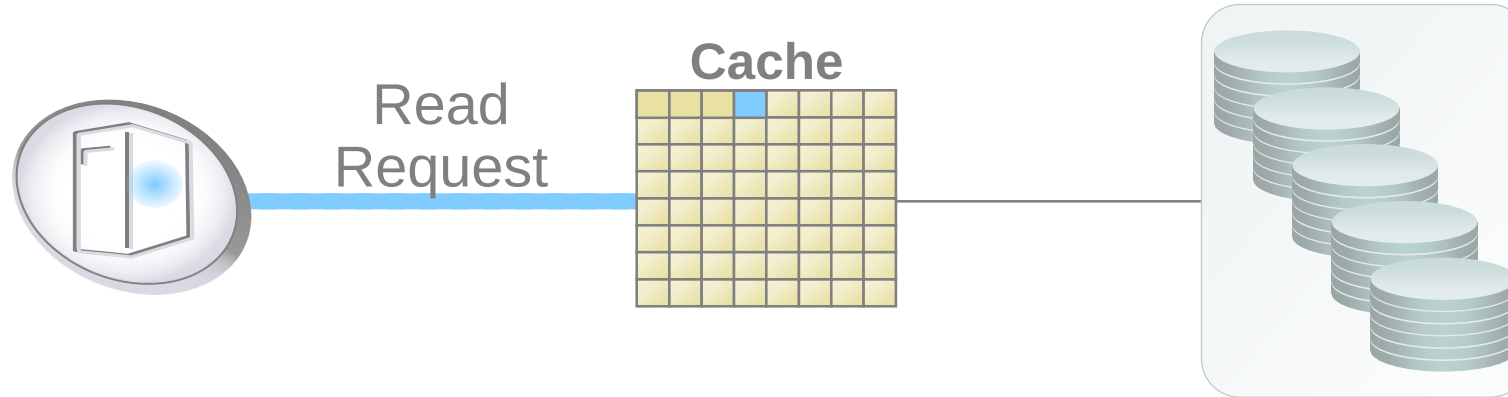


How Cache is Structured

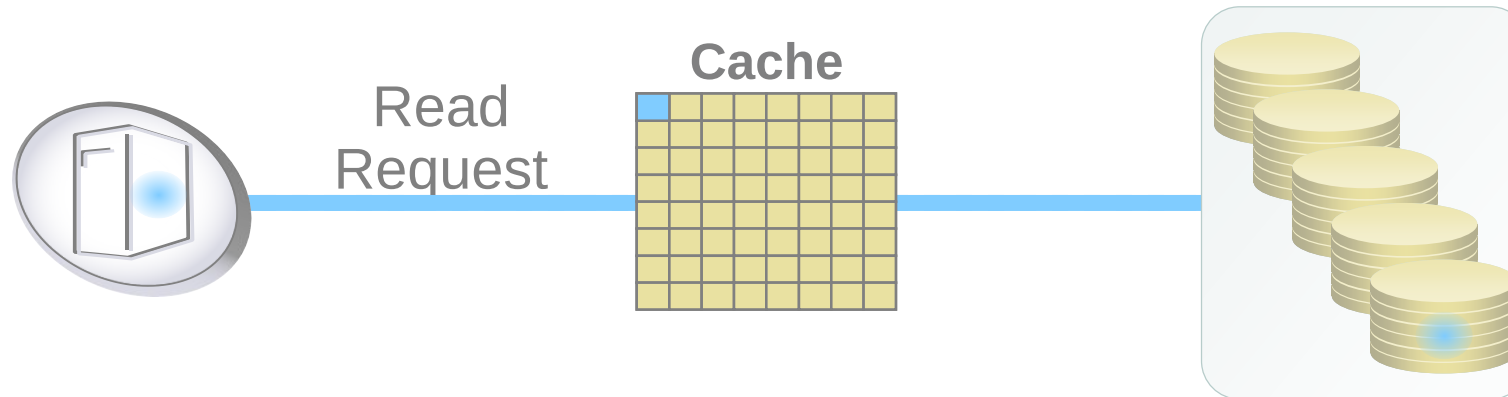


Read Cache 'Hits' and 'Misses'

Data found in cache = 'Hit'

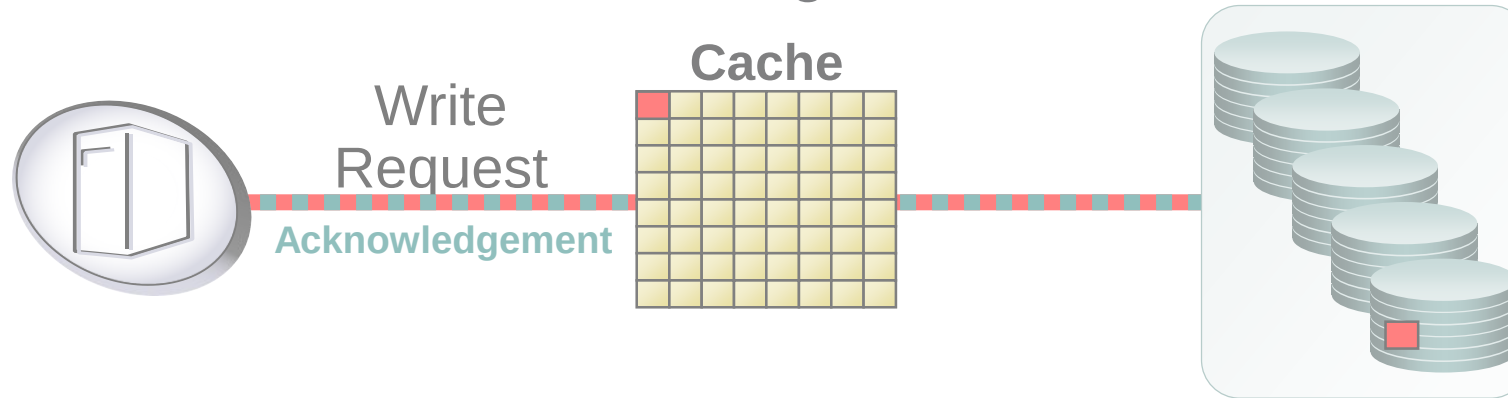


No data found = 'Miss'

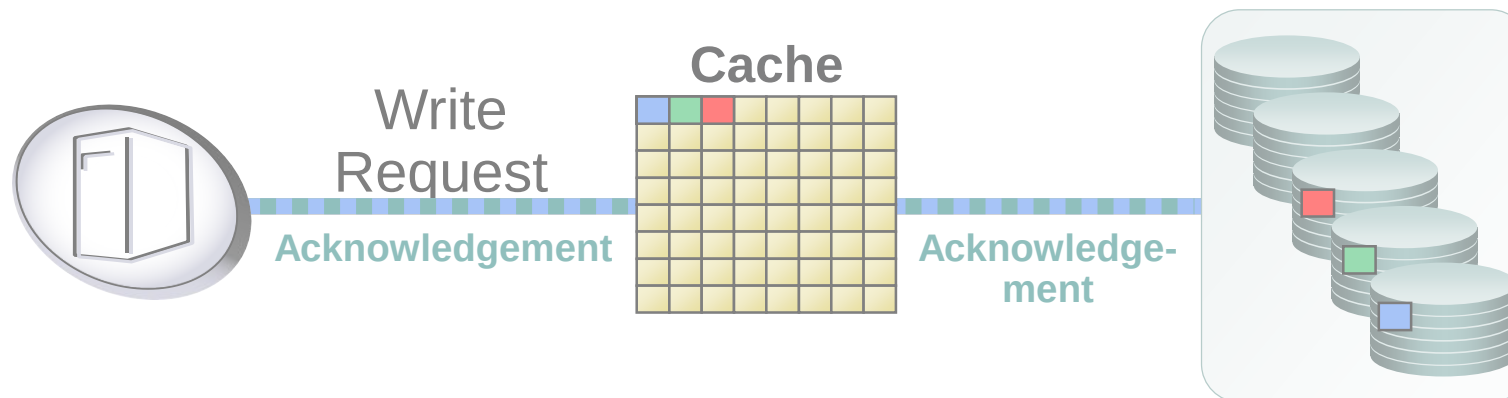


Write Algorithms

Write-through Cache

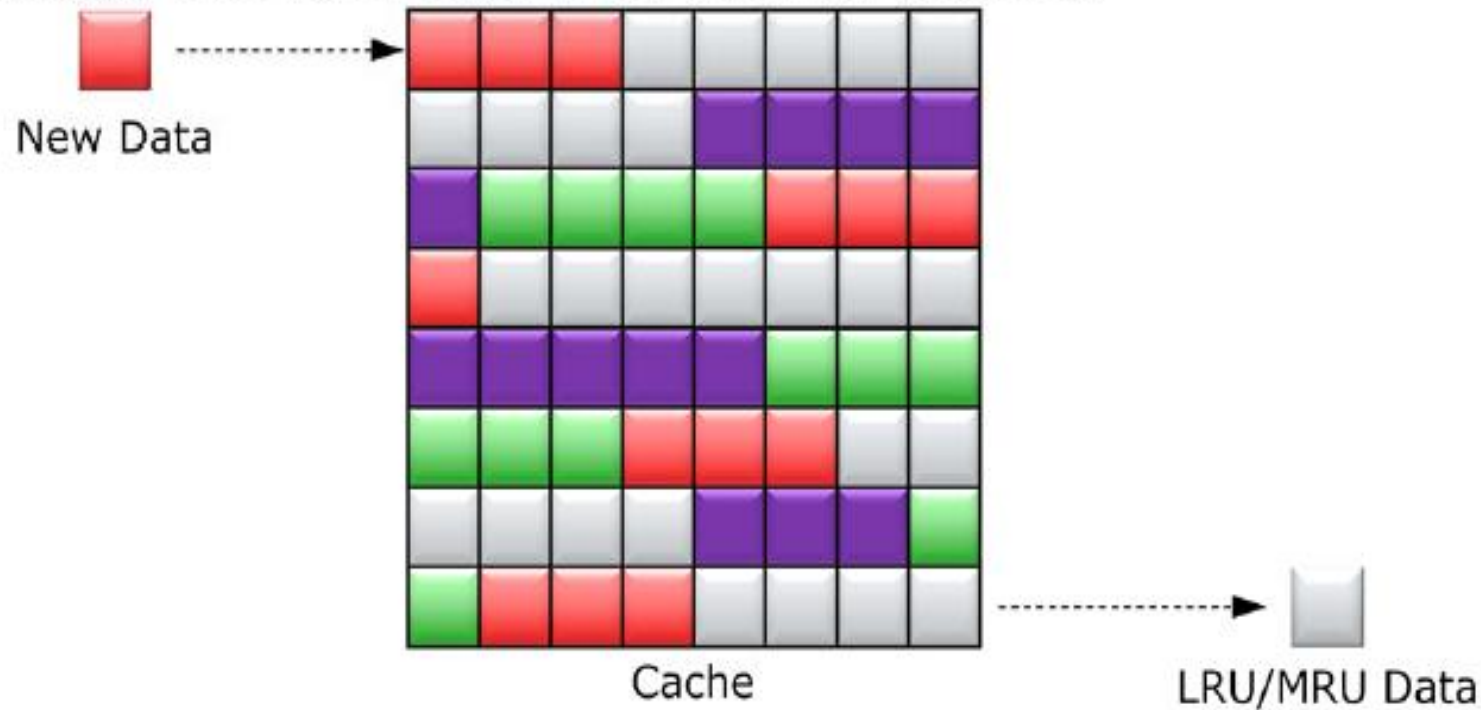


Write-back



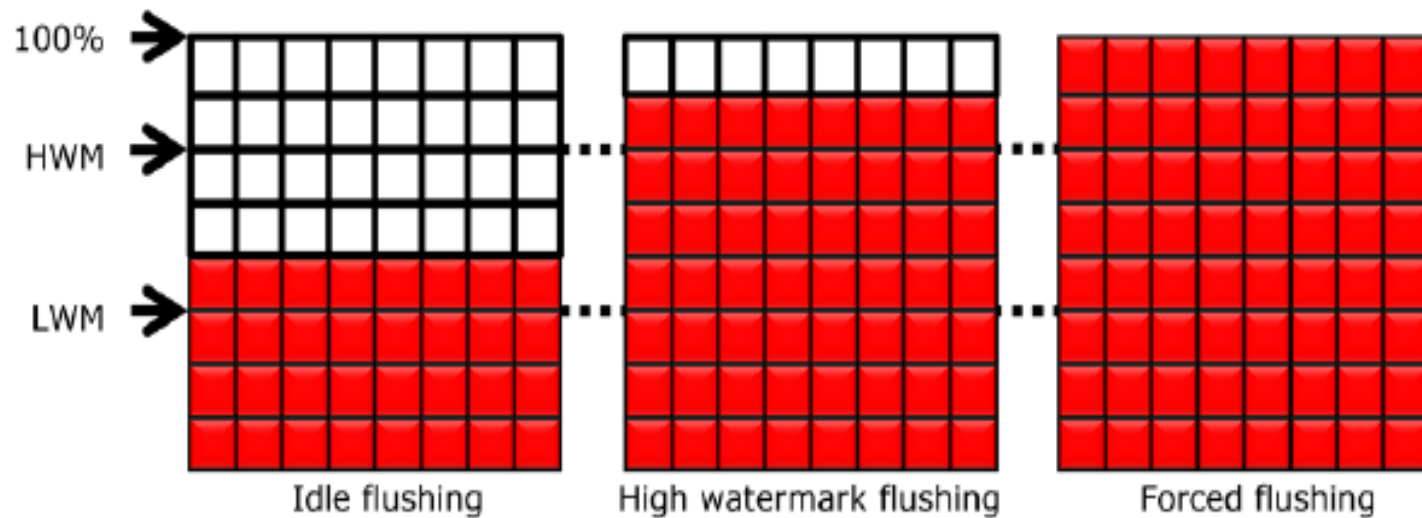
Cache Management: Algorithms

- Least recently used (LRU)
 - Discards data that have not been accessed for a long time
- Most recently used (MRU)
 - Discards data that have been most recently accessed



Cache Management: Watermarking

- Manages I/O burst through flushing process
 - Flushing is the process of committing data from cache to the storage drives
- Three modes of flushing to manage cache utilization are:
 - Idle flushing
 - High watermark flushing
 - Forced flushing



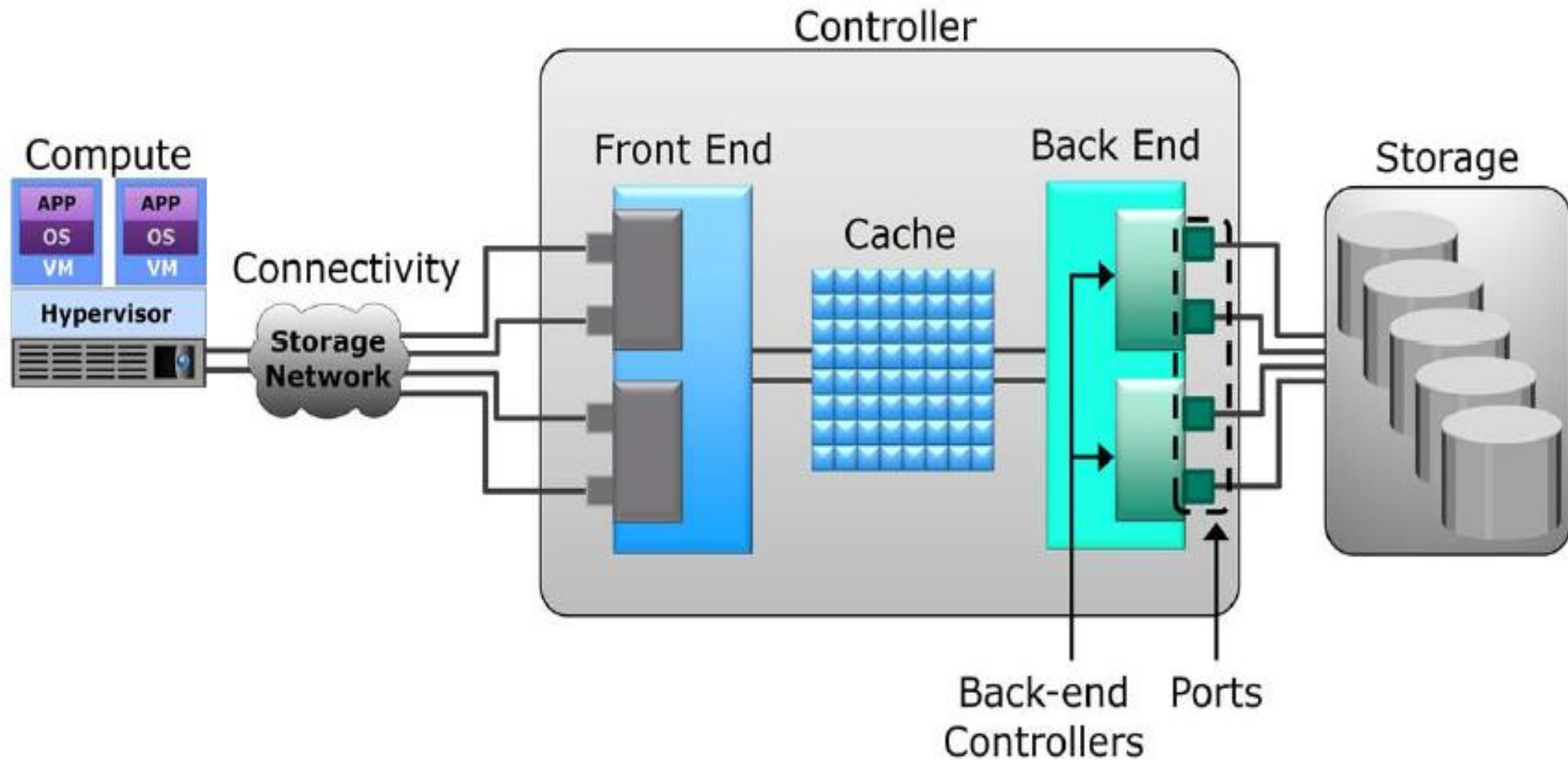
Cache Data Protection

- Protects data in the cache against power or cache failures:
 - Cache mirroring
 - Provides protection to data against cache failure
 - Each write to the cache is held in two different memory locations on two independent memory cards
 - Cache vaulting
 - Provides protection to data against power failure
 - In the event of power failure, uncommitted data is dumped to a dedicated set of drives called vault drives



Components of a Controller

Back End

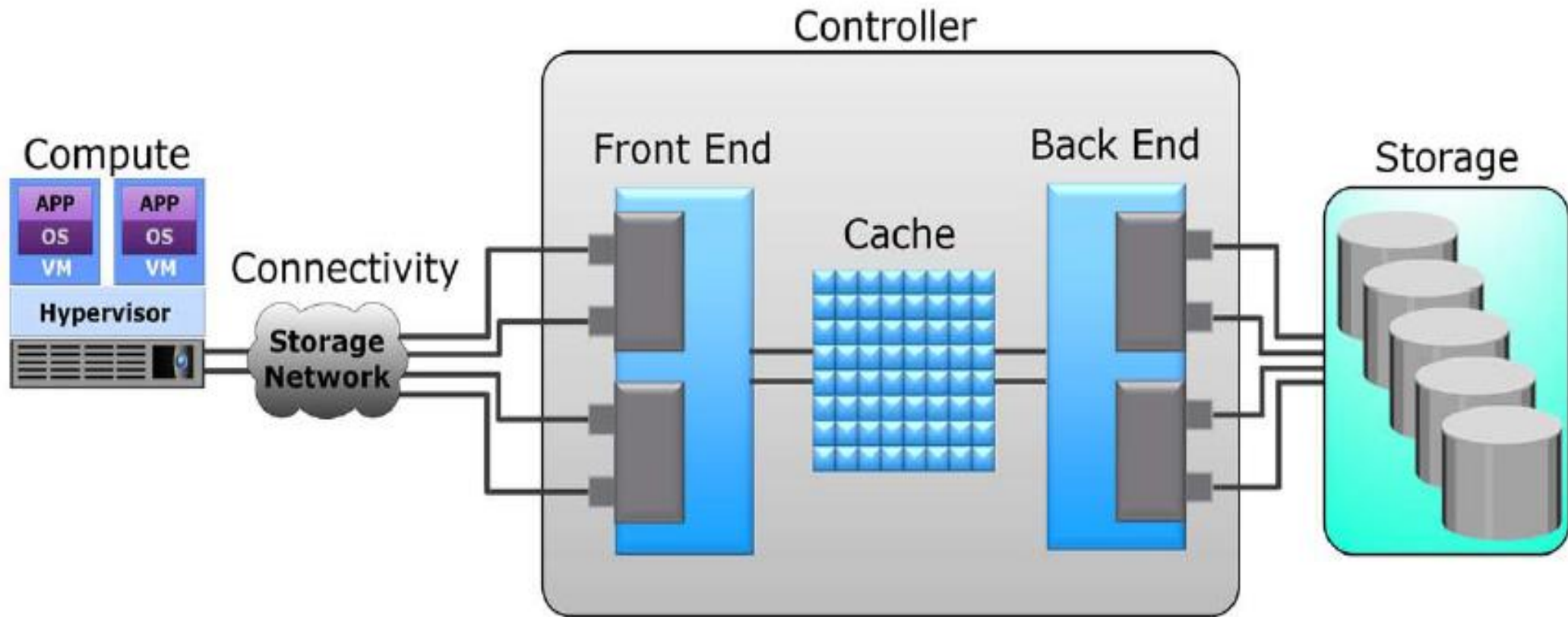


Storage Ports or Backend Ports

- ▶ Ports are used to connect the disk enclosures.
 - ▶ FC Ports
 - ▶ SAS Ports



Storage





Storage Management Interfaces



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Management PORT and Console

- ▶ All the storage controllers has Management port.
- ▶ This port is used to configure, manage and monitor the storage array
- ▶ This management port is not used for any host data communication
- ▶ Direct Consoles (Hyper terminals)
- ▶ CLI
- ▶ GUI
- ▶ API



Administrative interfaces



Unisphere GUI

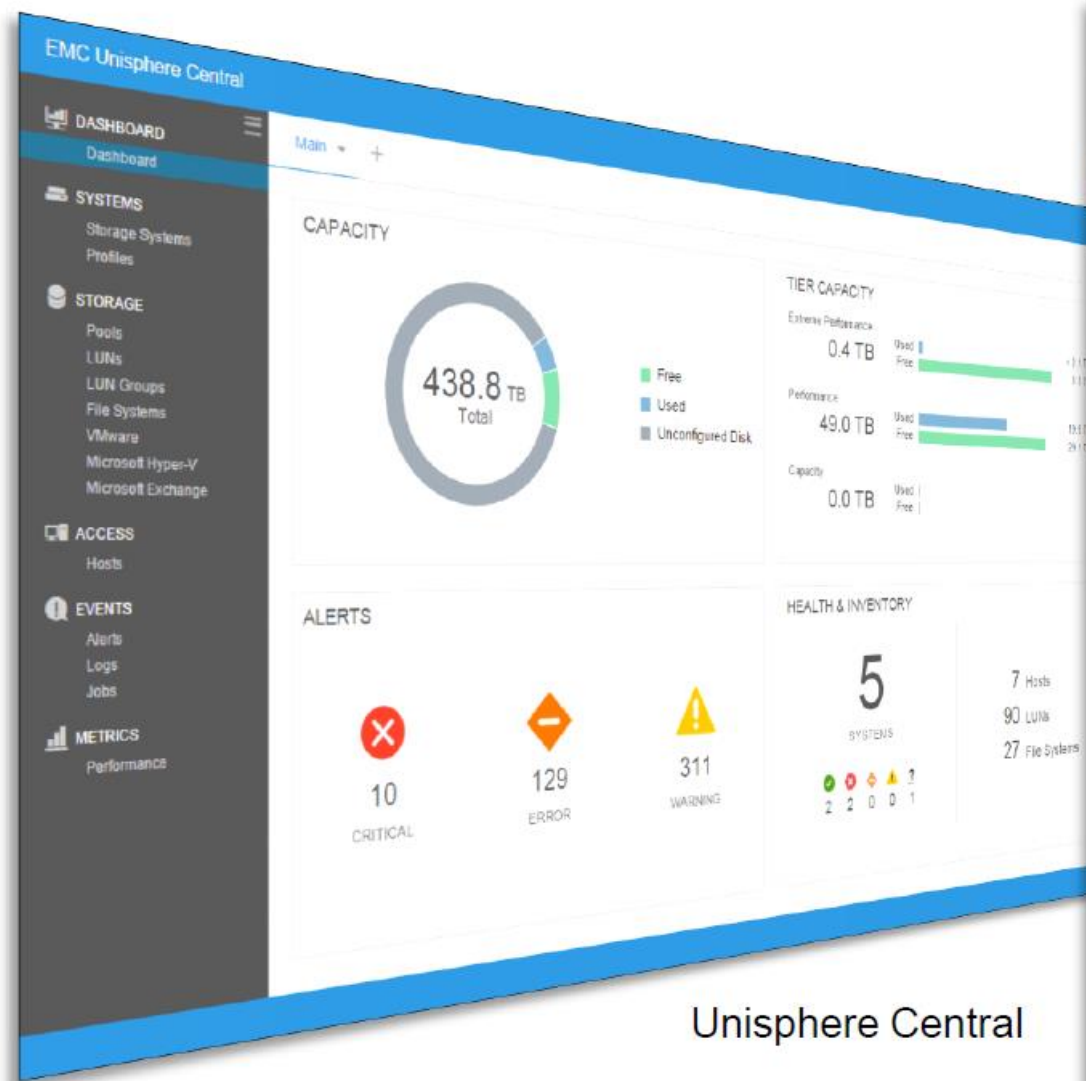


Unisphere CLI



REST API

Centralized management



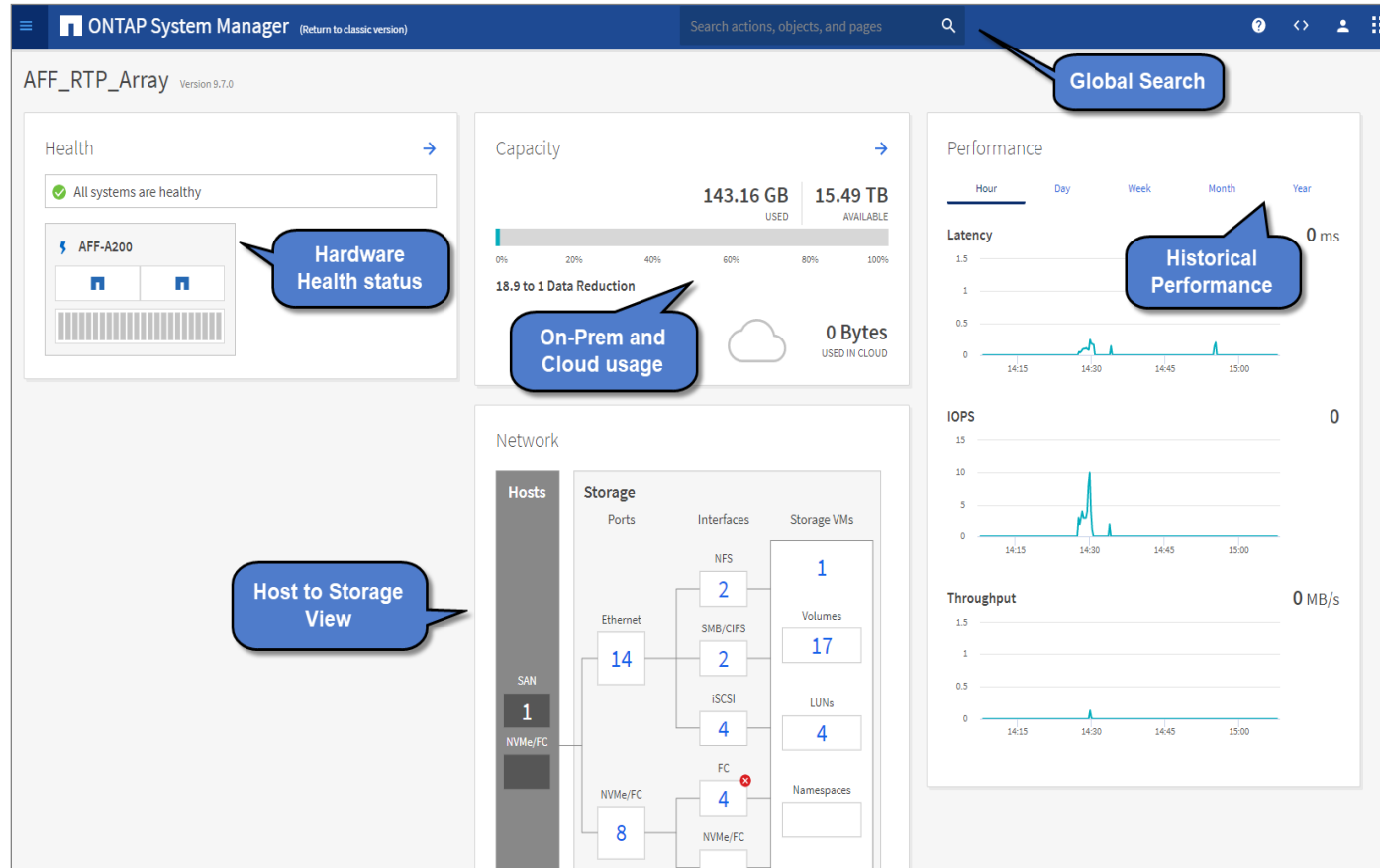
Unisphere Central



CloudIQ

NetApp ONTAP System Manager Dashboard

Storage infrastructure at a glance view



- Hardware Health Visualization
- Host to Storage visualization
- Comprehensive on-premises and in-cloud capacity reporting
- View granular performance graphs for different time periods



Storage & Data Management



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Storage Provisioning



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Storage Pools

- ▶ A Storage is provisioned from storage pools or RAID Array. A Pool is a collection of disks that are dedicated to create LUNs, Consistency Groups, File Systems, and VMware datastores.
- ▶ Storage pools provide optimized storage for a particular set of applications or conditions. The storage pool configuration defines the types and capacities of the disks in the pool. A user can defined the RAID configuration (RAID types and stripe widths) when selecting a tier to build a storage pool.



Overview of Storage Provisioning



Definition: Storage Provisioning

The process of assigning storage resources to compute system based on capacity, availability, and performance requirements.

- Storage provisioning can be performed in two ways:
 - Traditional
 - Virtual
- Virtual provisioning leverages virtualization technology for provisioning storage for applications



Logical Unit Number (LUN)



Definition: LUN

Each logical unit created from the RAID set is assigned a unique ID, called a LUN. A LUN is also referred to as a volume, partition, or device.

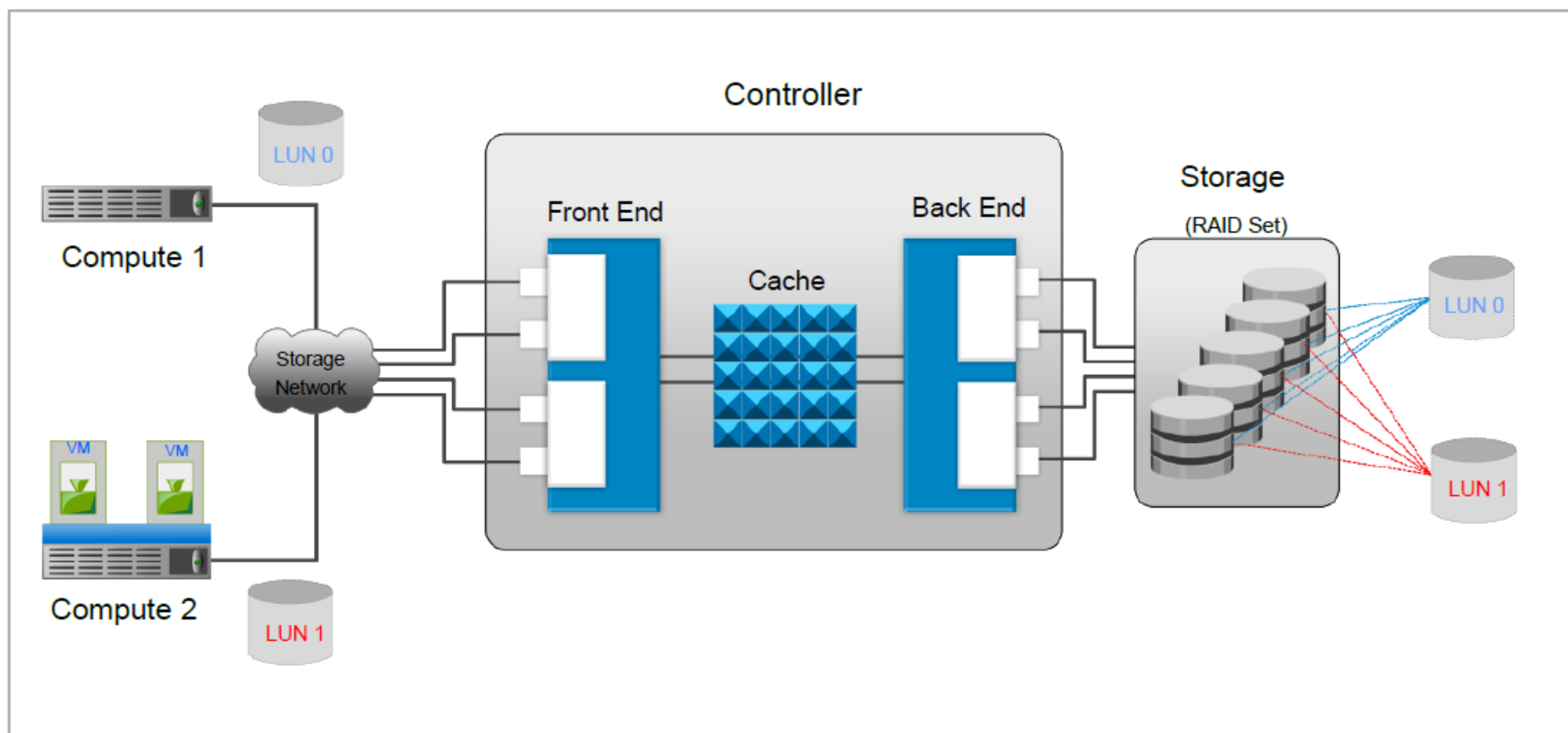
- LUNs hide the organization and composition of the RAID set from the compute systems
- LUNs created by traditional storage provisioning methods are also referred to as thick
- Once allocated, a LUN appears to a host as an internal physical disk



Traditional Provisioning

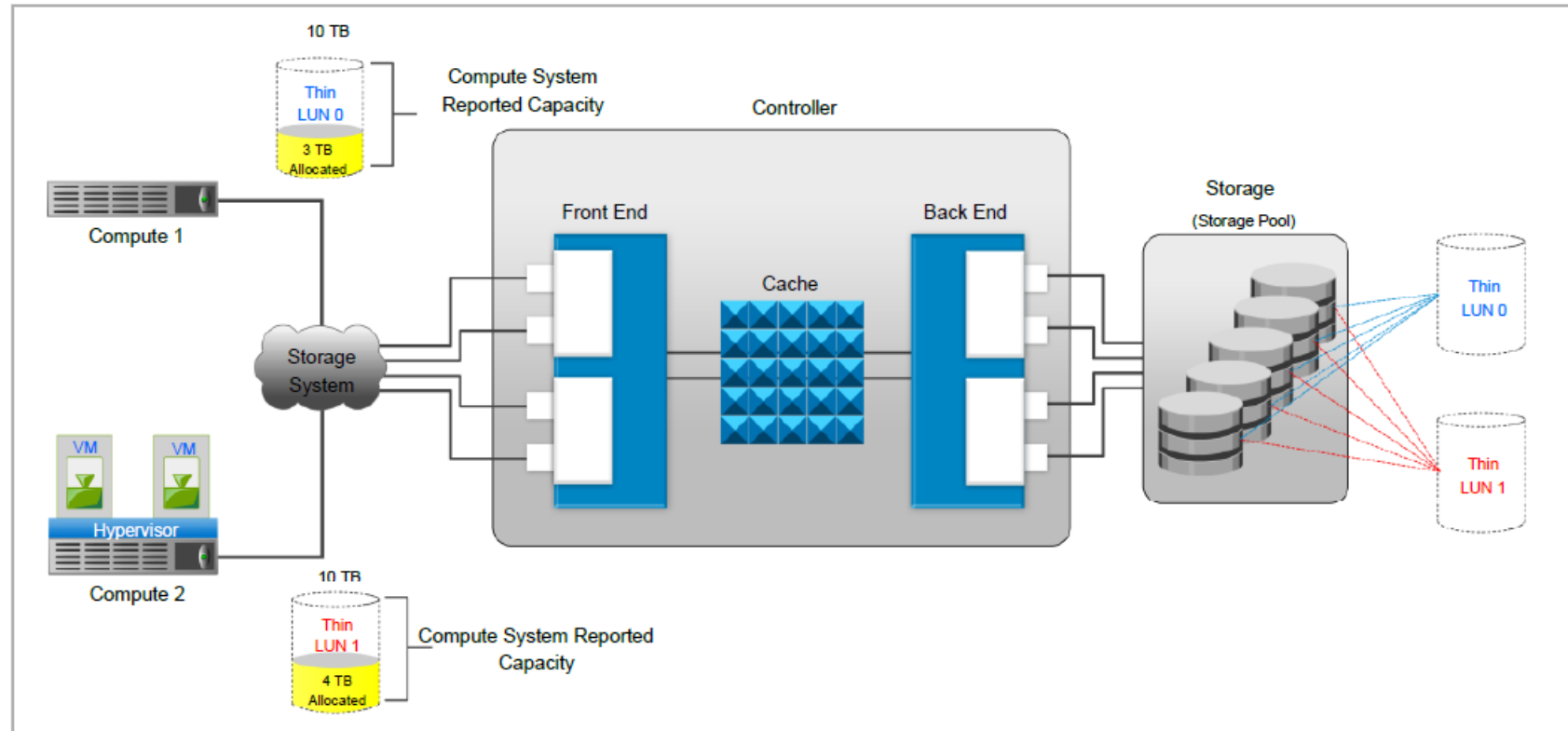
In traditional storage provisioning, physical storage drives are logically grouped together on which a required RAID level is applied to form a set, called RAID set.

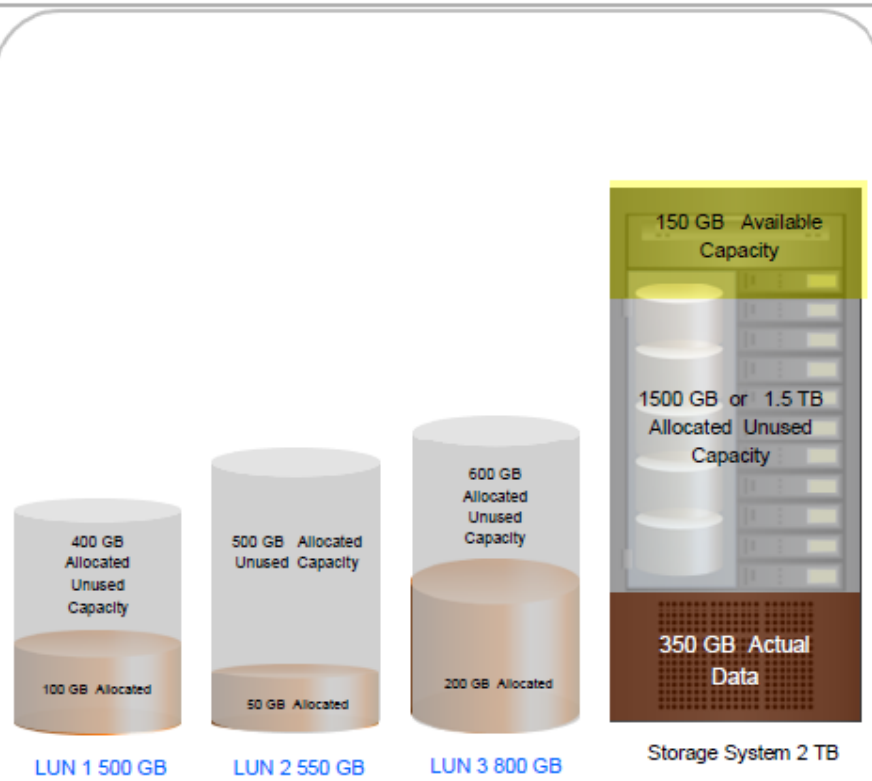
The illustration shows a RAID set consisting of five storage drives that have been sliced or partitioned into two LUNs: LUN 0 and LUN 1. These LUNs are then assigned to Compute 1 and Compute 2 for their storage requirements.



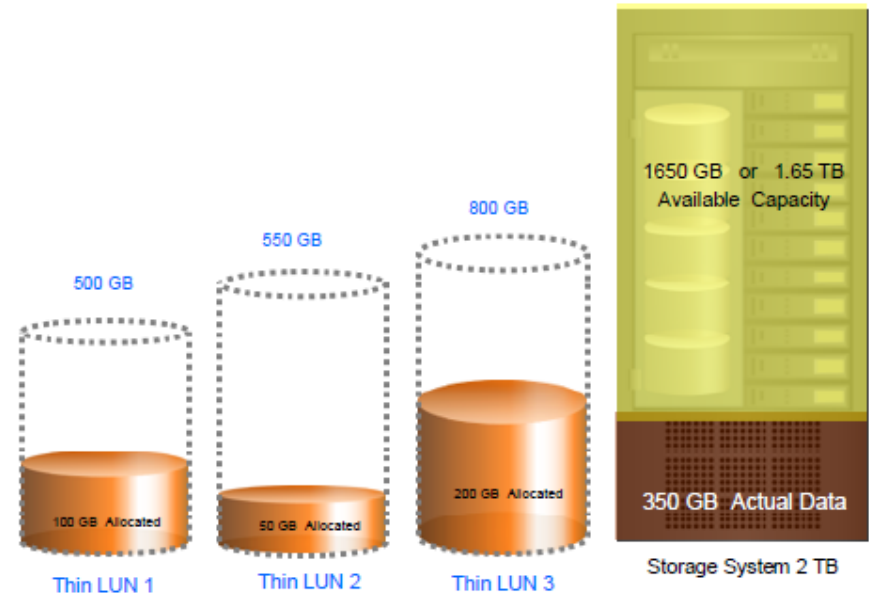
Virtual Provisioning

- Virtual provisioning enables creating and presenting a LUN with more capacity than is physically allocated to it on the storage system
- The LUN created using virtual provisioning is called a thin LUN to distinguish it from the traditional LUN
- Thin LUNs do not require physical storage to be completely allocated to them at the time they are created and presented to a compute system





Traditional Provisioning



Virtual Provisioning



Traditional Provisioning vs. Virtual Provisioning

- Administrators typically allocate storage capacity based on anticipated storage requirements.
 - This generally results in the over provisioning of storage capacity, which then leads to higher costs and lower capacity utilization.
- Administrators often over-provision storage to an application for various reasons such as to:
 - Avoid frequent provisioning of storage if the LUN capacity is exhausted
 - Reduce disruption to application availability
- Virtual provisioning:
 - Addresses these challenges
 - Improves storage capacity utilization and simplifies storage management



LUN Masking



Definition: LUN Masking

A process that provides data access control by defining which LUNs a compute system can access.

- Implemented on a storage system
- Prevents unauthorized or accidental use of LUNs in a shared environment



Storage Tiering Overview



Definition: Storage Tiering

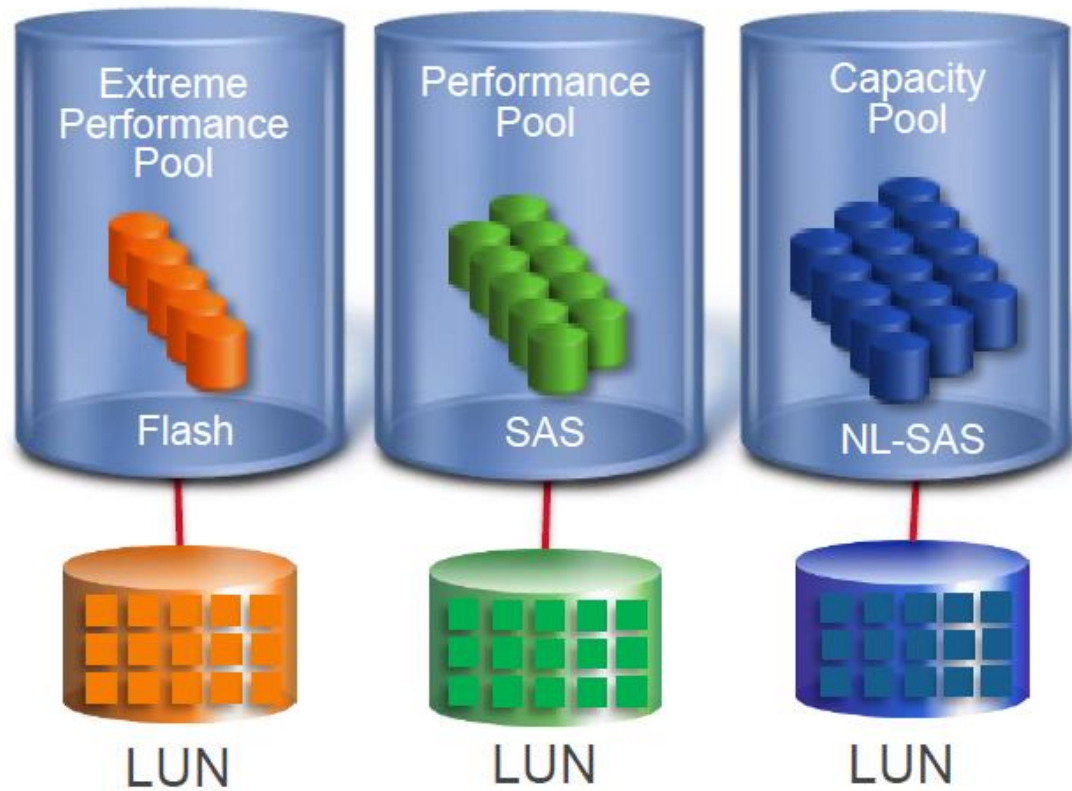
A technique of establishing a hierarchy of storage types and identifying the candidate data to relocate to the appropriate storage type to meet service level requirements at a minimal cost.

- Definition: Storage Tiering
 - A technique of establishing a hierarchy of storage types and identifying the candidate data to relocate to the appropriate storage type to meet service level requirements at a minimal cost
- Each tier has different levels of protection, performance, and cost
- Efficient storage tiering requires defining tiering policies

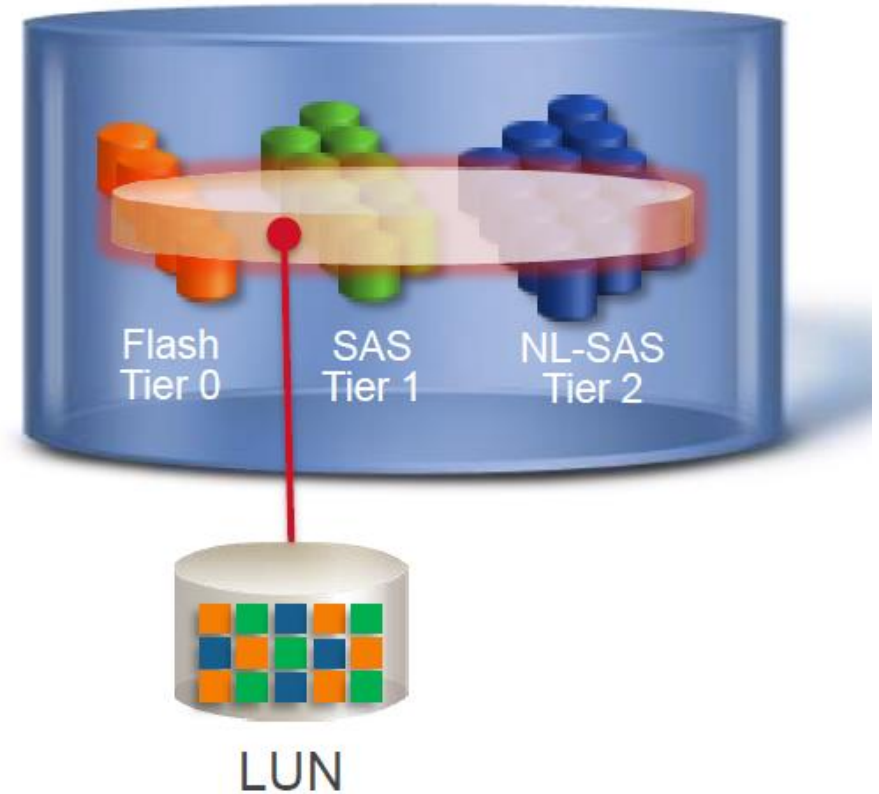


Storage pool characteristics

Homogeneous Pools



Heterogeneous Pool



LUN tiering

- Moves entire LUN from one tier to another.
- Does not give effective cost and performance benefits.

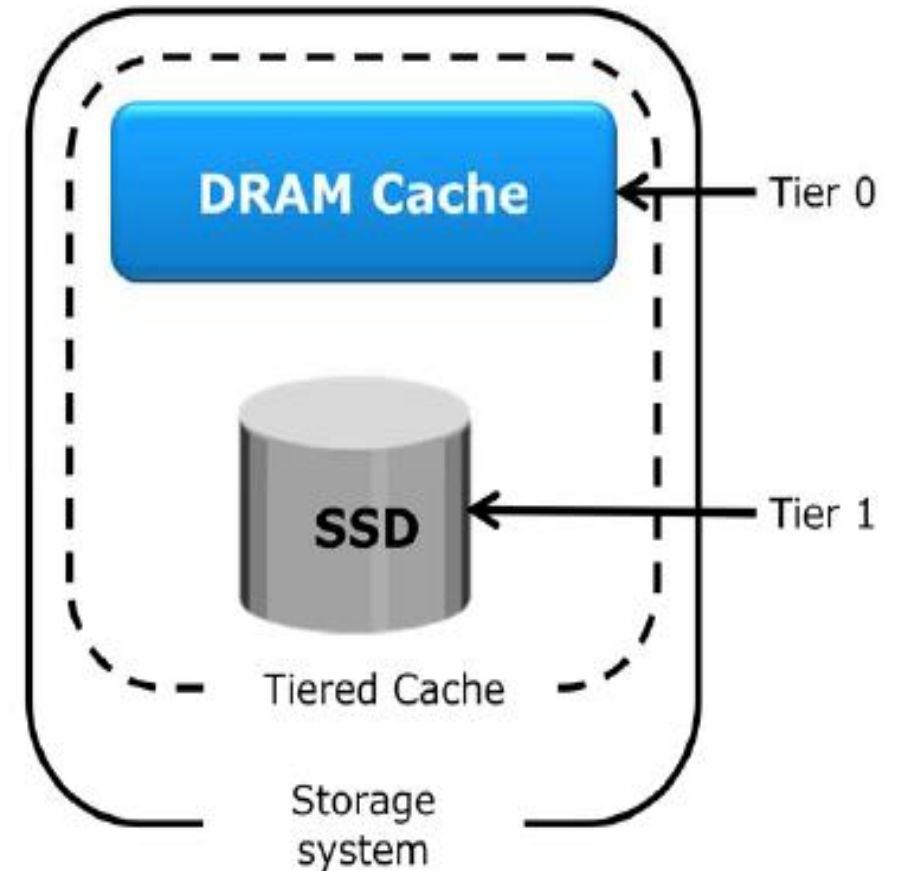
Sub-LUN Tiering

- A LUN is broken down into smaller segments and tiered at that level.
- Provides effective cost and performance benefits



Cache Tiering

- Enables creation of a large capacity secondary cache using SSDs
- Enables tiering between DRAM cache and SSDs (secondary cache)
- Most reads are served directly from high performance tiered cache
- Benefits
 - Enhances performance during peak workload
 - Non-disruptive and transparent to applications





Business Continuity



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Why Business Continuity?

- Information is an organization's most important asset
- Continuous access to information ensures smooth functioning of business operations
- Cost of unavailability of information to an organization is greater than ever



What is Business Continuity?

Business Continuity

It is a process that prepares for, responds to, and recovers from a system outage that can adversely affects business operations.

- An integrated and enterprise-wide process that includes set of activities to ensure “information availability”
- BC involves proactive measures and reactive countermeasures
- In a virtualized environment, BC solutions need to protect both physical and virtualized resources



Information Availability

Information Availability

It is the ability of an IT infrastructure to function according to business expectations, during its specified time of operation.

- Information availability can be defined with the help of:
 - ▶ Accessibility
 - ▶▶ Information should be accessible to the right user when required
 - ▶ Reliability
 - ▶▶ Information should be reliable and correct in all aspects
 - ▶ Timeliness
 - ▶▶ Defines the time window during which information must be accessible



Impact of Downtime

Lost Productivity

- Number of employees impacted x hours out x hourly rate

Know the downtime costs (per hour, day, two days, and so on.)

Lost Revenue

- Direct loss
- Compensatory payments
- Lost future revenue
- Billing losses
- Investment losses

Damaged Reputation

- Customers
- Suppliers
- Financial markets
- Banks
- Business partners

Financial Performance

- Revenue recognition
- Cash flow
- Lost discounts (A/P)
- Payment guarantees
- Credit rating
- Stock price

Other Expenses

- Temporary employees, equipment rental, overtime costs, extra shipping costs, travel expenses, and so on.



Availability Measurement – Levels of ‘9s’ Availability

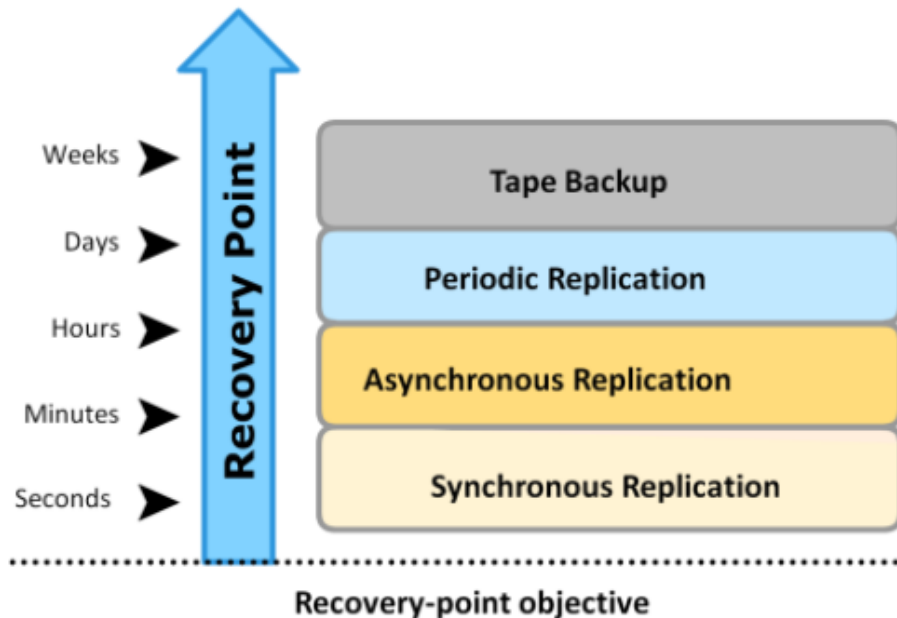
Uptime (%)	Downtime (%)	Downtime per Year	Downtime per Week
98	2	7.3 days	3hrs, 22 minutes
99	1	3.65 days	1 hr, 41 minutes
99.8	0.2	17 hrs, 31 minutes	20 minutes, 10 secs
99.9	0.1	8 hrs, 45 minutes	10 minutes, 5 secs
99.99	0.01	52.5 minutes	1 minute
99.999	0.001	5.25 minutes	6 secs
99.9999	0.0001	31.5 secs	0.6 secs



BC Terminologies – 2

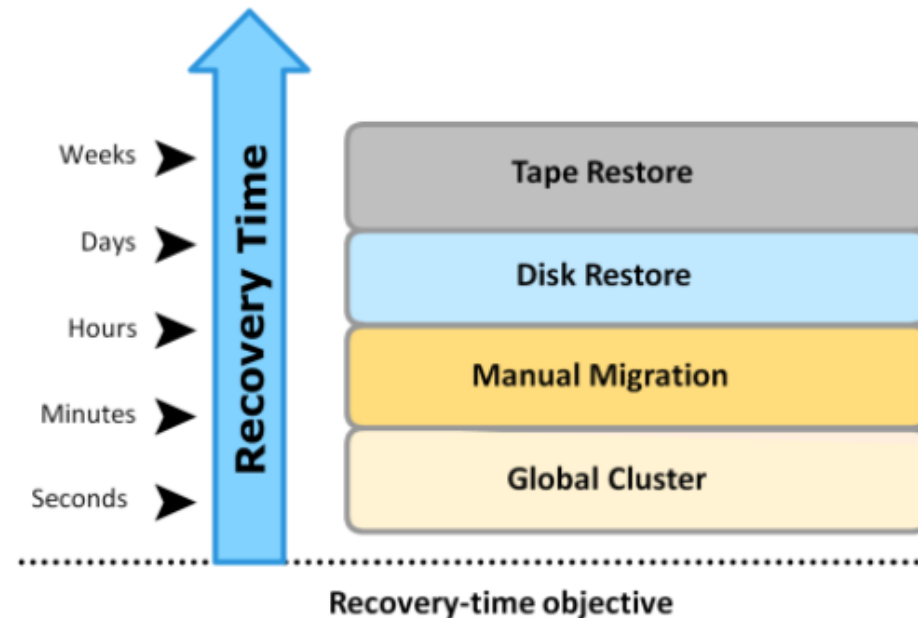
Recovery-Point Objective (RPO)

- Point-in-time to which systems and data must be recovered after an outage
- Amount of data loss that a business can endure



Recovery-Time Objective (RTO)

- Time within which systems and applications must be recovered after an outage
- Amount of downtime that a business can endure and survive



BC Technology Solutions

- Solutions that enable BC are:
 - ▶ Resolving single points of failure
 - ▶ Multipathing software
 - ▶ Backup and replication
 - ▶▶ Backup
 - ▶▶ Local replication
 - ▶▶ Remote replication



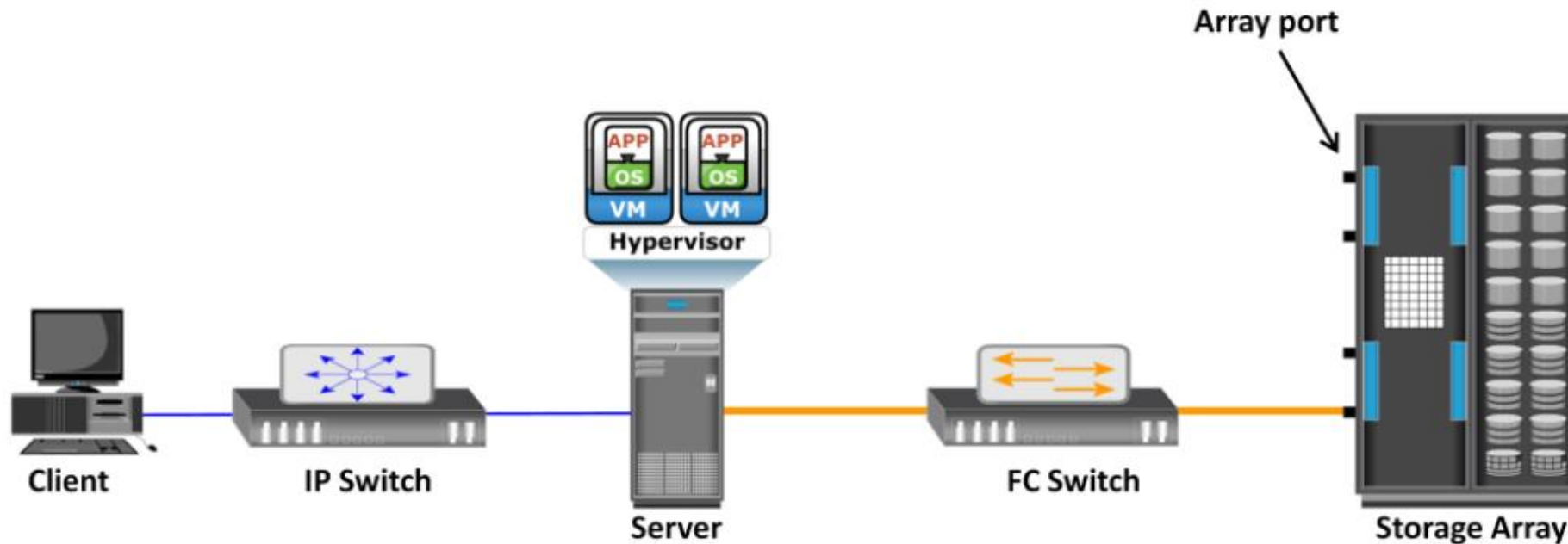
Single Point of Failure (SPOF)

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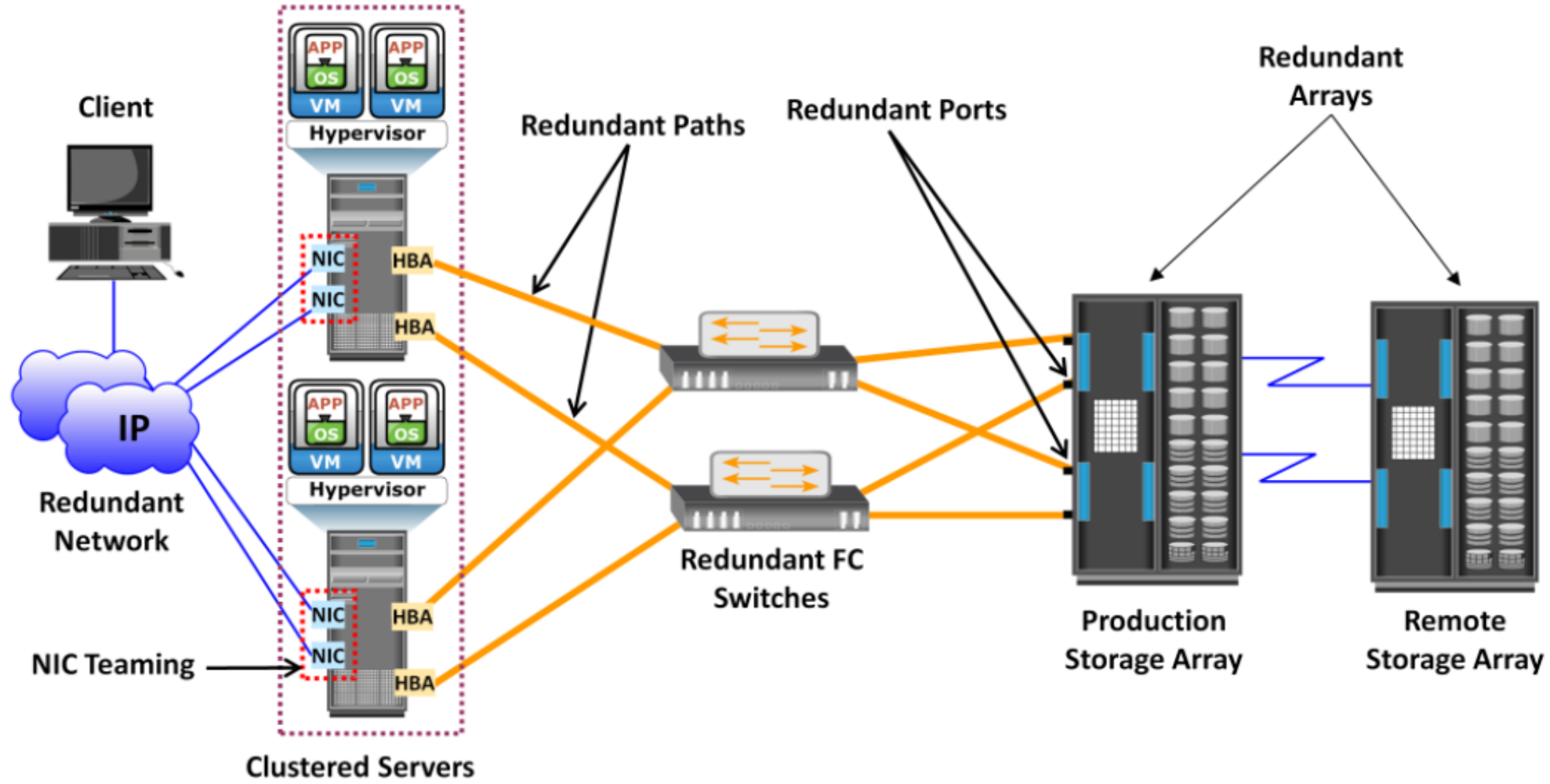
Single Points of Failure

Single Points of Failure

It refers to the failure of a component of a system that can terminate the availability of the entire system or IT service.



Resolving Single Points of Failure



Multipathing Software

- Recognizes and utilizes alternate I/O path to data
- Provides load balancing by distributing I/Os to all available, active paths:
 - ▶ Improves I/O performance and data path utilization
- Intelligently manages the paths to a device by sending I/O down the optimal path:
 - ▶ Based on the load balancing and failover policy setting for the device



What is Backup?

Backup

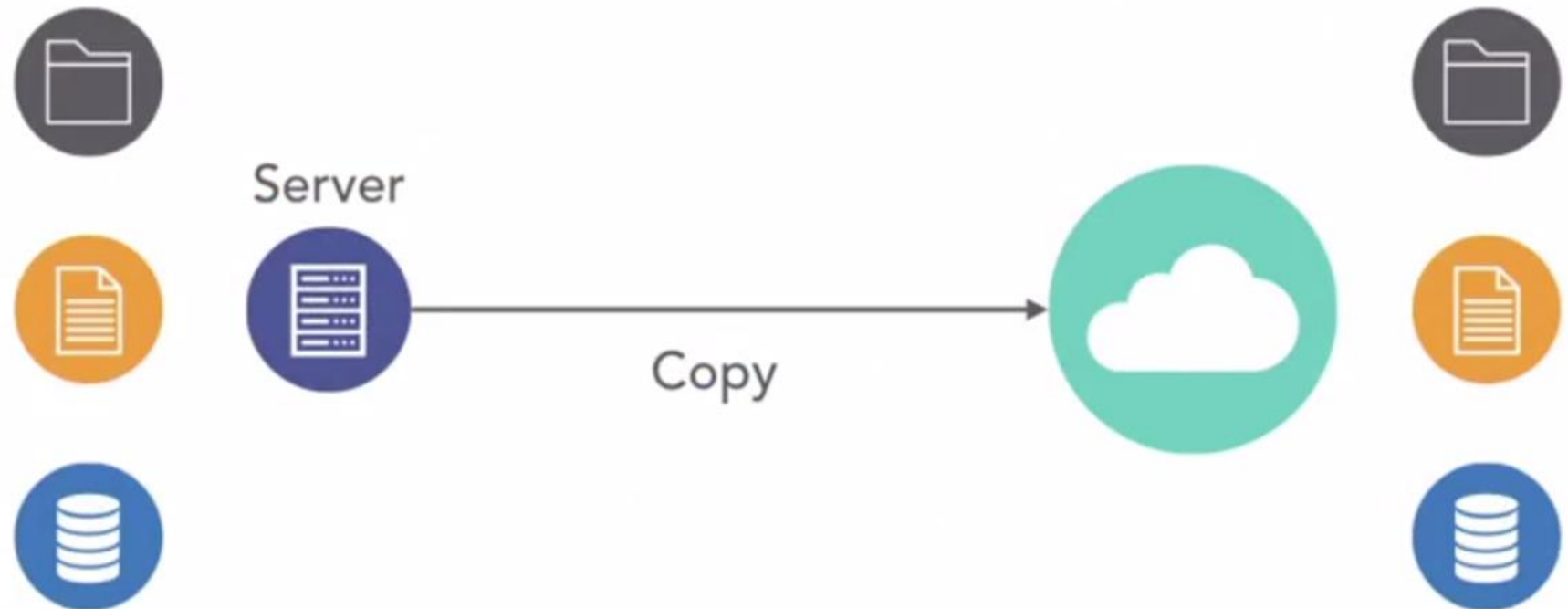
It is an additional copy of production data that is created and retained for the sole purpose of recovering lost or corrupted data.

- Organization also takes backup to comply with regulatory requirements
- Backups are performed to serve three purposes:
 - ▶ Disaster recovery
 - ▶ Operational recovery
 - ▶ Archive



What Is Backup?

- Backup is an additional copy of data that can be used for restore and recovery purposes
- The backup copy is used when the primary copy is lost or corrupted



What Is a Restore?

- ▶ A restore is a process that recreates the original data from a backup copy. This process consists of the preparation and actual restore of data, and some post-restore actions that make that data ready for use.

Restore Process



Backup Methods

- Two methods of backup, based on the state of the application when the backup is performed
 - ▶ Hot or Online
 - ▶▶ Application is up and running, with users accessing their data during backup
 - ▶▶ Open file agent can be used to backup open files
 - ▶ Cold or Offline
 - ▶▶ Requires application to be shutdown during the backup process
- Bare-metal recovery
 - ▶ OS, hardware, and application configurations are appropriately backed up for a full system recovery
 - ▶ Server configuration backup (SCB) can also recover a server onto dissimilar hardware



Why Do You Need Backups?

- Hardware failures
- Human factors
- Application failures
- Security breaches
- Disasters
- Regulatory and business requirements



Backup Media

- Tape
 - ▶ Sequential access
 - ▶ No protection
- Virtual tape
 - ▶ Advantages of disk with tape emulation
- Traditional disk
 - ▶ Random access
 - ▶ Protected by the storage array (RAID, hot spare, etc.)
- Purpose-built disk
 - ▶ EMC Avamar
 - ▶ EMC Data Domain



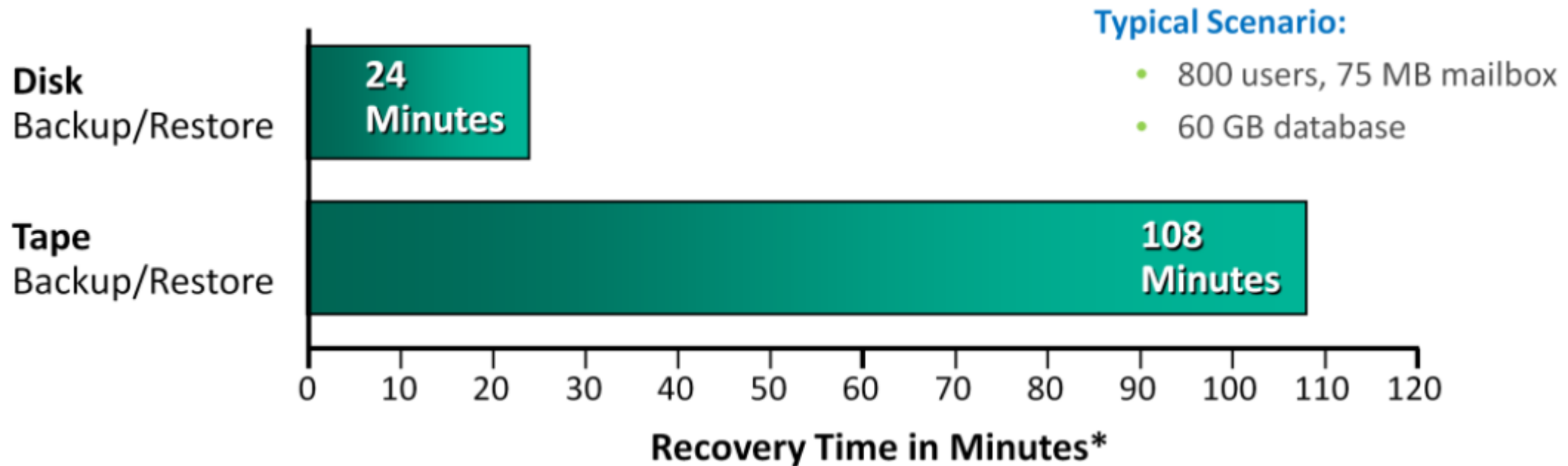
Backup to Tape

- Traditionally low cost solution
- Tape drives are used to read/write data from/to a tape
- Sequential/linear access
- Multiple streaming to improve media performance
 - ▶ Writes data from multiple streams on a single tape
- Limitation of tape
 - ▶ Backup and recovery operations are slow due to sequential access
 - ▶ Wear and tear of tape
 - ▶ Shipping/handling challenges
 - ▶ Controlled environment is required for tape storage



Backup to Disk

- Enhanced overall backup and recovery performance
 - ▶ Random access
- More reliable
- Can be accessed by multiple hosts simultaneously





Backup Types



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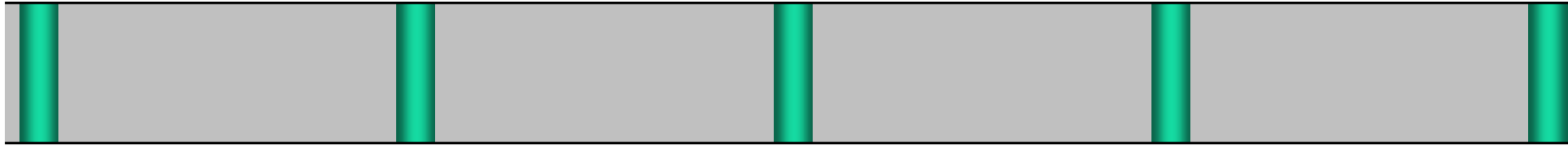
Backup Types

- ▶ Full Backup
- ▶ Incremental Backup
- ▶ Differential or Cumulative Backup

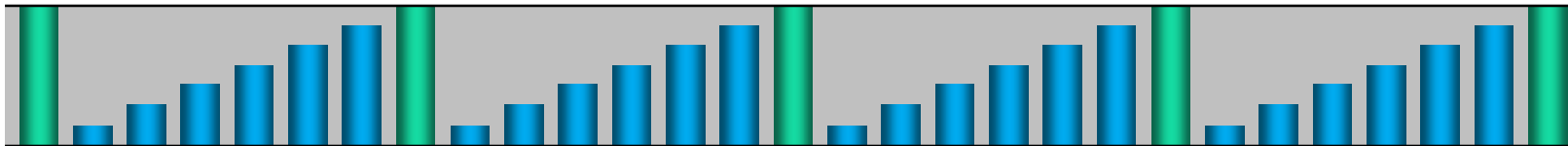


Backup Granularity and Levels

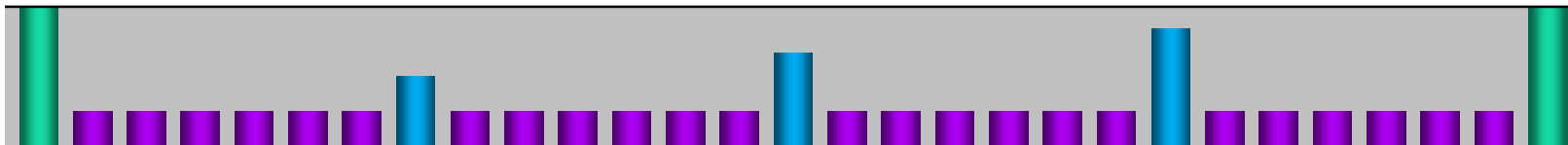
Full Backup



Cumulative (Differential)



Incremental



Full



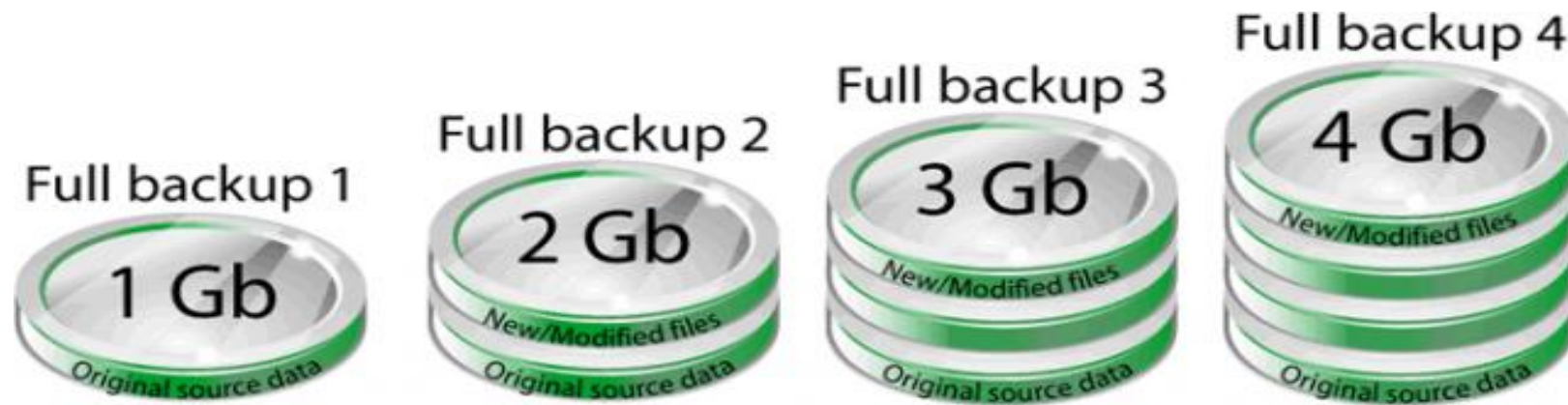
Cumulative



Incremental

Full Backup

- ▶ Full backup is the starting point for all other types of backup and contains all the data in the folders and files that are selected to be backed up. Because full backup stores all files and folders, frequent full backups result in faster and simpler restore operations.



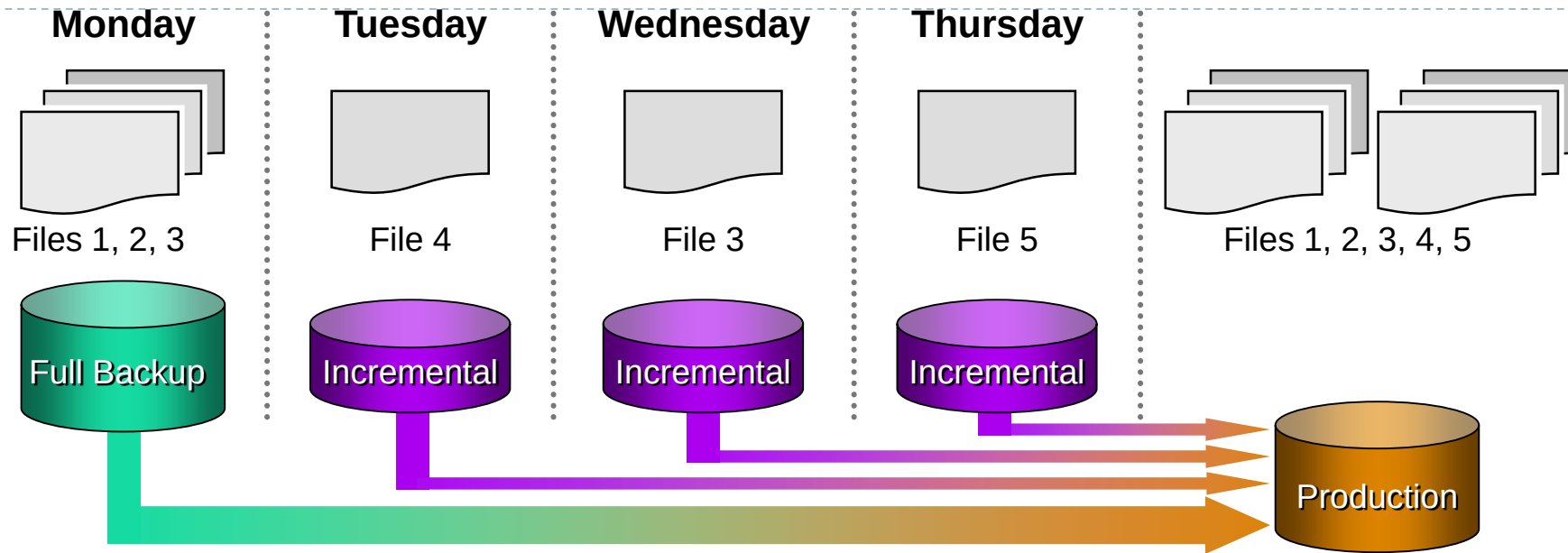
Note: Full Backup will always back-up the entire source data. If you don't delete/exclude sources (only add/modify) it will always grow in size because it backs up everything.

Incremental Backup

- ▶ Incremental backup stores all files that have changed since the last FULL, DIFFERENTIAL OR INCREMENTAL backup.
- ▶ The advantage of an incremental backup is that it takes the least time to complete.



Restoring an Incremental Backup

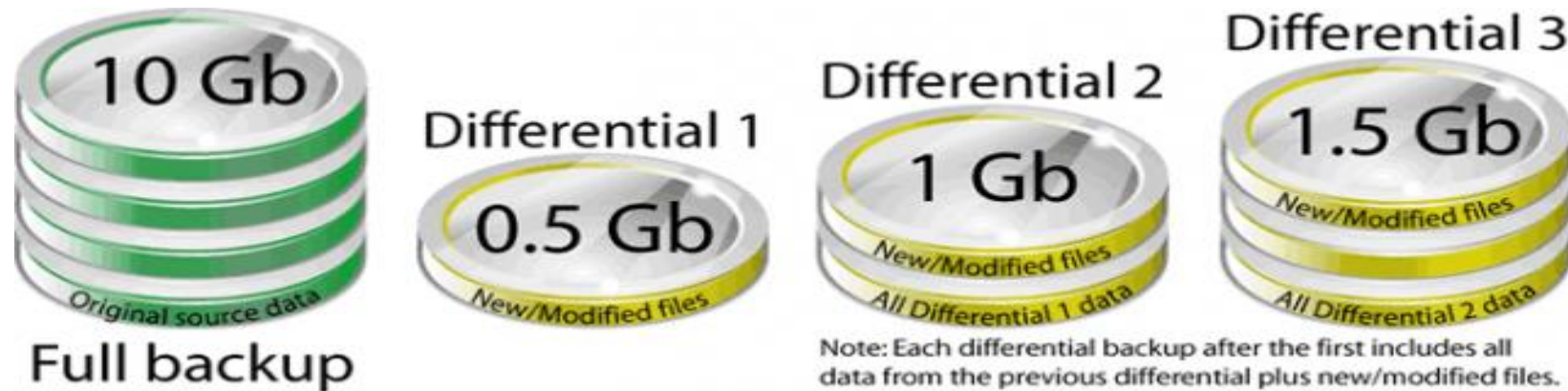


► Key Features

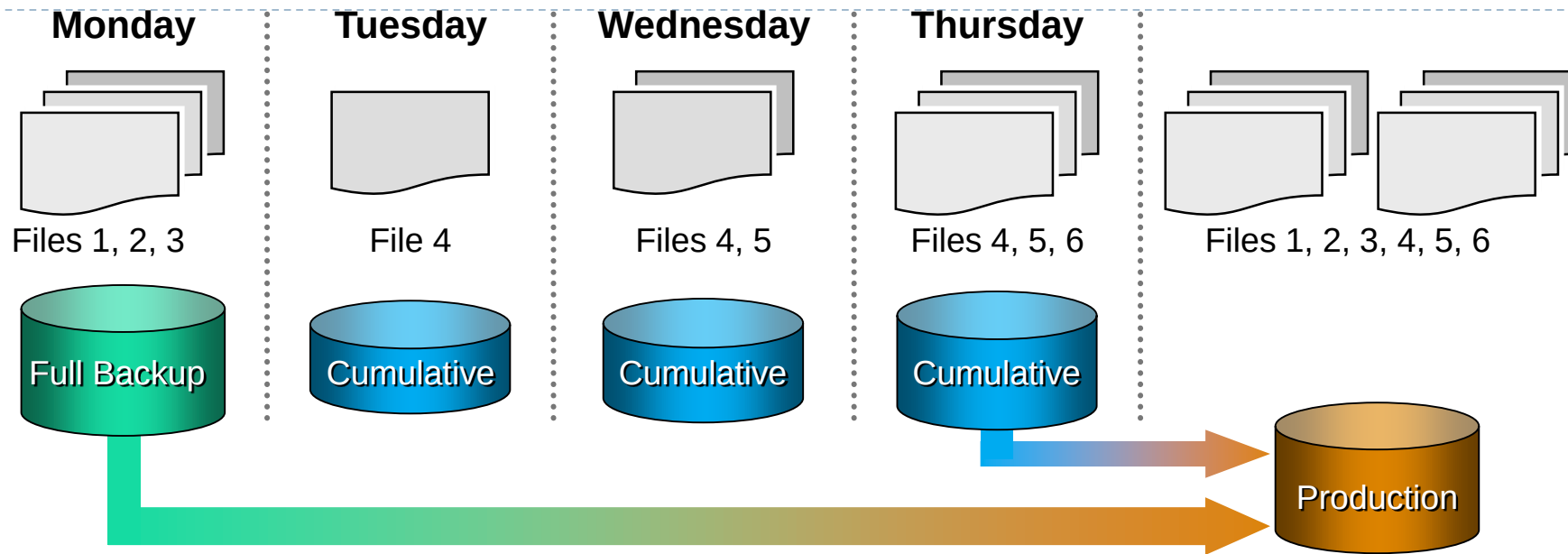
- Files that have changed since the last full or incremental backup are backed up
- Fewest amount of files to be backed up, therefore faster backup and less storage space
- Longer restore because last full and all subsequent incremental backups must be applied

Differential Backup

- ▶ Differential backup contains all files that have changed since the last FULL backup.
- ▶ The advantage of a differential backup is that it shortens restore time compared to a full backup or an incremental backup.



Restoring a Cumulative Backup



► Key Features

- More files to be backed up, therefore it takes more time to backup and uses more storage space
- Much faster restore because only the last full and the last cumulative backup must be applied

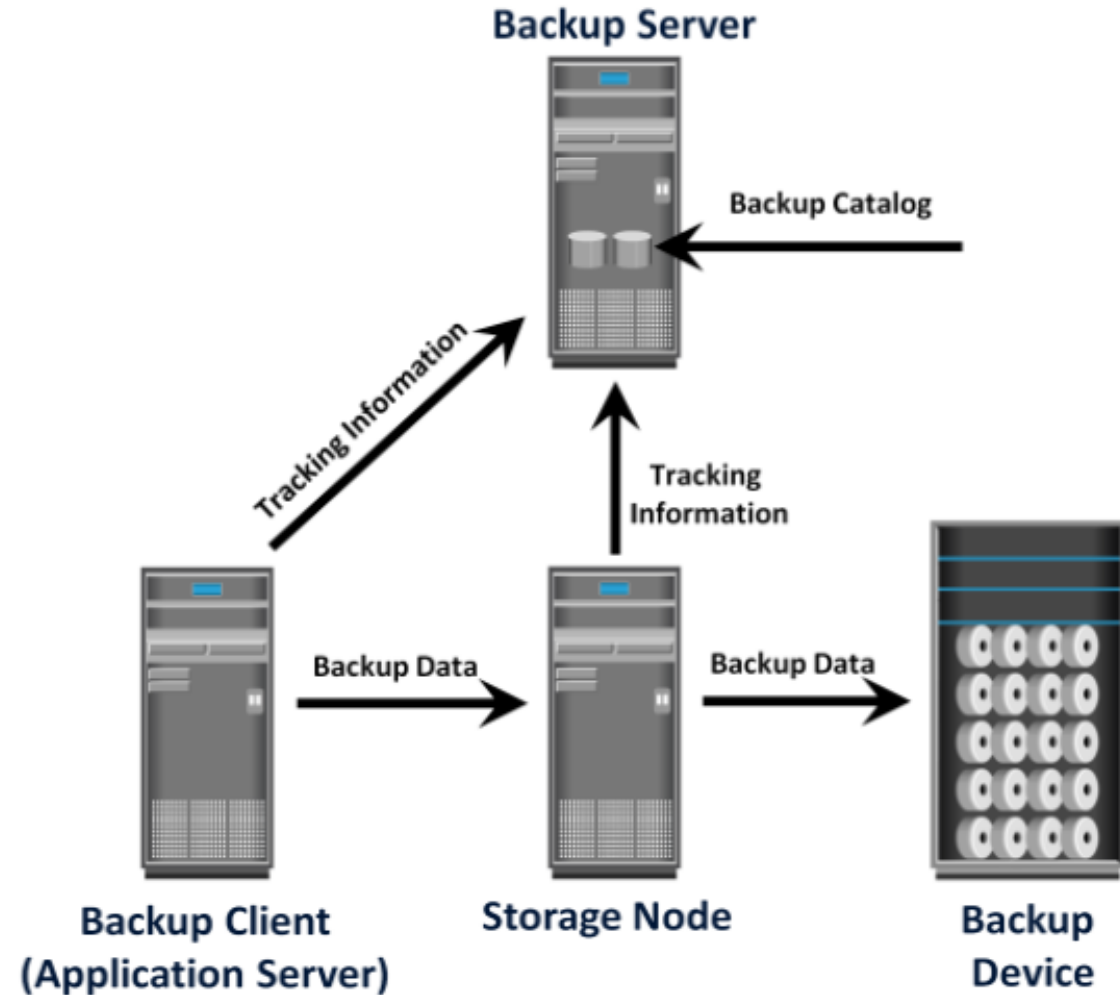


Backup Components

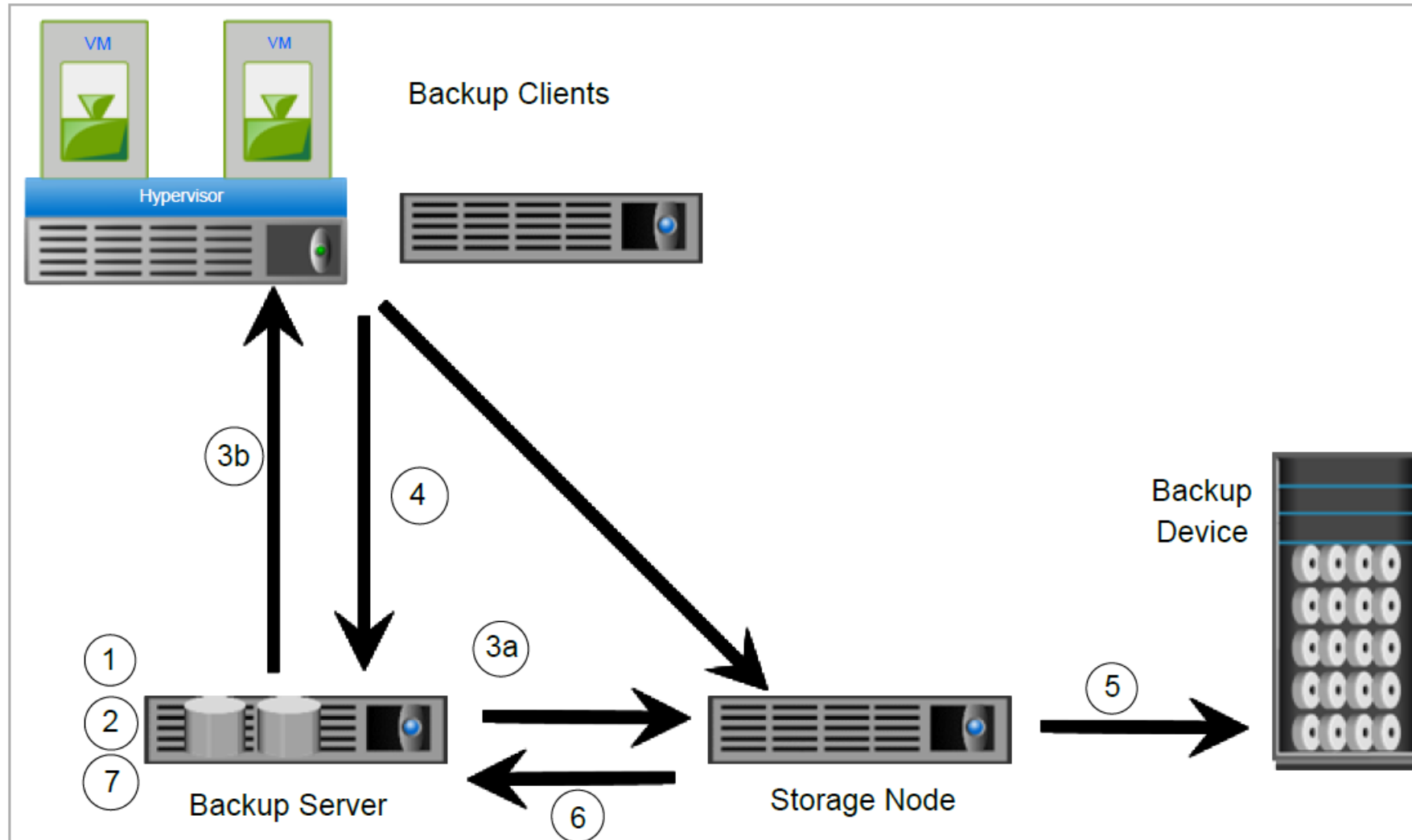


Backup Architecture

- Backup client
 - ▶ Gathers the data that is to be backed up and send it to storage node
- Backup server
 - ▶ Manages backup operations and maintains backup catalog
- Storage node
 - ▶ Responsible for writing data to backup device
 - ▶ Manages the backup device



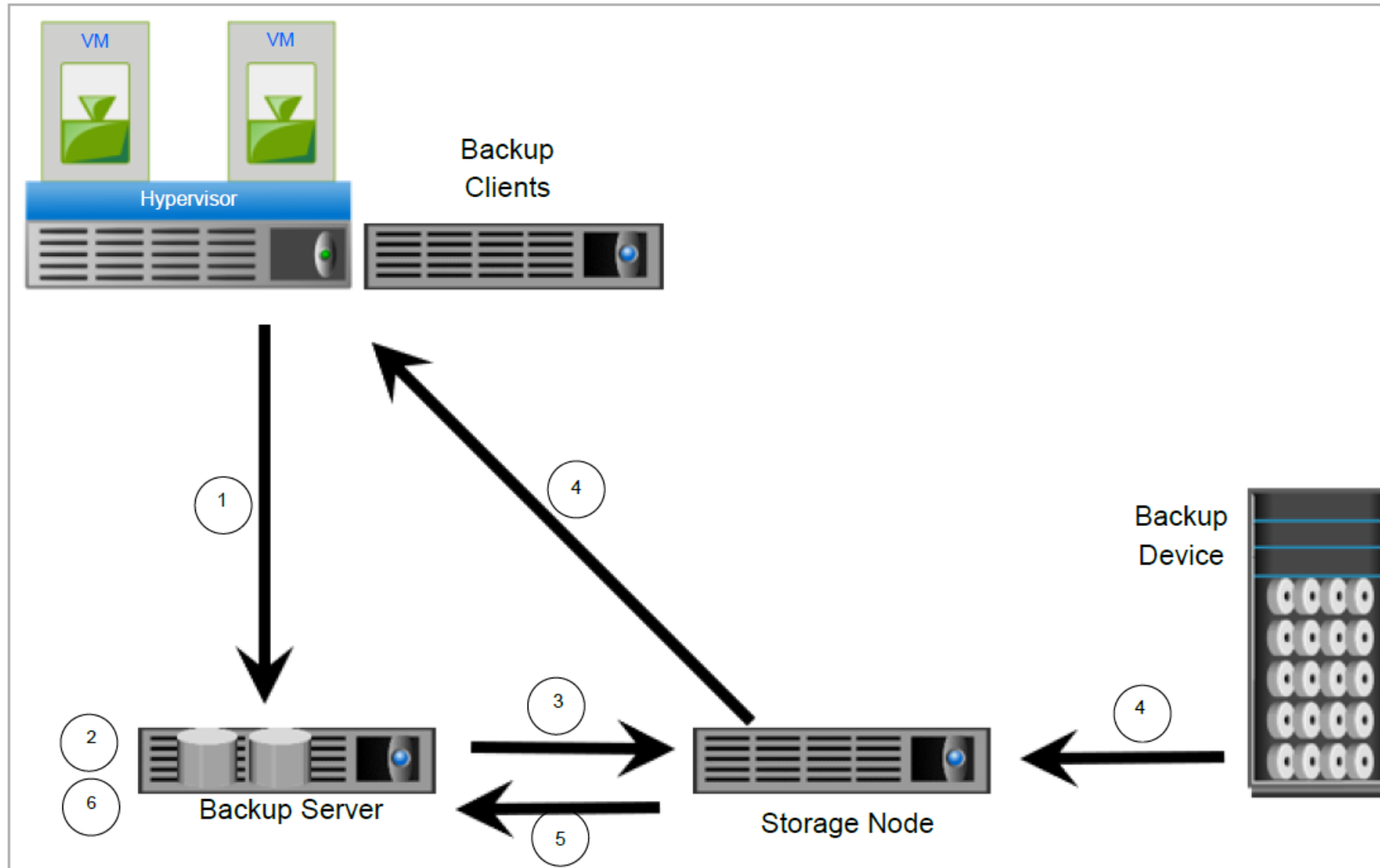
Backup Operation



- (1) Backup server initiates scheduled backup process.
- (2) Backup server retrieves backup-related information from the backup catalog.
- (3a) Backup server instructs storage node to load backup media in the backup device.
- (3b) Backup server instructs backup clients to send data to be backed up to the storage node.
- (4) Backup clients send data to storage node and update the backup catalog on the backup server.
- (5) Storage node sends data to the backup device
- (6) Storage node sends metadata and media information to the backup server
- (7) Backup server updates the backup catalog



Recovery Operation



- (1) Backup client requests backup server for data restore
- (2) Backup server scans backup catalog to identify data to be restored and the client that will receive data
- (3) Backup server instructs storage node to load backup media in the backup device
- (4) Data is then read and sent to the backup client
- (5) Storage node sends restore metadata to the backup server
- (6) Backup server updates the backup catalog





Backup Architecture



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Standalone Backup

Storing files locally so that the data are accessible when the computer is not connected to the network.



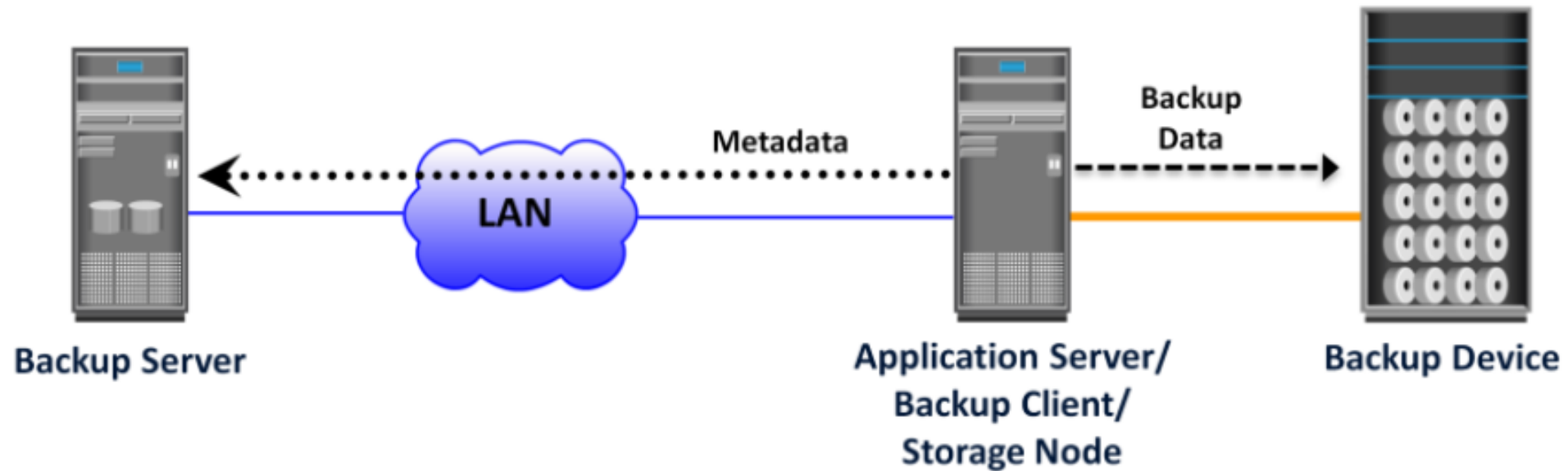
Backup manager
... and has a backup catalog
... and a backup agent



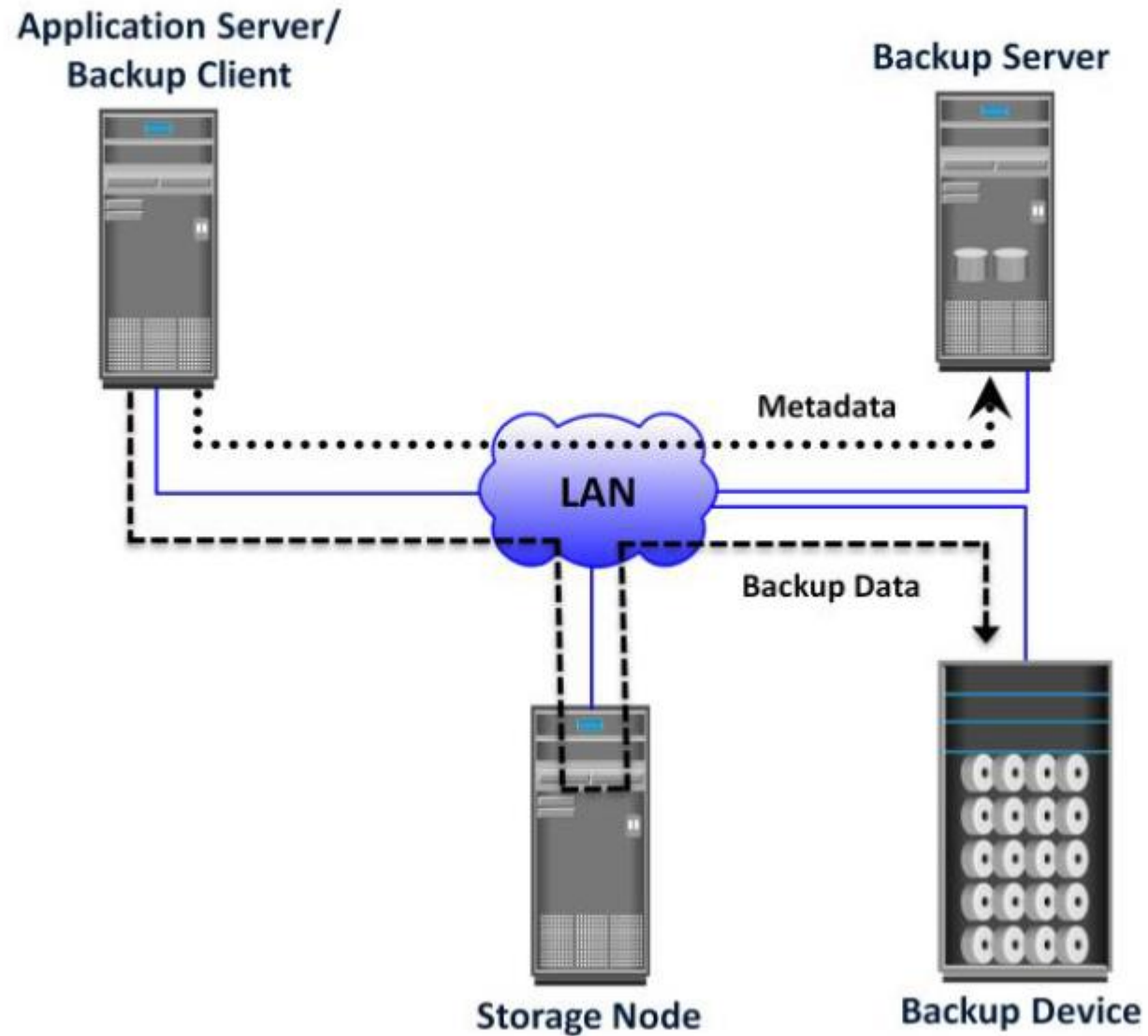
Backup storage
(target)



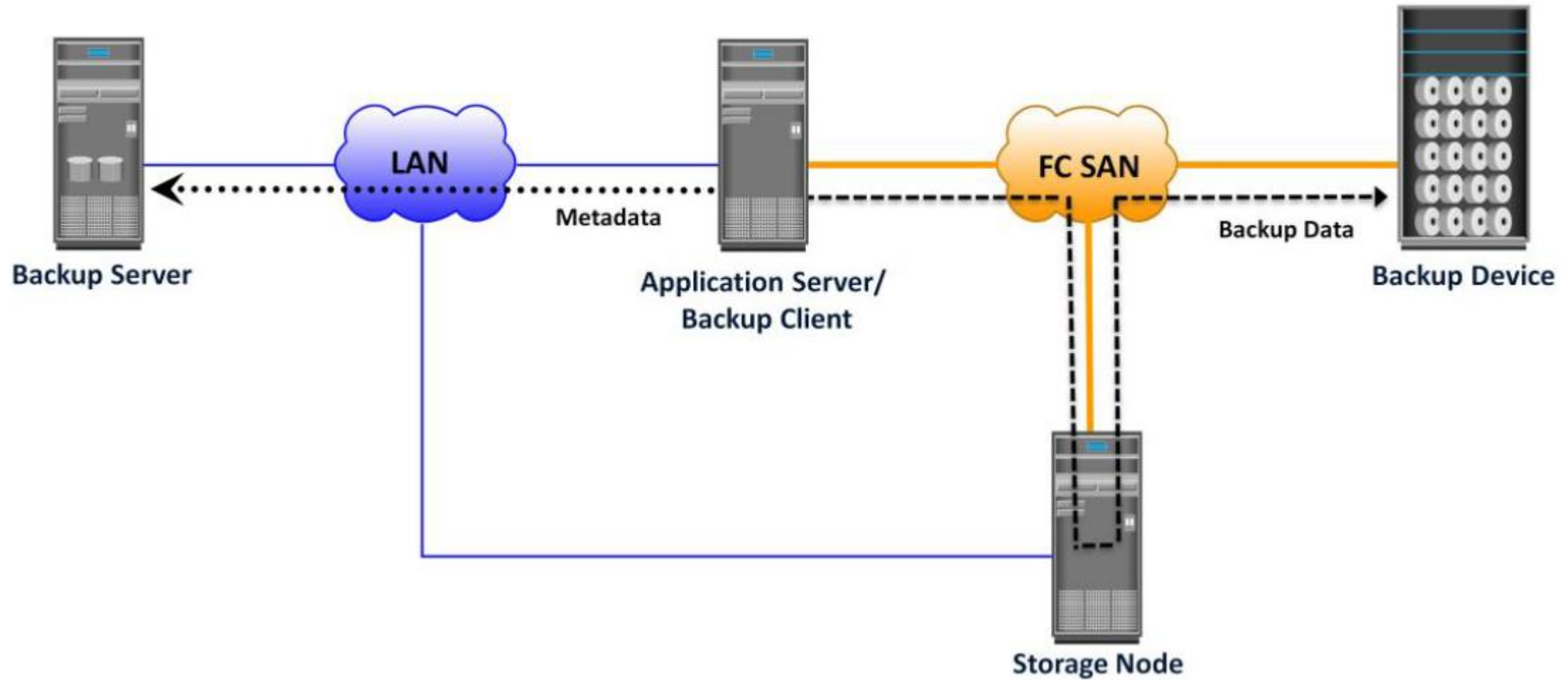
Direct-Attached Backup



LAN-based Backup



SAN-based Backup



NDMP Backup (NAS)

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NDMP-Based Backup

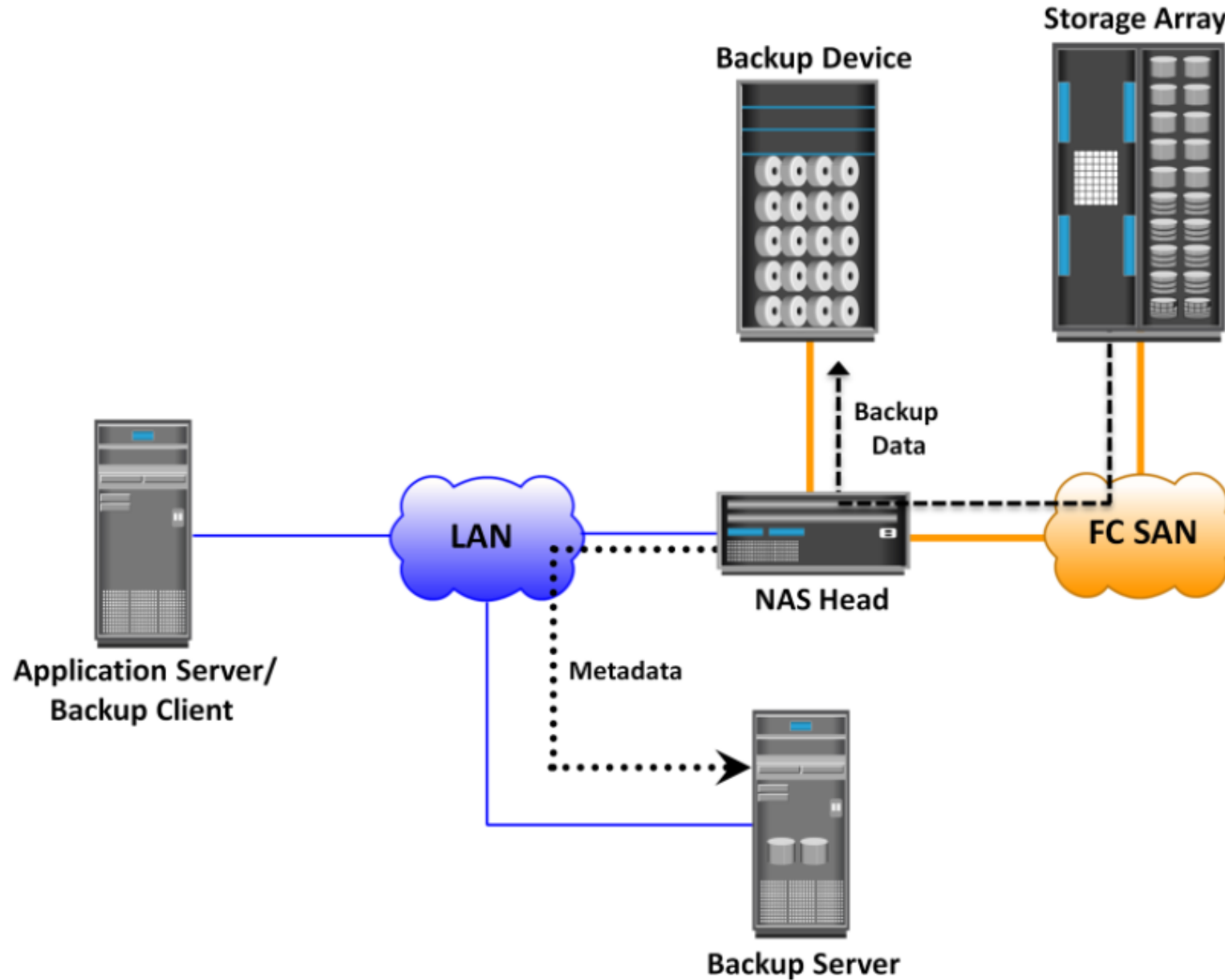
**Definition: NDMP**

An open standard TCP/IP-based protocol that is designed for a backup in a NAS environment.

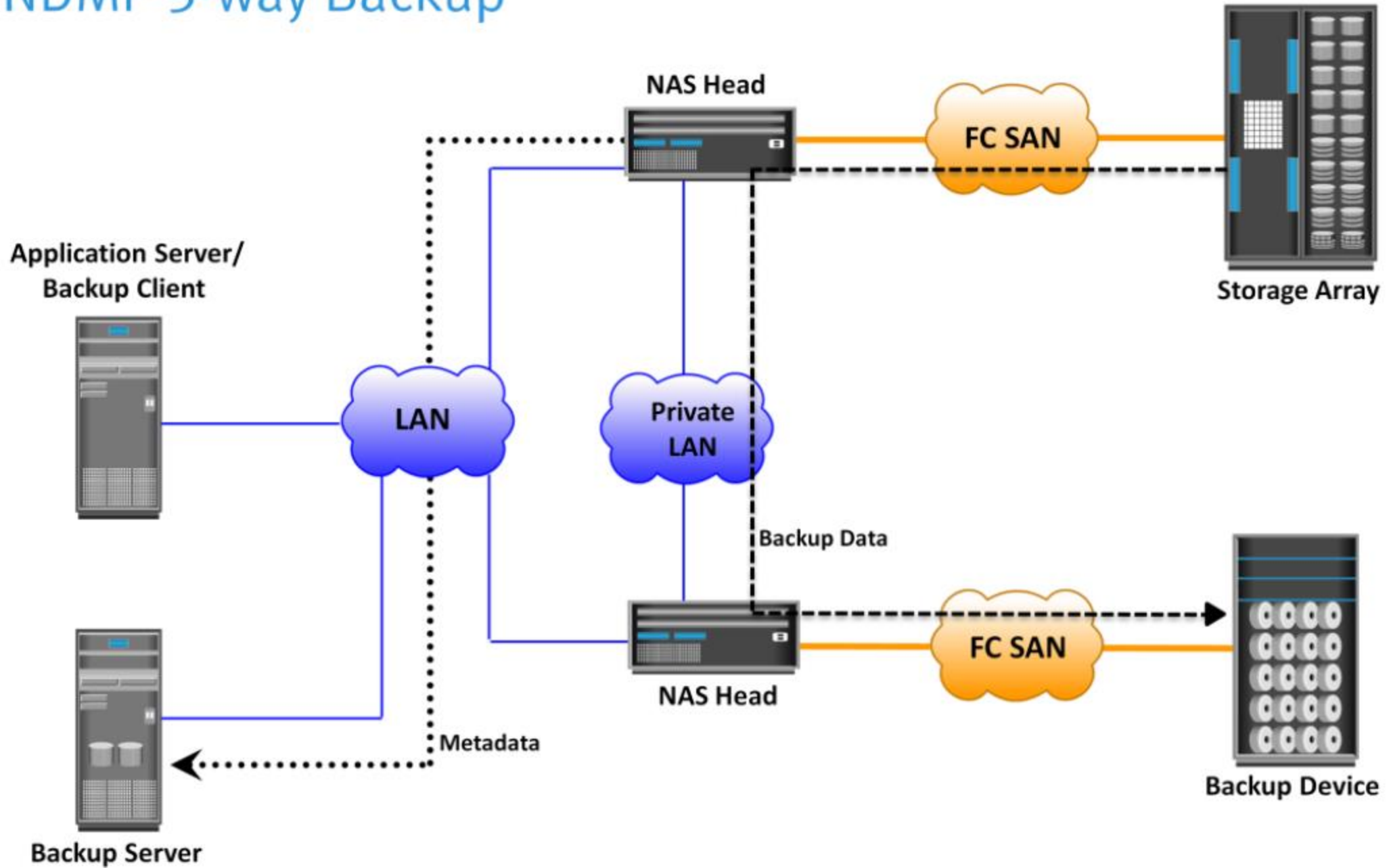
- Data can be backed up using NDMP regardless of the operating system or platform
- Backup data is sent directly from NAS to the backup device
 - No longer necessary to transport data through application servers
- Backs up and restores data while preserving security attributes of file system (NFS and CIFS) and maintains data integrity



NDMP 2-way Backup



NDMP 3-way Backup





Data Duplication



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What is Data Deduplication?

Data Deduplication

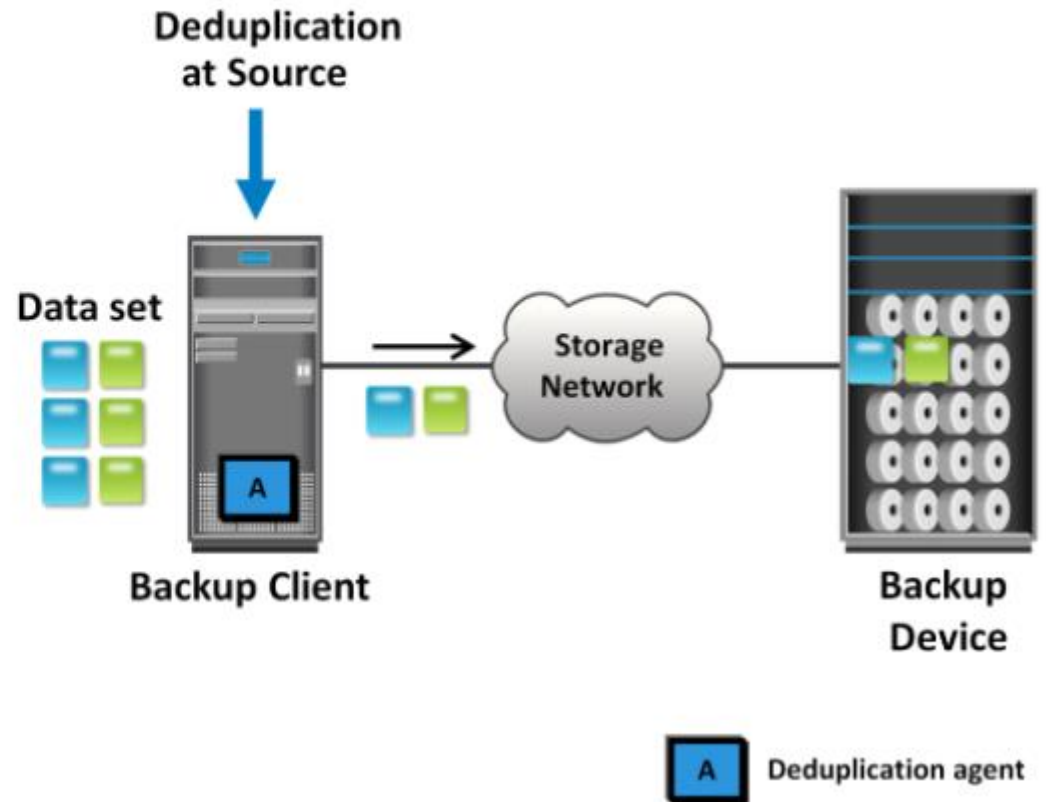
It is a process of identifying and eliminating redundant data.

- Deduplication methods
 - ▶ File level
 - ▶ Subfile level
- Deduplication implementations
 - ▶ Source-based
 - ▶ Target-based



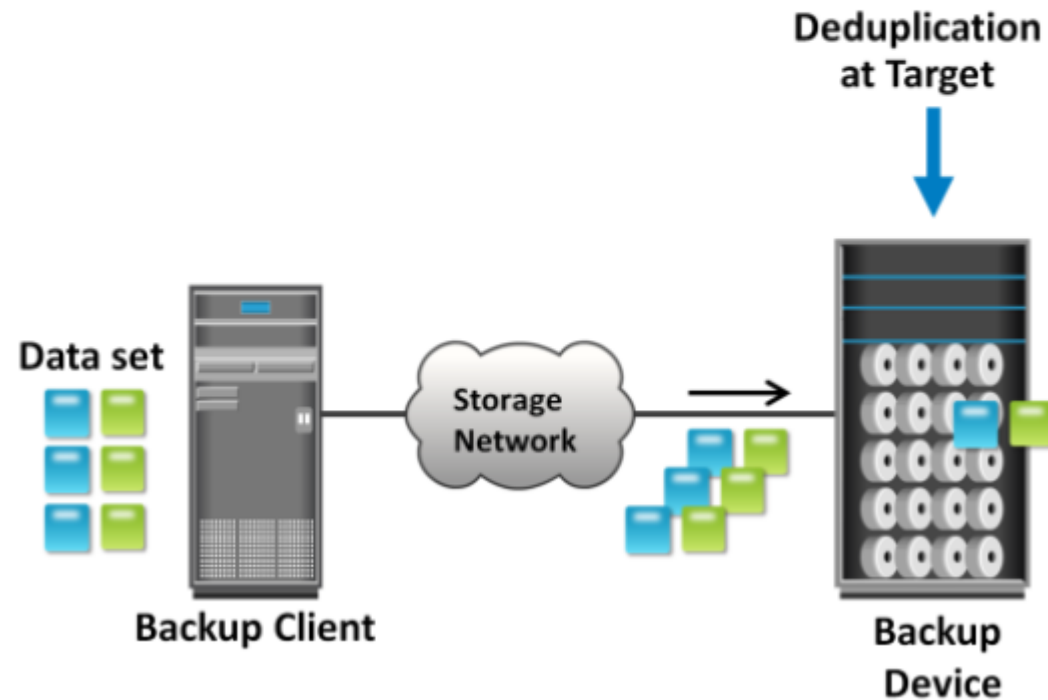
Data Deduplication Implementation – Source-based

- Data is deduplicated at the source (backup client)
- Backup client sends only new, unique segments across the network
- Reduced storage capacity and network bandwidth requirements
- Increased overhead on the backup client



Data Deduplication Implementation – Target-based

- Data is deduplicated at the target
 - ▶ Inline
 - ▶ Post-process
- Offloads the backup client from deduplication process
- All the backup data traverse the network





Disaster Recovery (DR)



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Key BC Concepts: Disaster Recovery



Definition: Disaster Recovery (DR)

A part of BC process, which involves a set of policies and procedures for restoring IT infrastructure, including data that is required to support ongoing IT services, after a natural or human-induced disaster occurs.



What is Replication?

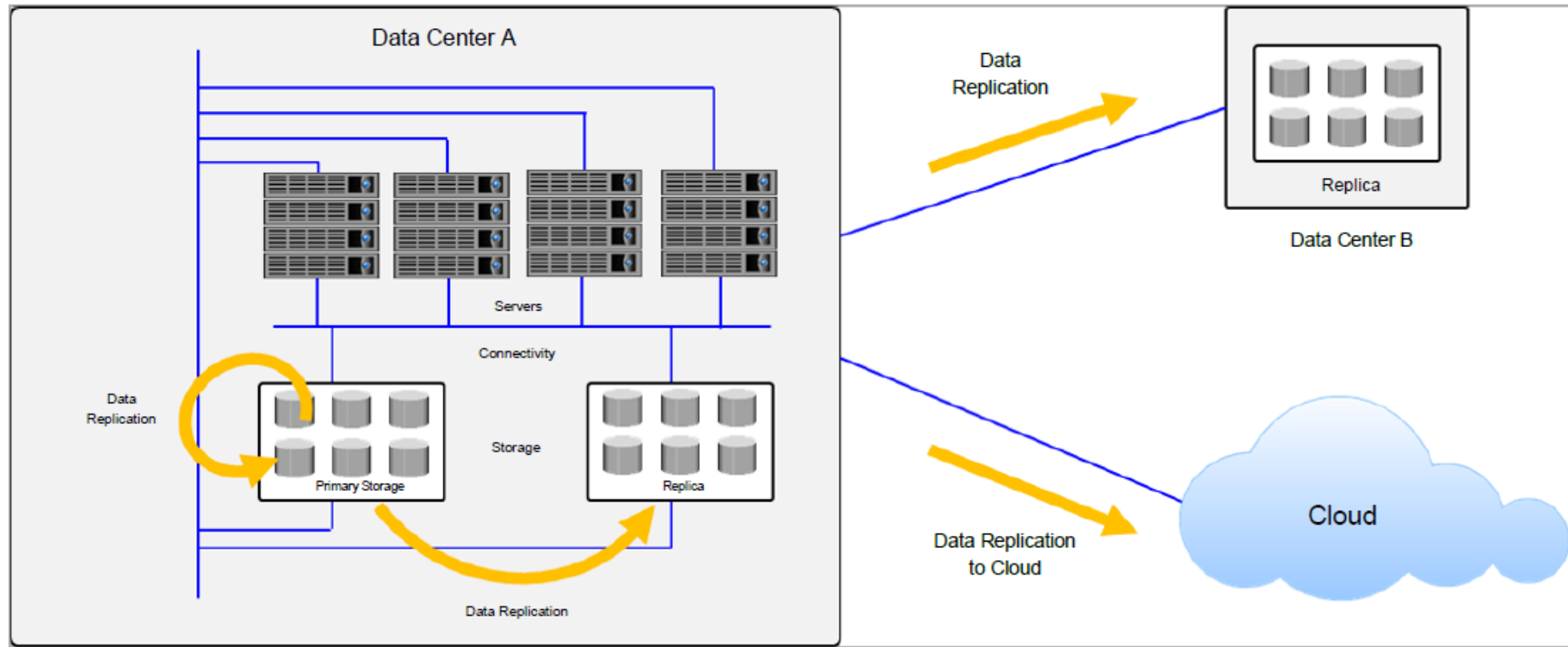
Replication

It is a process of creating an exact copy (replica) of data.

- Replication can be classified as
 - ▶ Local replication
 - ▶▶ Replicating data within the same array or data center
 - ▶ Remote replication
 - ▶▶ Replicating data at remote site



Replication



Uses of Local Replica

- Alternate source for backup
- Fast recovery
- Decision support activities
- Testing platform
- Data Migration



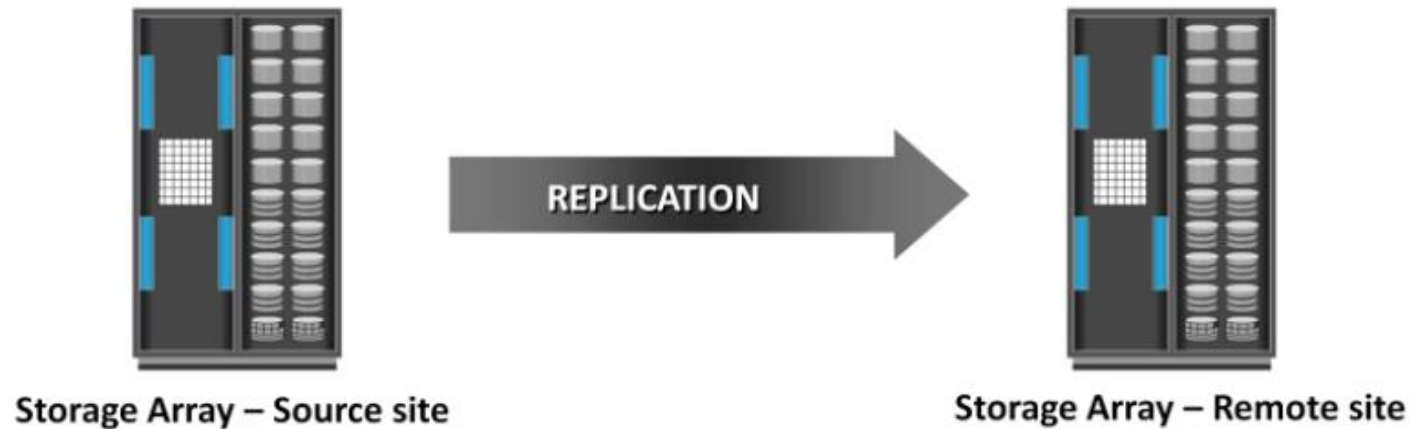
Replica Characteristics

- Recoverability/Restartability
 - ▶ Replica should be able to restore data on the source device
 - ▶ Restart business operation from replica
- Consistency
 - ▶ Replica must be consistent with the source
- Choice of replica tie back into RPO
 - ▶ Point-in-Time (PIT)
 - ▶▶ Non-zero RPO
 - ▶ Continuous
 - ▶▶ Near-zero RPO



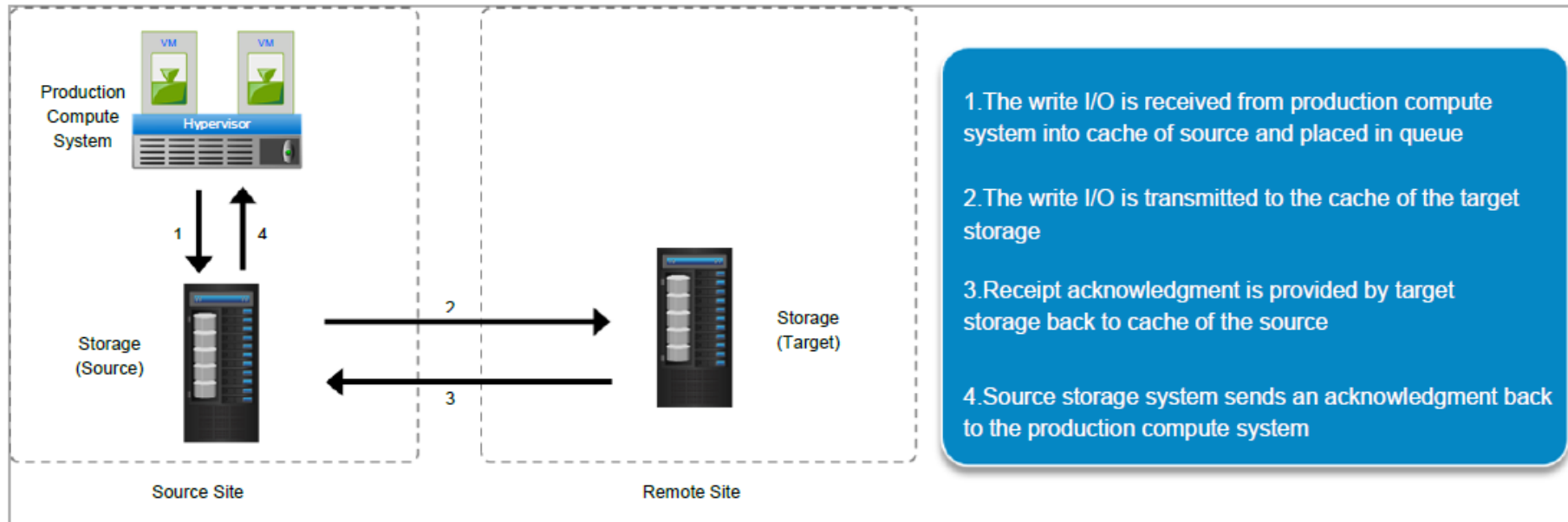
What is Remote Replication?

- Process of creating replicas at remote sites
 - ▶ Addresses risk associated with regionally driven outages
- Modes of remote replication
 - ▶ Synchronous
 - ▶ Asynchronous



Remote Replication: Synchronous

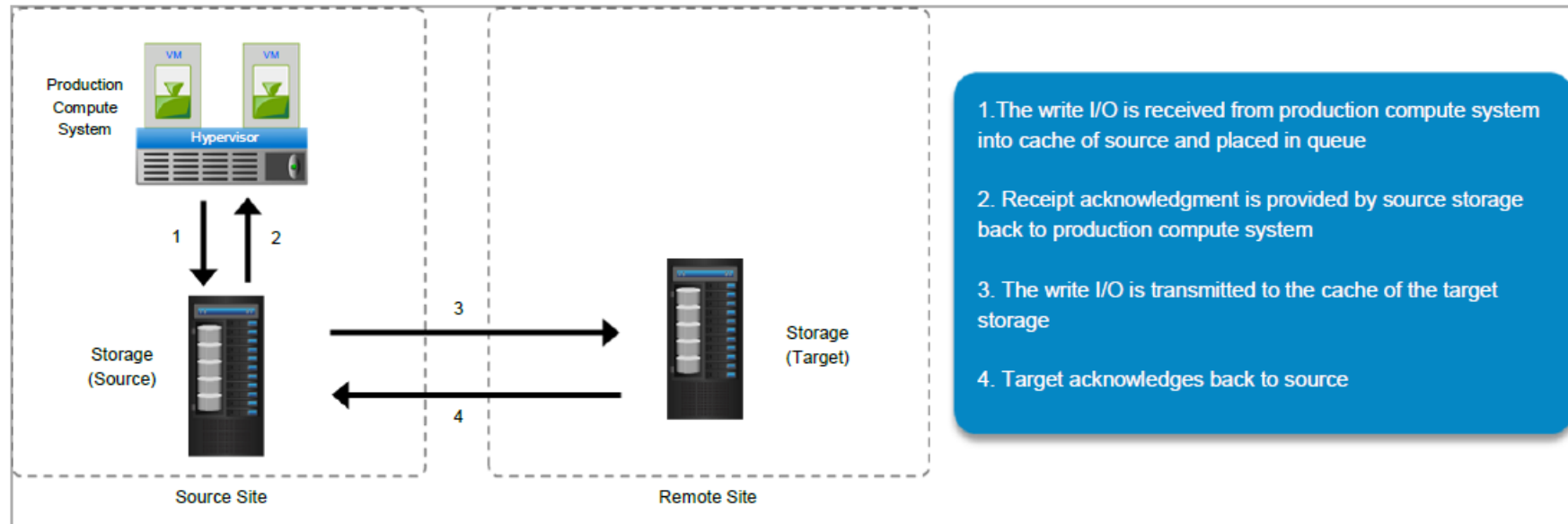
- Write is committed to both the source and the remote replica before it is acknowledged to the compute system
- Enables to restart business operations at a remote site with zero data loss; Provides near zero RPO



Remote Replication: Asynchronous

A write is committed to the source and immediately acknowledged to the compute system:

- Data is buffered at the source and sent to the remote site periodically
- Applications write response time is not dependent on the latency of the link
- Replica is behind the source by a finite amount (finite RPO)



Local Replication: Clone

- Cloning provides the ability to create fully populated point-in-time copies of LUNs within a storage system or create a copy of an existing VM
- Clone of a storage volume
 - Initial synchronization is performed between the source LUN and the replica (clone)
 - Changes made to both the source and the replica can be tracked at some predefined granularity



Pointer-based Virtual Replication – Snapshot

- An instantaneous virtual copy of a source LUN
 - Snapshot is not a real LUN, it is made up of pointers
 - No data is copied at creation time
- Snapshot is immediately accessible when the session is started
 - Allows secondary compute system to view PIT copy of the source
- Two mechanisms are used to preserve a PIT snapshot – CoFW and RoW
- Requires a small fraction of the size of the source volumes
 - Snapshot is recommended, if the changes to the source are typically less than 30%



Three-site Replication

- Data from source site is replicated to two remote sites
 - ▶ Replication is synchronous to one of the remote sites and asynchronous or disk buffered to the other remote site
- Mitigates the risk in two site replication
 - ▶ No DR protection after source or remote site failure
- Implemented in two ways:
 - ▶ Cascade/multihop
 - ▶ Triangle/multitarget





Introduction to NAS



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File Sharing Environment

- File sharing enables users to share files with other users
- Creator or owner of a file determines the type of access to be given to other users
- File sharing environment ensures data integrity when multiple users access a shared file at the same time
- Examples of file sharing methods:
 - File Transfer Protocol (FTP)
 - Peer-to-Peer (P2P)
 - Network File System (NFS) and Common Internet File System (CIFS)
 - Distributed File System (DFS)



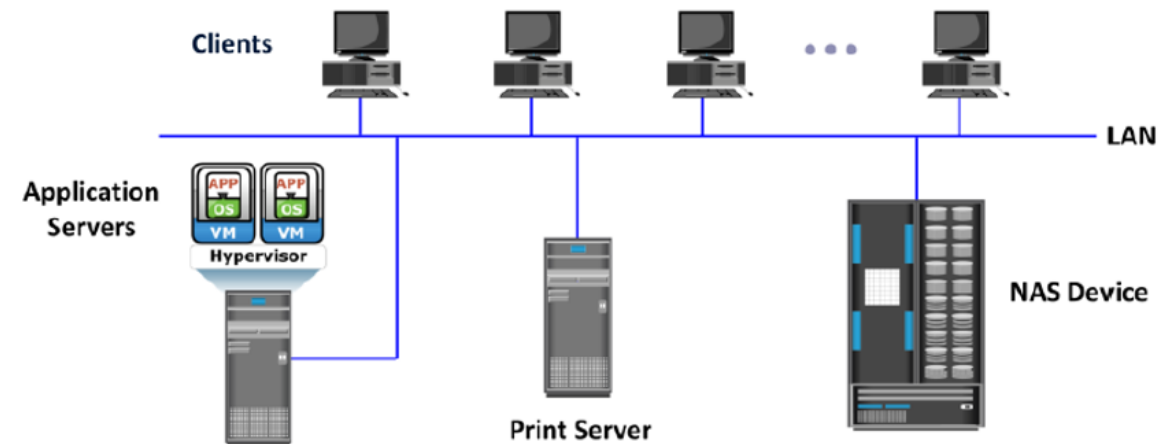
What is NAS?



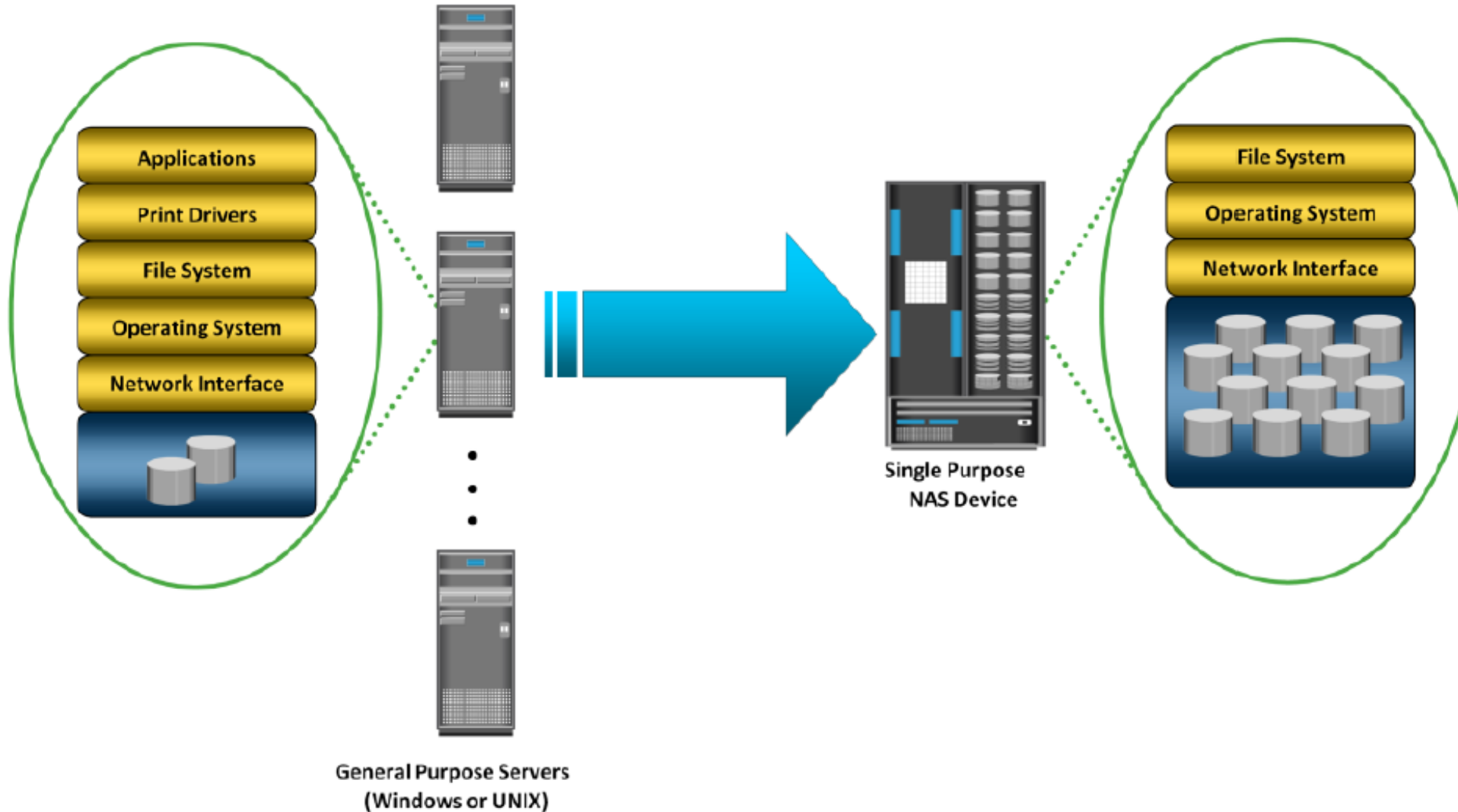
Definition: NAS

An IP-based, dedicated, high-performance file sharing and storage device.

- Enables NAS clients to share files over IP network
- Uses specialized operating system that is optimized for file I/O
- Enables both UNIX and Windows users to share data



General Purpose Servers Vs. NAS Devices

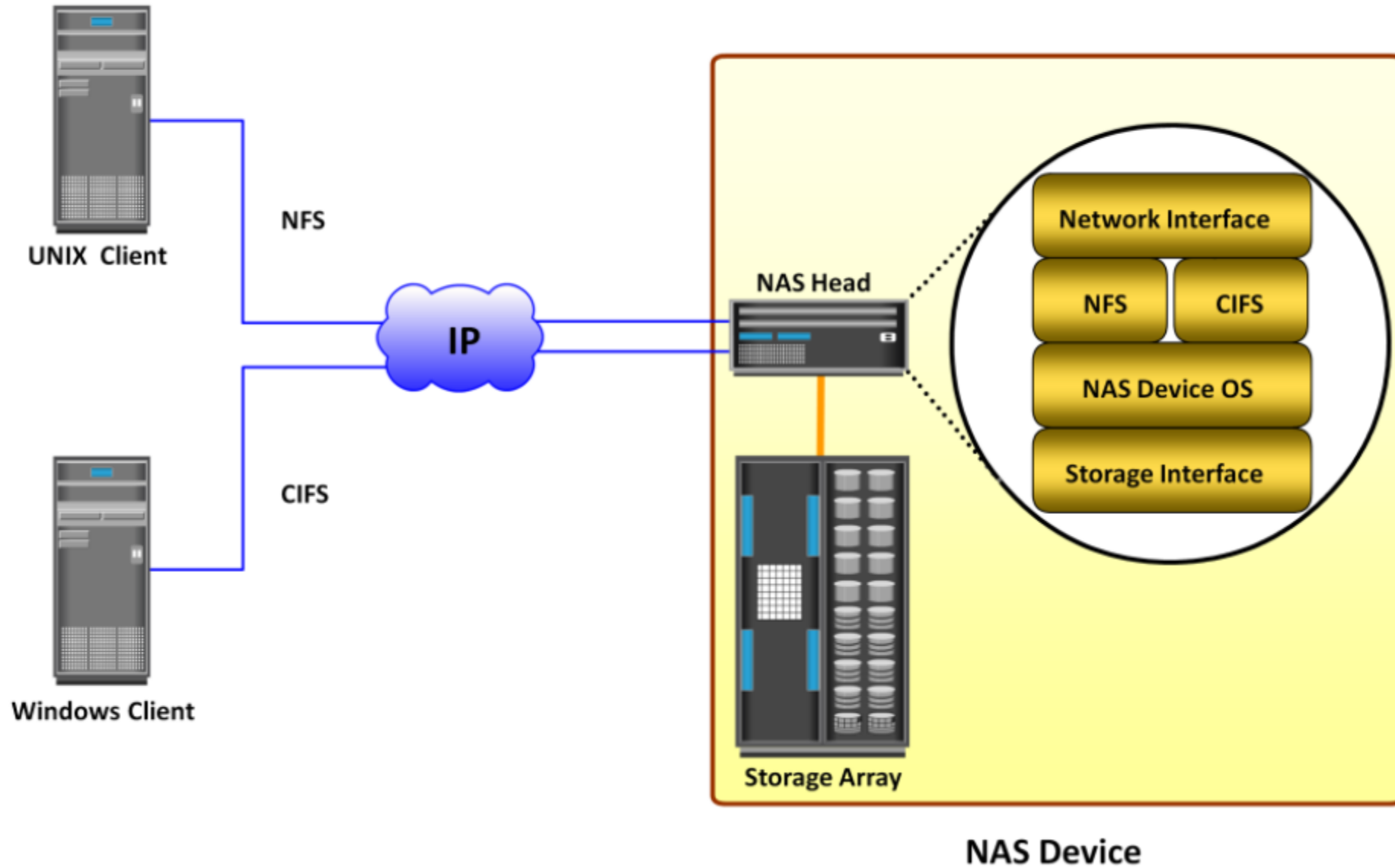


Benefits of NAS

- Improved efficiency
- Improved flexibility
- Centralized storage
- Simplified management
- Scalability
- High availability – through native clustering and replication
- Security – authentication, authorization, and file locking in conjunction with industry-standard security
- Low cost
- Ease of deployment



Components of NAS



- CPU and memory
- One or more network interface cards (NICs), which provide connectivity to the client network. Examples of network protocols supported by NIC include Gigabit Ethernet, Fast Ethernet, ATM, and Fiber Distributed Data Interface (FDDI)
- An optimized operating system for managing the NAS functionality. It translates file-level requests into block-storage requests and further converts the data supplied at the block level to file data
- NFS, CIFS, and other protocols for file sharing
- Industry-standard storage protocols and ports to connect and manage physical disk resources



NAS File Sharing Protocols

- Two common NAS file sharing protocols are:
 - ▶ Common Internet File System (CIFS)
 - ▶ Network File System (NFS)



Common Internet File System

- Client-server application protocol
 - ▶ An open variation of the Server Message Block (SMB) protocol
- Enables clients to access files that are on a server over TCP/IP
- Stateful Protocol
 - ▶ Maintains connection information regarding every connected client
 - ▶ Can automatically restore connections and reopen files that were open prior to interruption

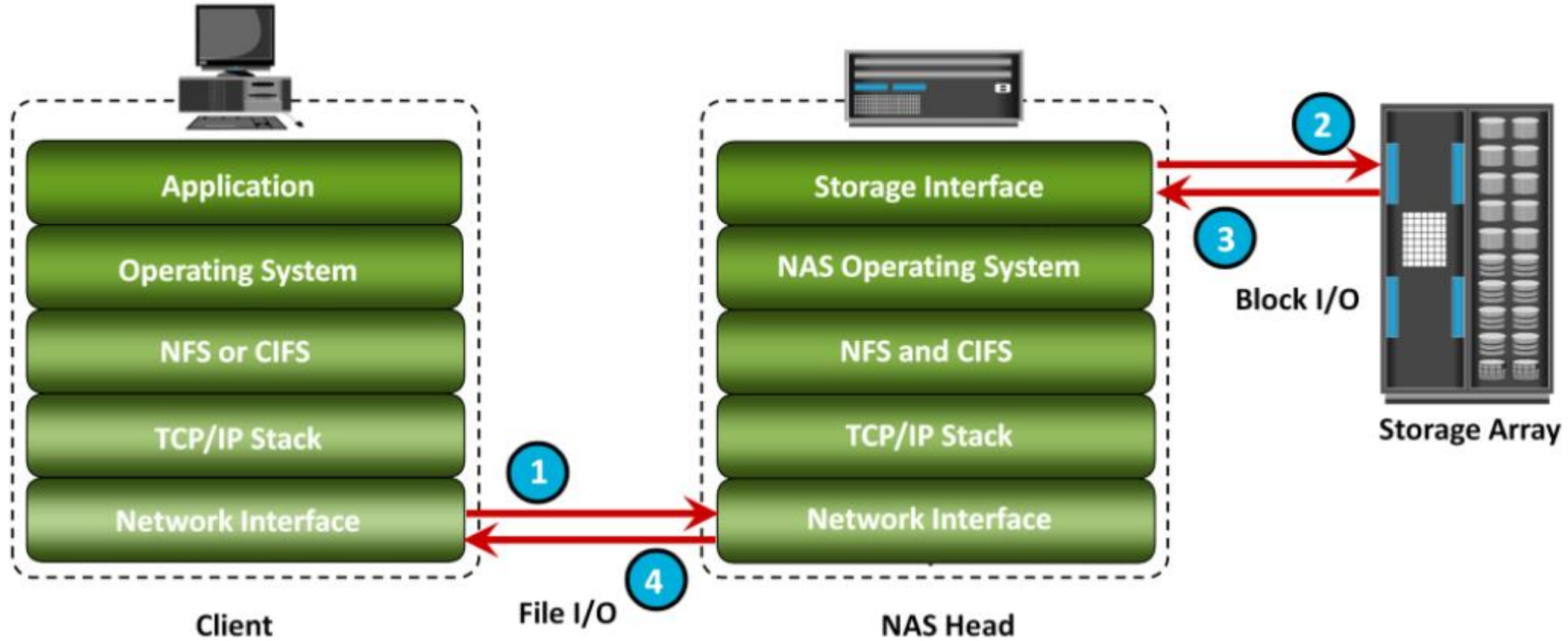


Network File System

- Client-server application protocol
- Enables clients to access files that are on a server
- Uses Remote Procedure Call (RPC) mechanism to provide access to remote file system
- Currently, three versions of NFS are in use:
 - ▶ NFS v2 is stateless and uses UDP as transport layer protocol
 - ▶ NFS v3 is stateless and uses UDP or optionally TCP as transport layer protocol
 - ▶ NFS v4 is stateful and uses TCP as transport layer protocol



NAS I/O Operation

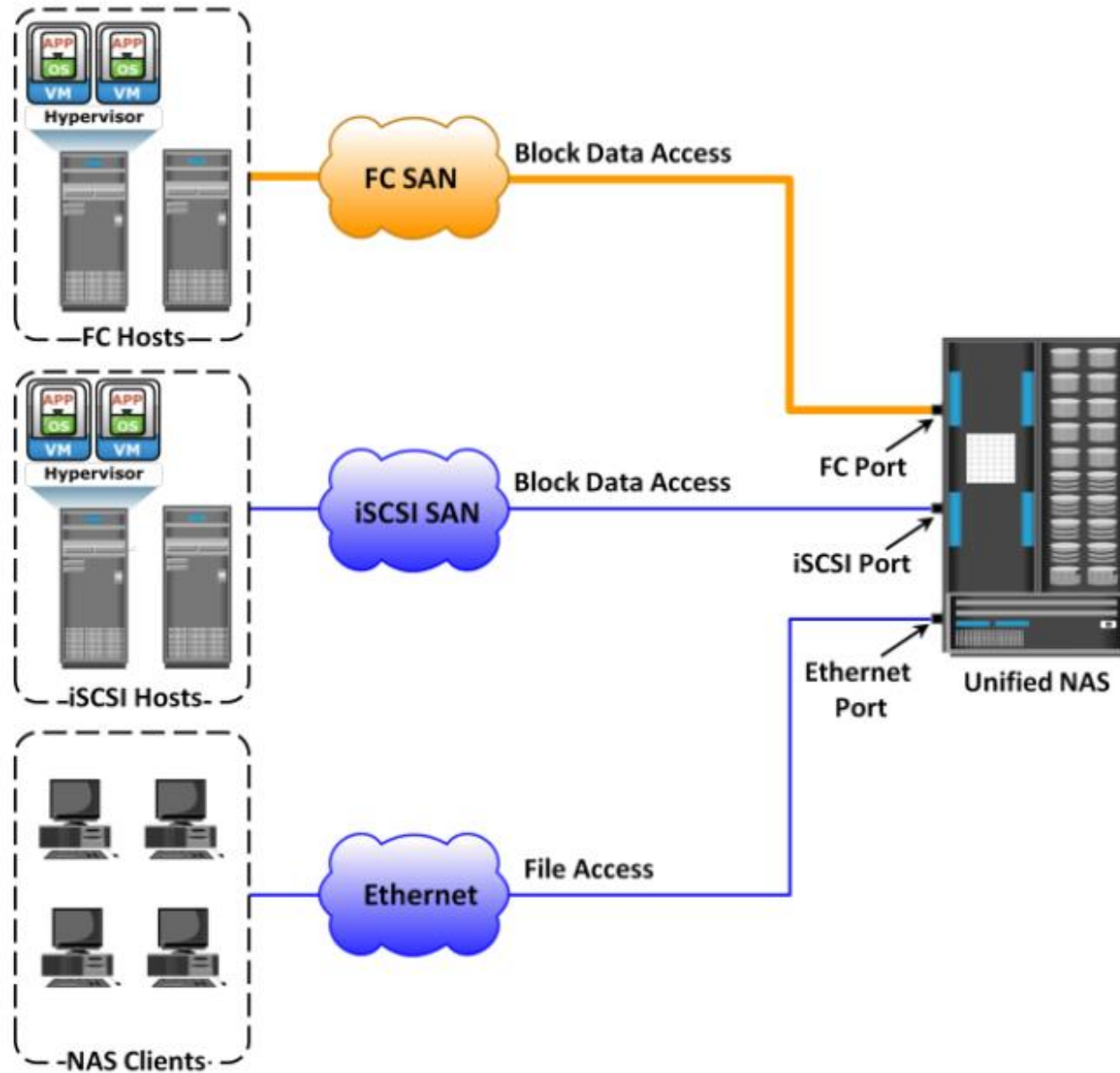


NAS Implementation – Unified NAS

- Consolidates NAS-based (file-level) and SAN-based (block-level) access on a single storage platform
- Supports both CIFS and NFS protocols for file access and iSCSI and FC protocols for block level access
- Provides unified management for both NAS head and storage



Unified NAS Connectivity

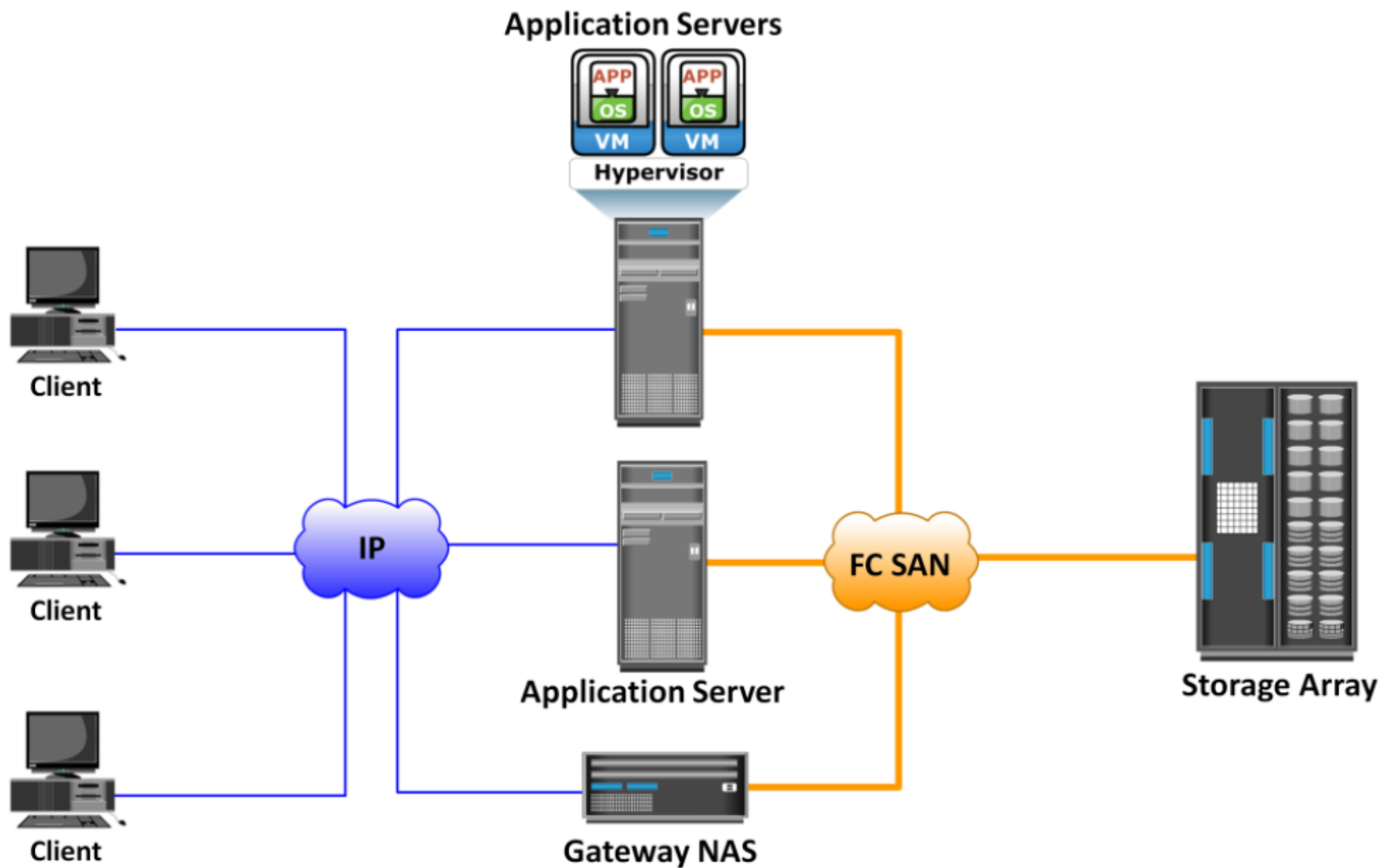


NAS Implementation – Gateway NAS

- Uses external and independently-managed storage
 - ▶ NAS heads access SAN-attached or direct-attached storage arrays
- NAS heads share storage with other application servers that perform block I/O
- Requires separate management of NAS head and storage



Gateway NAS Connectivity





Object Storage



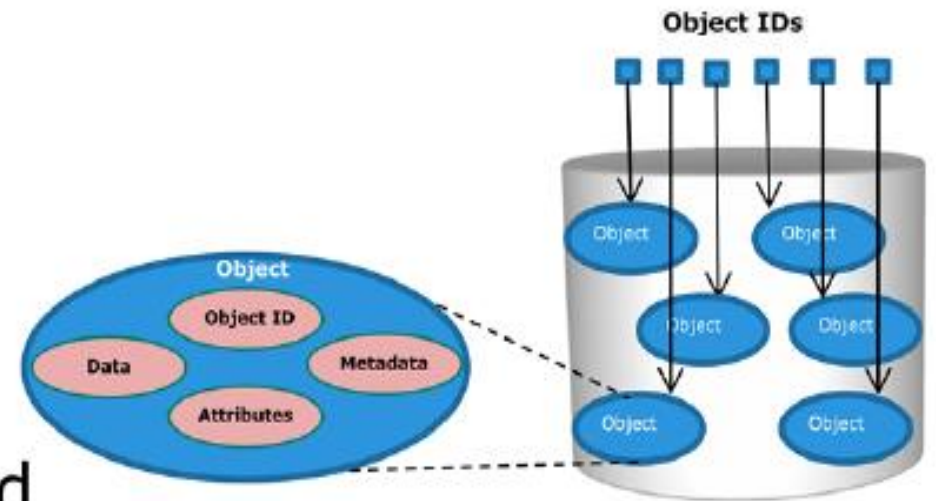
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Object-based Storage Device

Object-based Storage Device

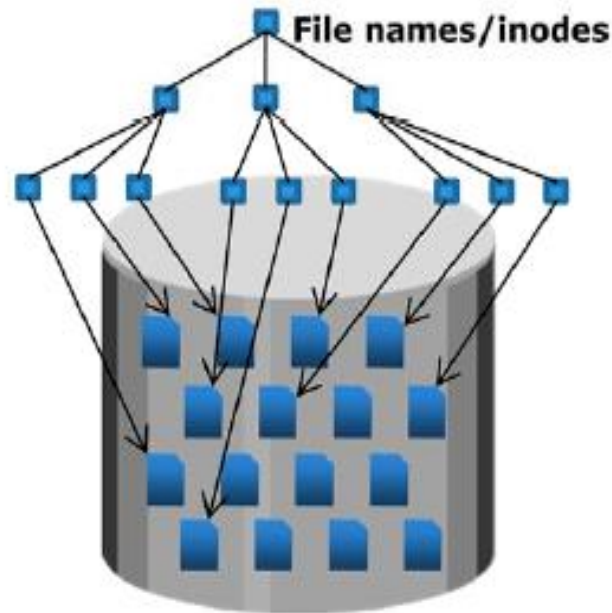
Stores data in the form of objects on flat address space based on its content and attributes rather than the name and location.

- Object contains user data, related metadata, and user-defined attributes
 - Objects are uniquely identified using object ID
- OSD provides APIs to integrate with software-defined data center and cloud

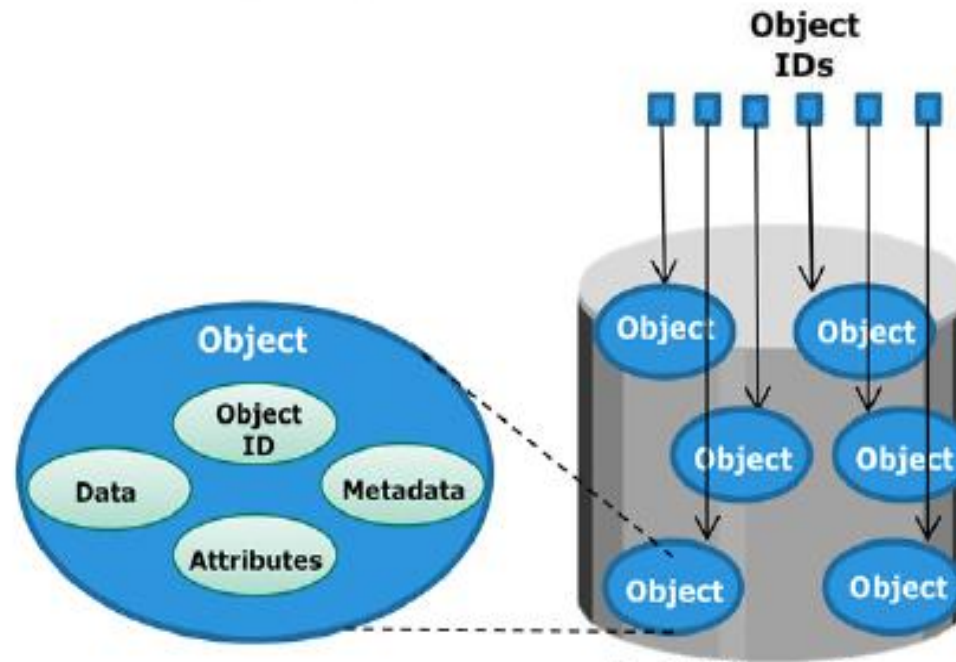


Hierarchical File System Vs. Flat Address Space

- Hierarchical file system organizes data in the form of files/directories
 - Limits the number of files that can be stored
- OSD uses flat address space that enables storing large number of objects
 - Enables the OSD to meet the scale-out storage requirement of third platform



Hierarchical File system



Flat Address Space

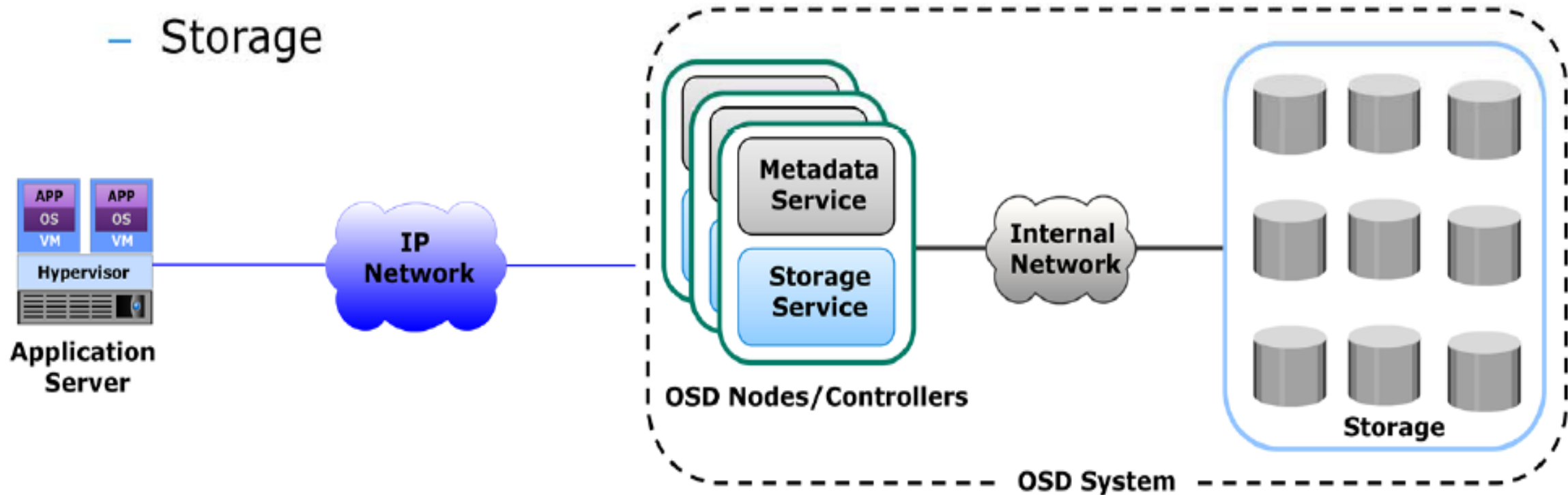
What is an Object?

Object is the atomically referenced unit of data storage and transmission

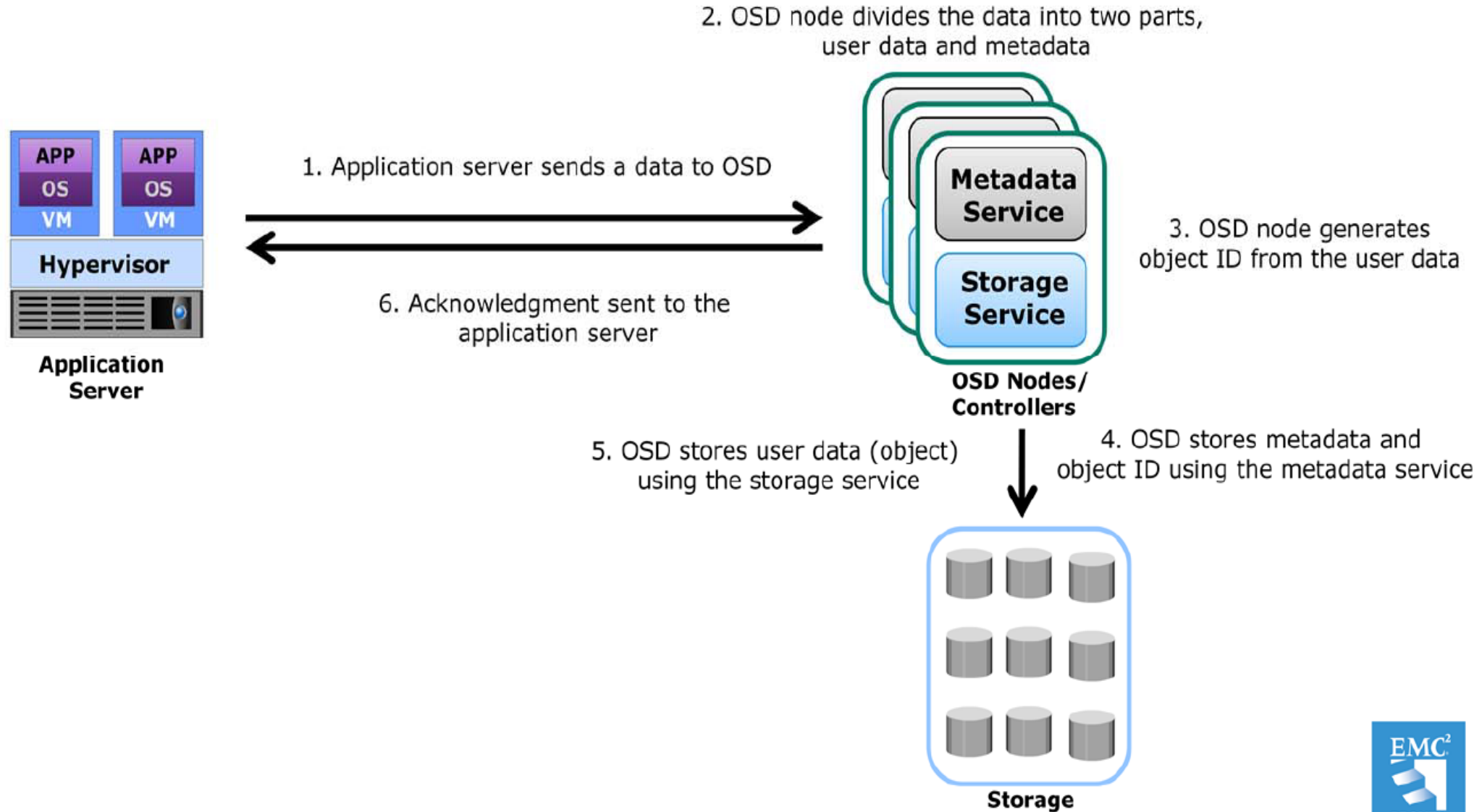


Components of Object-based Storage Device

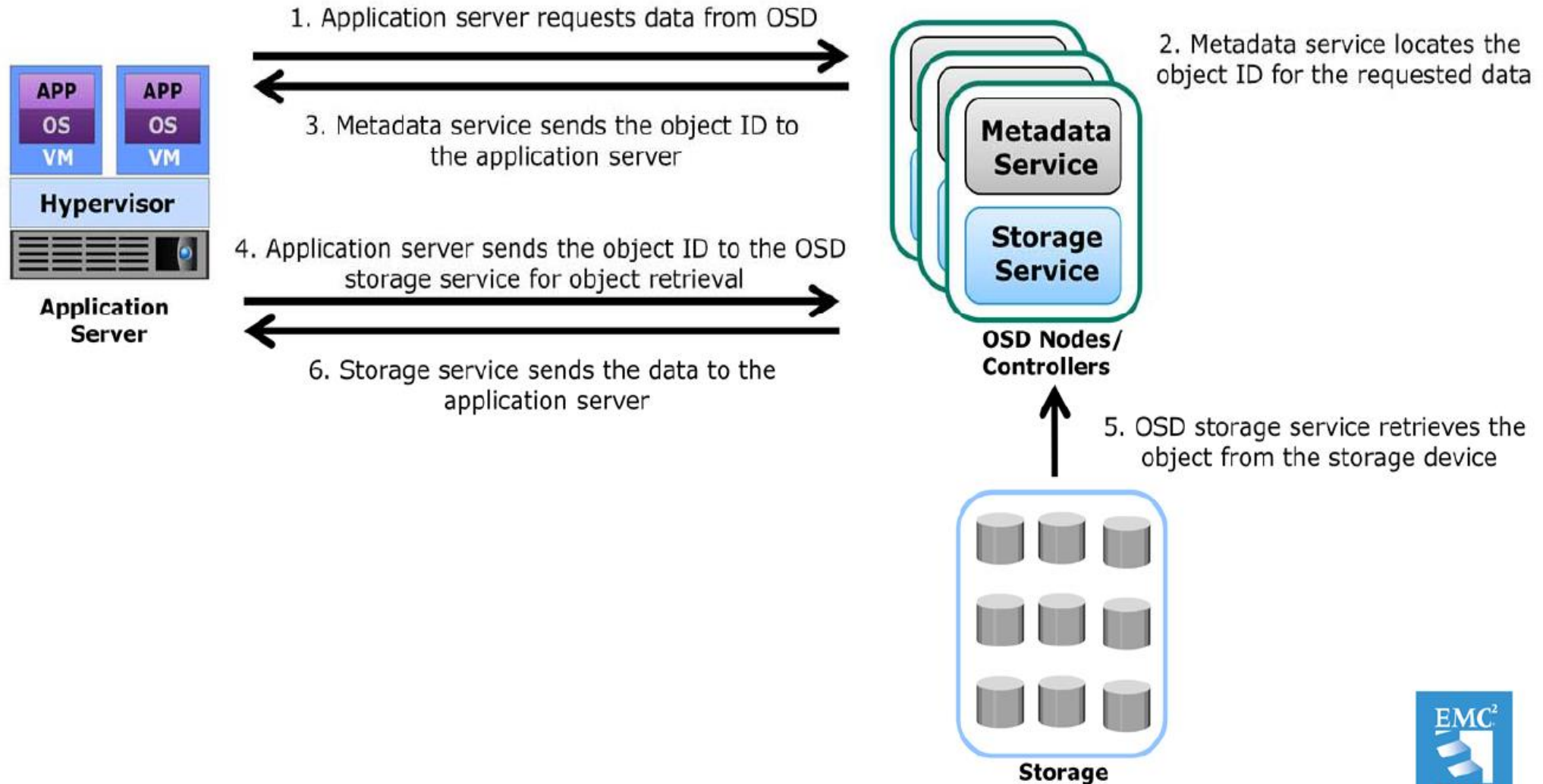
- OSD system typically comprises three key components:
 - OSD nodes (controllers)
 - Internal network
 - Storage



Process of Storing Object in OSD System



Process of Retrieving Object from OSD System



Unified Storage Overview

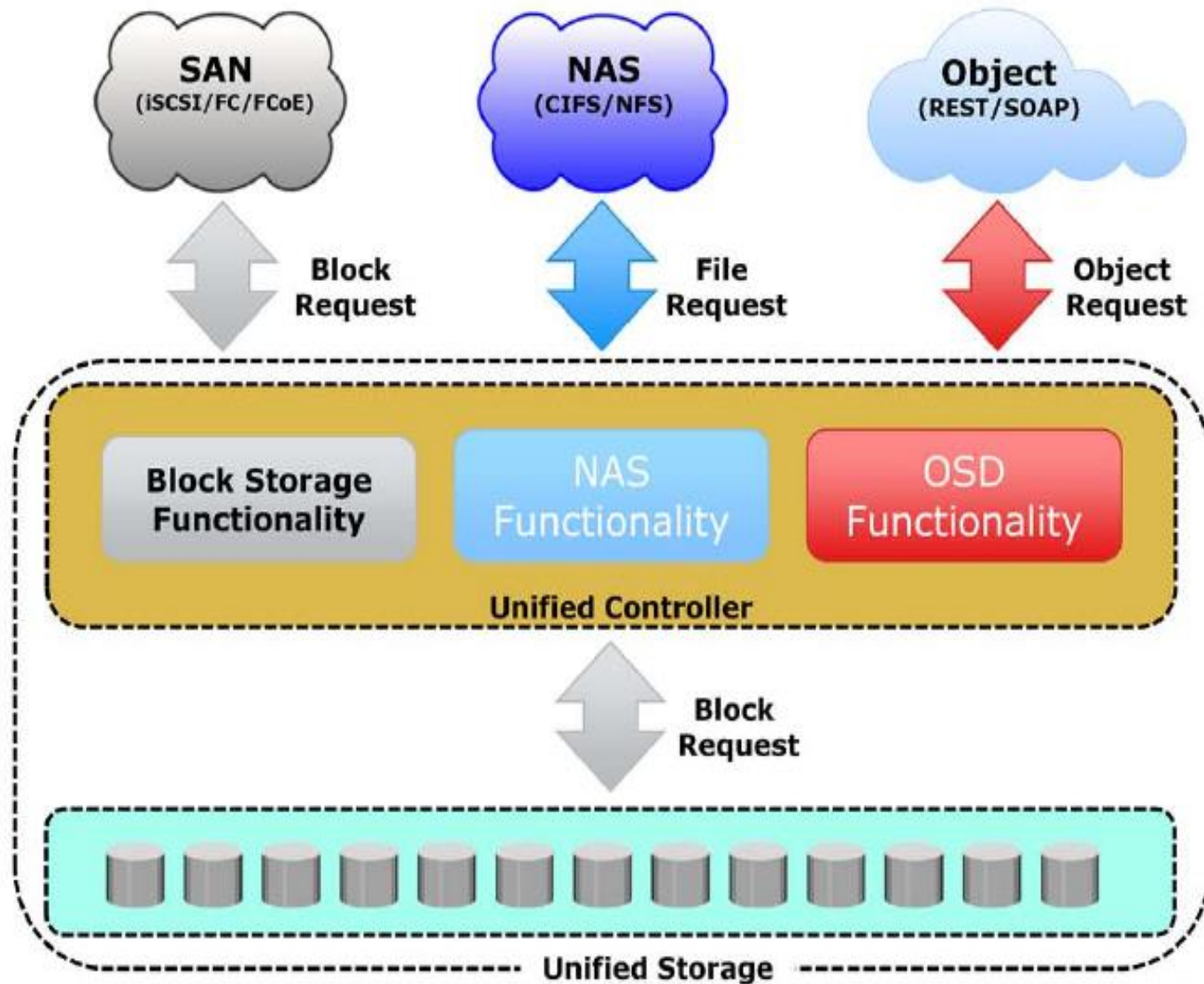
Unified Storage

A single integrated(converged) storage infrastructure that consolidates block (iSCSI, FC, FCoE), file,(CIFS, NFS), and object (REST, SOAP) access.

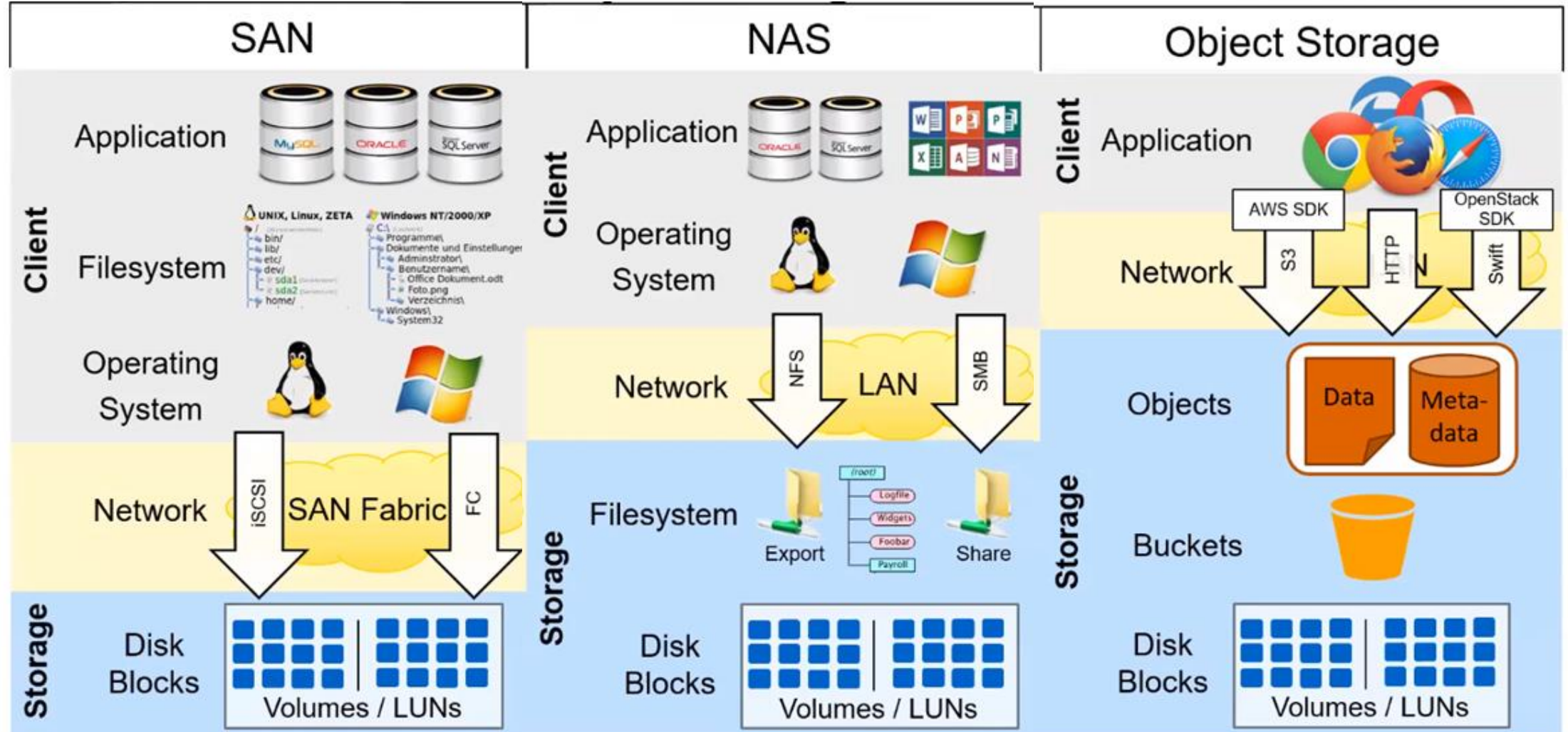
- Deploying unified storage provides following benefits
 - Reduces capital and operational expenses
 - Managed through single management interface
 - Increases storage utilization
- Integration with software-defined environment provides storage for mobile, cloud, big data, and social applications



Unified Storage Architecture



SAN/NAS/Object Comparison





IP SAN



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What is IP SAN

Uses Internet Protocol (IP) for the transport of storage traffic. It transports block I/O over an IP-based network.

Provides an efficient and dedicated point-to-point storage solution.

Typically runs over a standard IP-based network and uses the TCP/IP) for communication, commonly:

- Internet SCSI (iSCSI)
- Fibre Channel over IP (FCIP)



Drivers of IP SAN

The following are drivers have led to the adoption of IP SAN:

- Existing IP-based network infrastructure can be leveraged
 - Reduced cost compared to deploying new FC SAN infrastructure
- IP network makes it possible to extend or connect SANs over long distances
- Many long-distance disaster recovery solutions already leverage IP-based network
- Many robust and mature security options are available for IP network



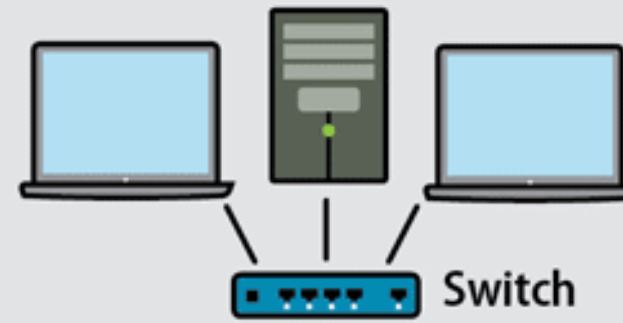
iSCSI Overview

- IP-based protocol that enables transporting SCSI data over an IP network
- Encapsulates SCSI I/O into IP packets and transports them using TCP/IP



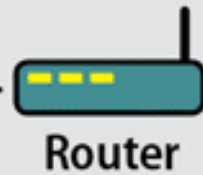
EXAMPLE 1

Clients connected to servers set up with iSCSI initiators over the Local Area Network



EXAMPLE 2

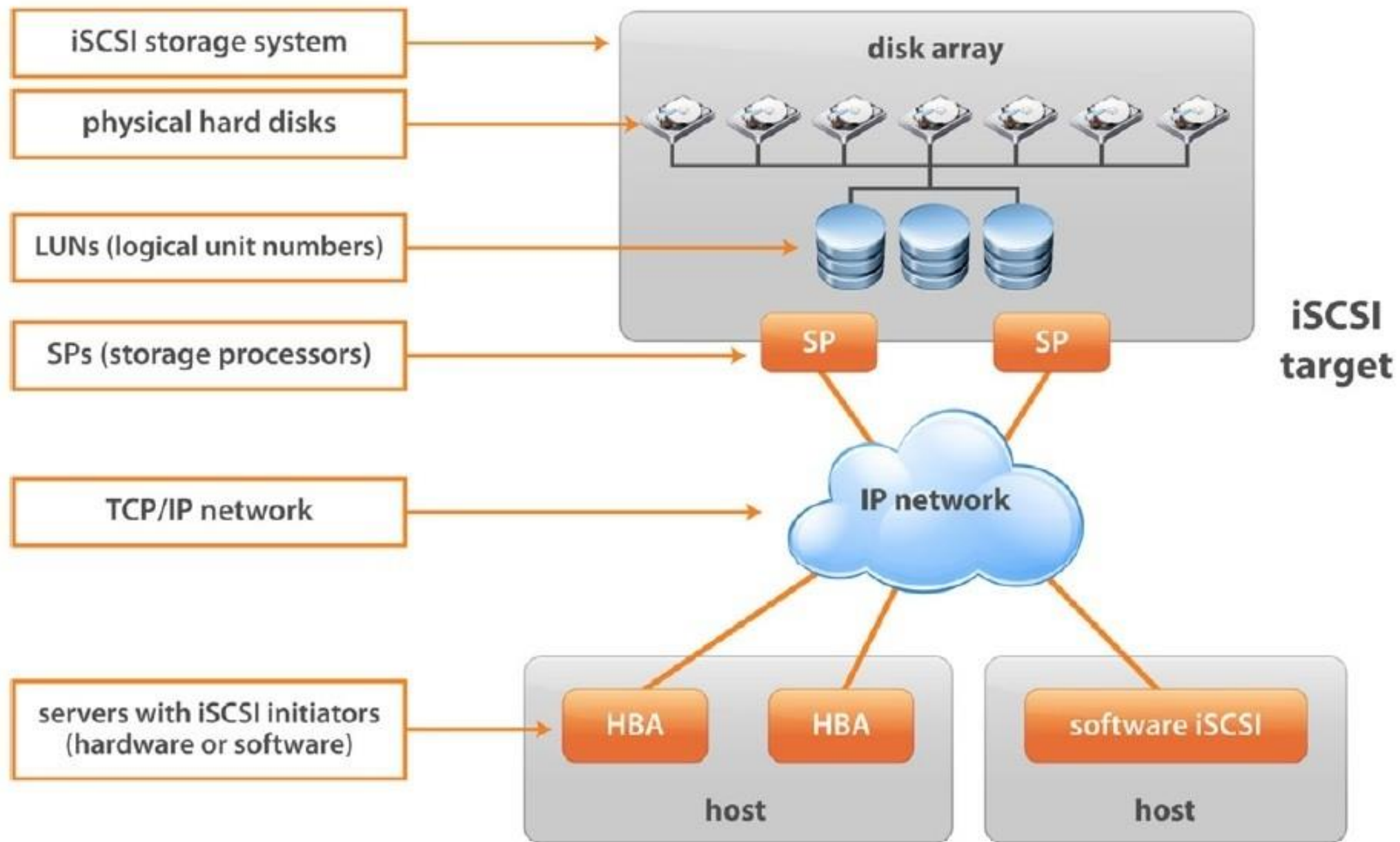
Clients with iSCSI initiators over the internet



iSCSI Appliances
(iSCSI Targets)

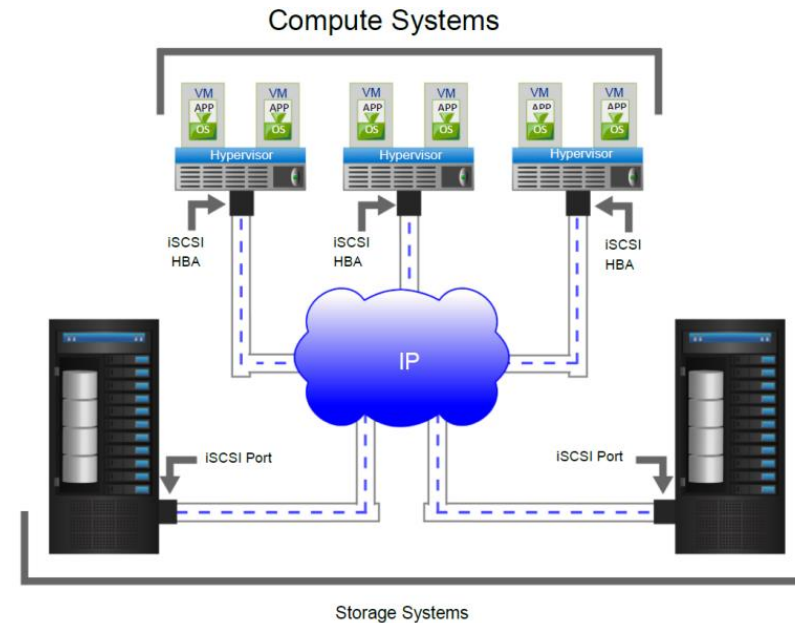
EXAMPLE 3

Clients with iSCSI initiator connected directly to gigabit switch over the Local Area Network



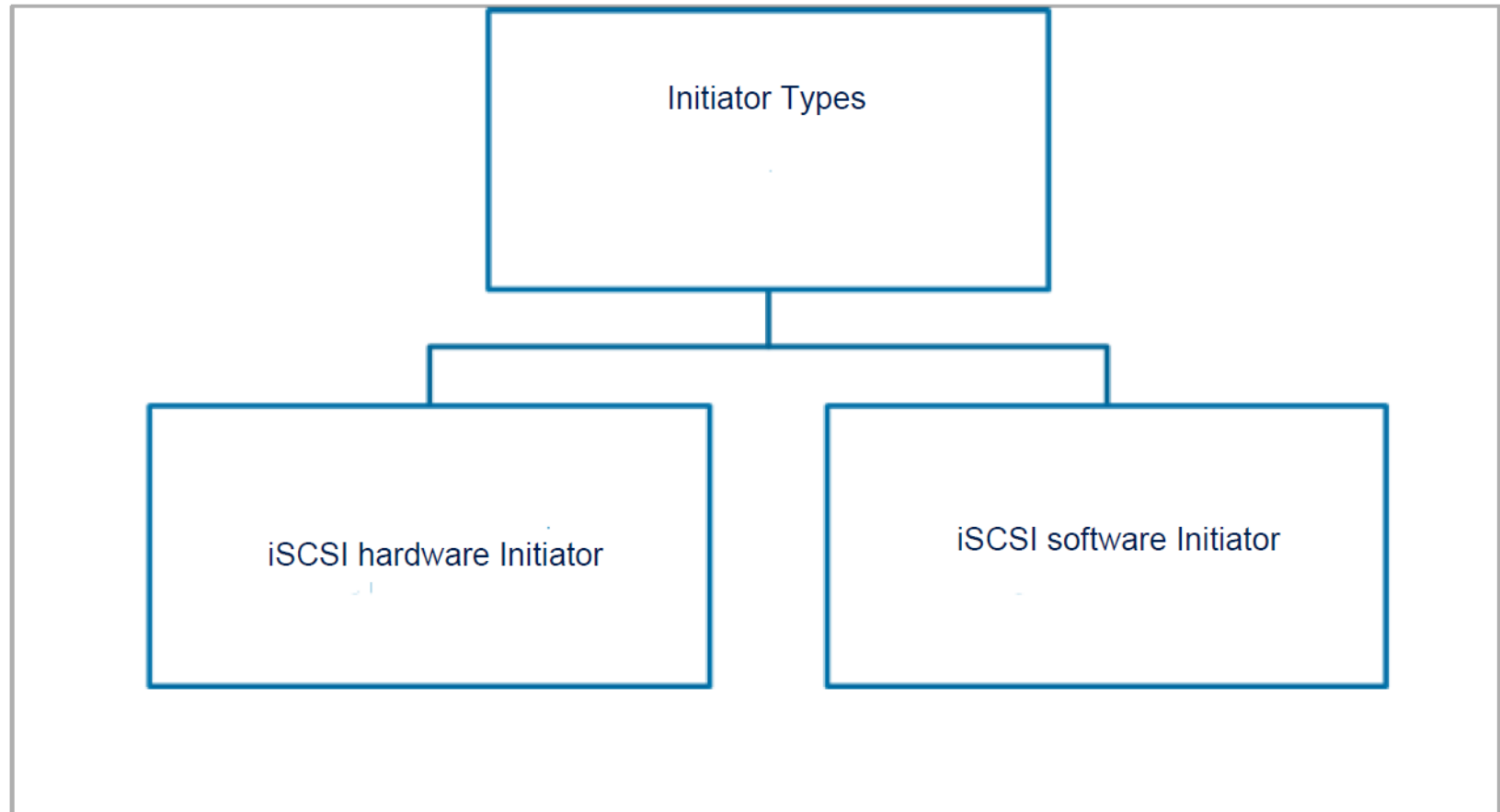
Components of iSCSI Network

- ▶ Key components for iSCSI communication are:
- ▶ iSCSI initiators
 - ▶ Example: iSCSI HBA
- ▶ iSCSI targets
 - ▶ Example: Storage system with iSCSI port
- ▶ IP-based network
 - ▶ Example: Gigabit Ethernet LAN



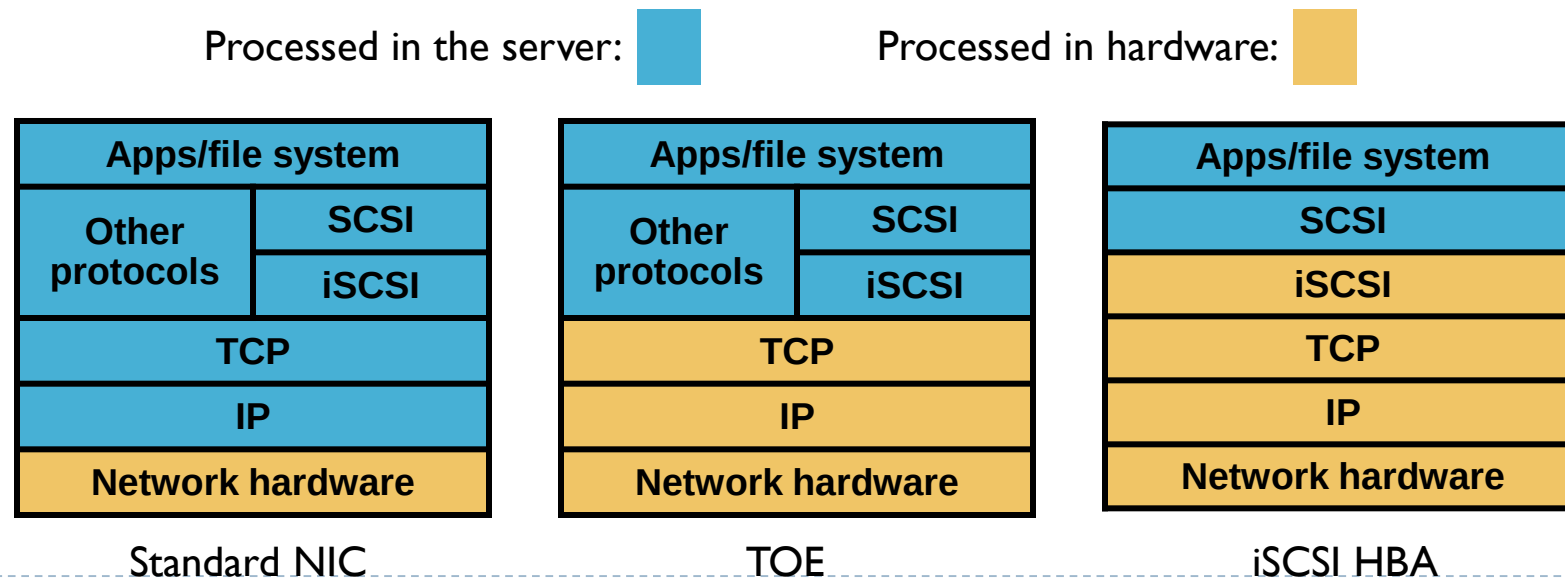
Types of iSCSI Initiator

- ▶ Hardware and software initiators are types of iSCSI initiators that are used by the host to access iSCSI targets.



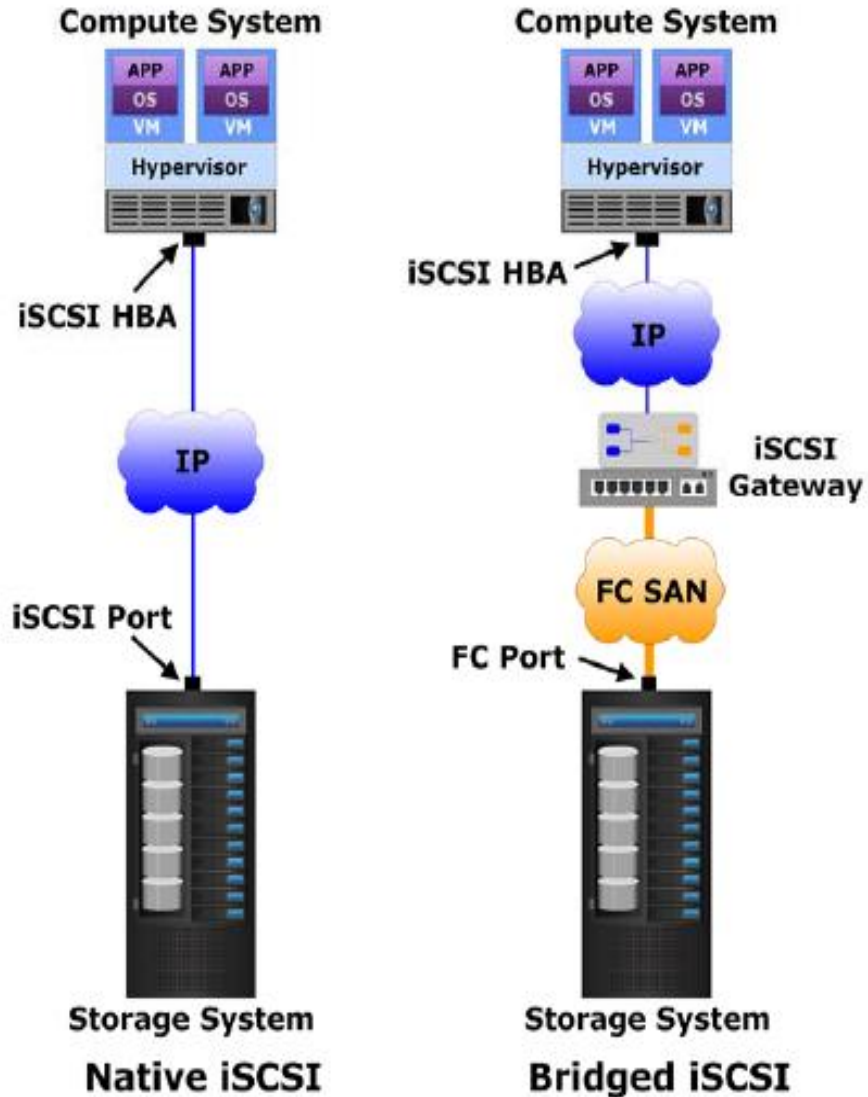
iSCSI Drivers and Offload Engines

- ▶ Different approaches to iSCSI initiators:
 - ▶ iSCSI driver with a standard network card
 - ▶ NIC with a TOE
 - ▶ HBAs that offload both TCP and iSCSI



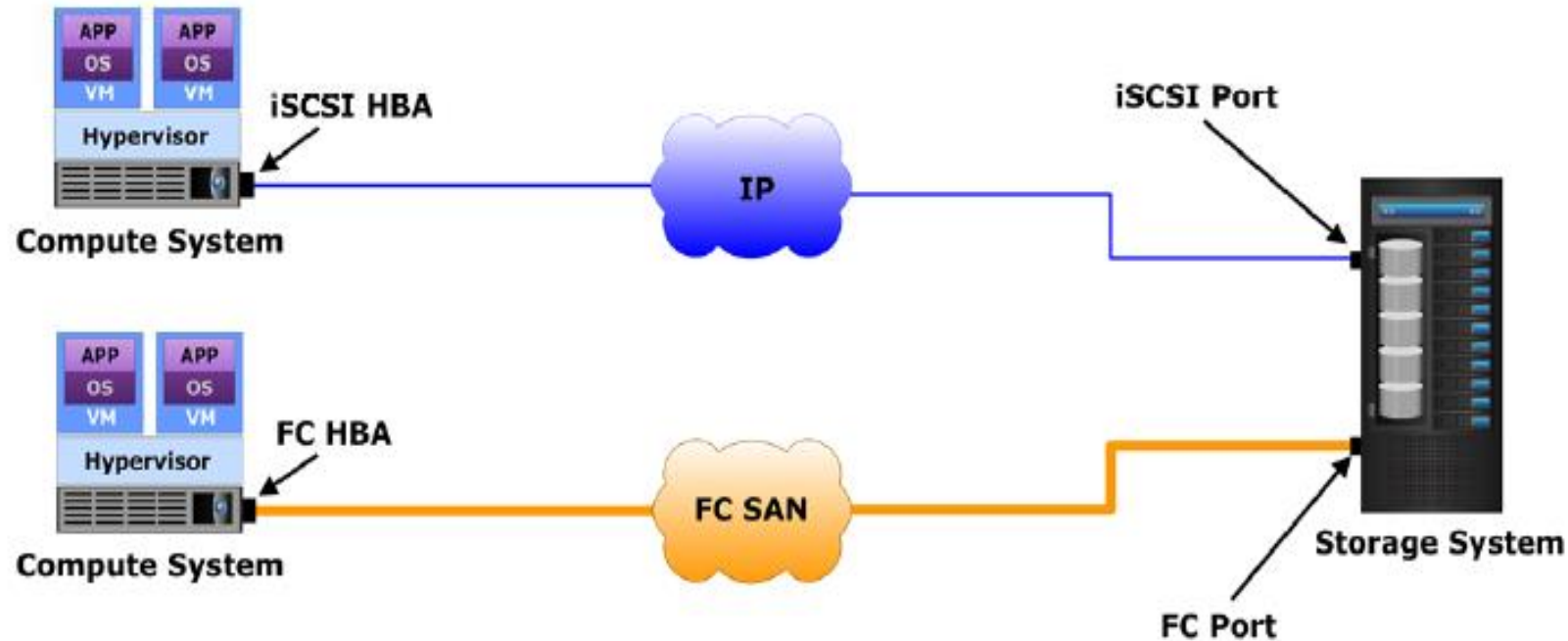
iSCSI Connectivity

- Native
 - iSCSI initiators connect to iSCSI targets directly/through IP network
 - No FC component
- Bridged
 - iSCSI initiators are attached to IP network
 - Storage systems are attached to FC SAN
 - iSCSI gateway provides bridging functionality



Combining FC and Native iSCSI Connectivity

- Storage system provides both FC and iSCSI ports
 - Enable iSCSI and FC connectivity in the same environment
 - No bridge device is needed





Storage Virtualization



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Benefits of Consolidation

Data Center Challenges

- Security
- Regulatory compliance
- Controlling costs
- Organizational structure
- Time to market for new applications
- Information management:
 - Dealing with growth, and so on
- Heat, energy, cooling, and space



Benefits of Consolidation

- ▶ **Reduced operational costs:**
 - ▶ By reducing infrastructure and virtualizing resources
- ▶ **Reduced space:**
 - ▶ By consolidating servers, storage and switches
- ▶ **Reduced downtime:**
 - ▶ Integrated management provides more effective control
- ▶ **More agile:**
 - ▶ Virtualized resources are quickly redeployed when required
- ▶ **More effective management:**
 - ▶ Centralized and integrated management of SAN resources



Consolidation Methods

- ▶ **Server consolidation:**

- ▶ Consolidate many physical servers into small number of high end servers hosting virtual machines, for example VMWare, zSeries, and so on
- ▶ NPIV allows several virtual servers to share a single switch port while maintaining their identity

- ▶ **Network consolidation:**

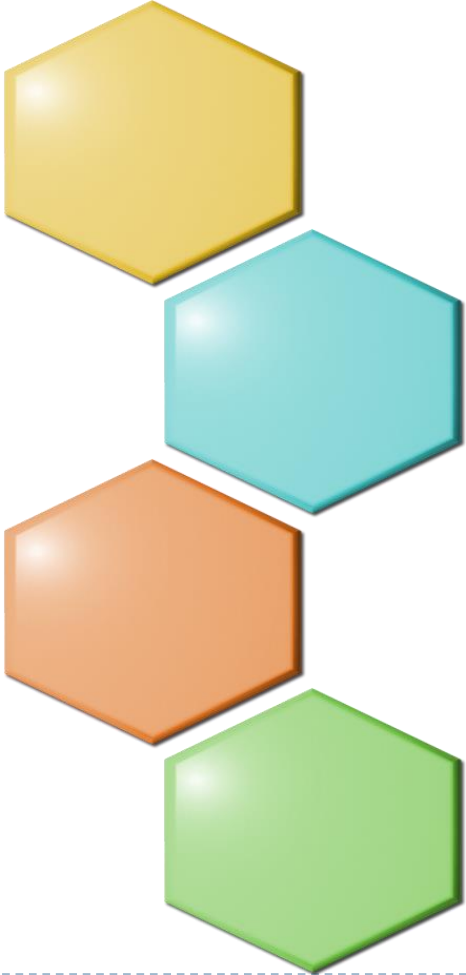
- ▶ VSANs allow consolidation of many physical fabrics or SAN Islands onto one or more MDS switches, creating logical fabrics, to maintain isolation
- ▶ Trunking allows frames belonging to different VSANs to share a single ISL or Port Channel between switches

- ▶ **Storage virtualization:**

- ▶ RAID groups combine several physical disks into LUNs for better performance and redundancy
- ▶ Volume Manager combines LUNs for mirroring or to create larger volumes



Defining Virtualization



Virtualization provides logical views of physical resources while preserving the usage interfaces for those resources

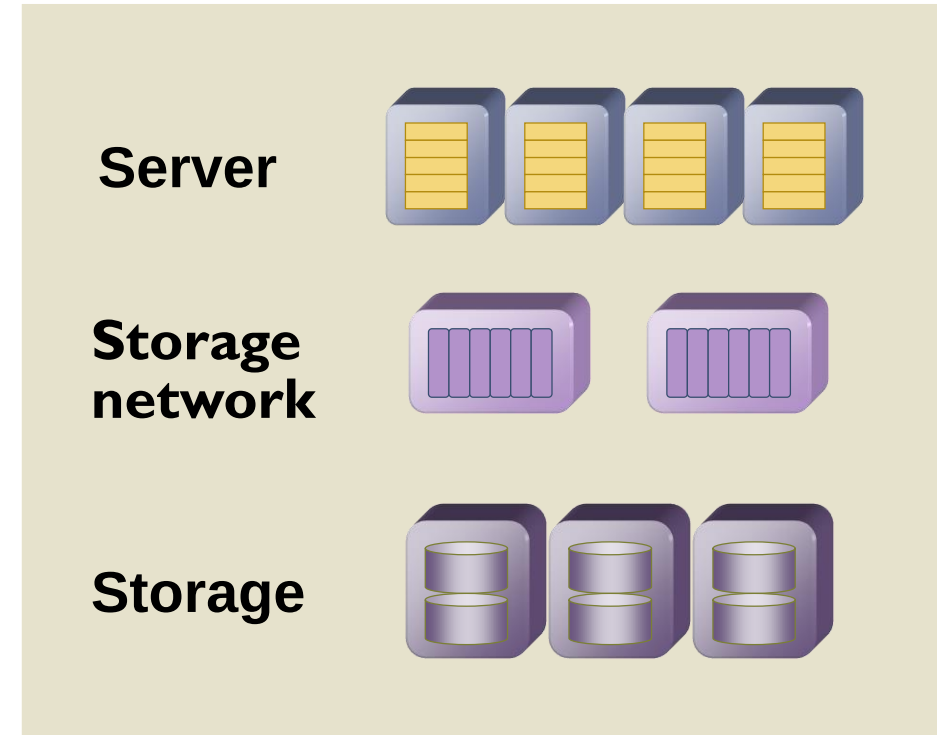
Virtualization removes physical resource limits and improves resource utilization



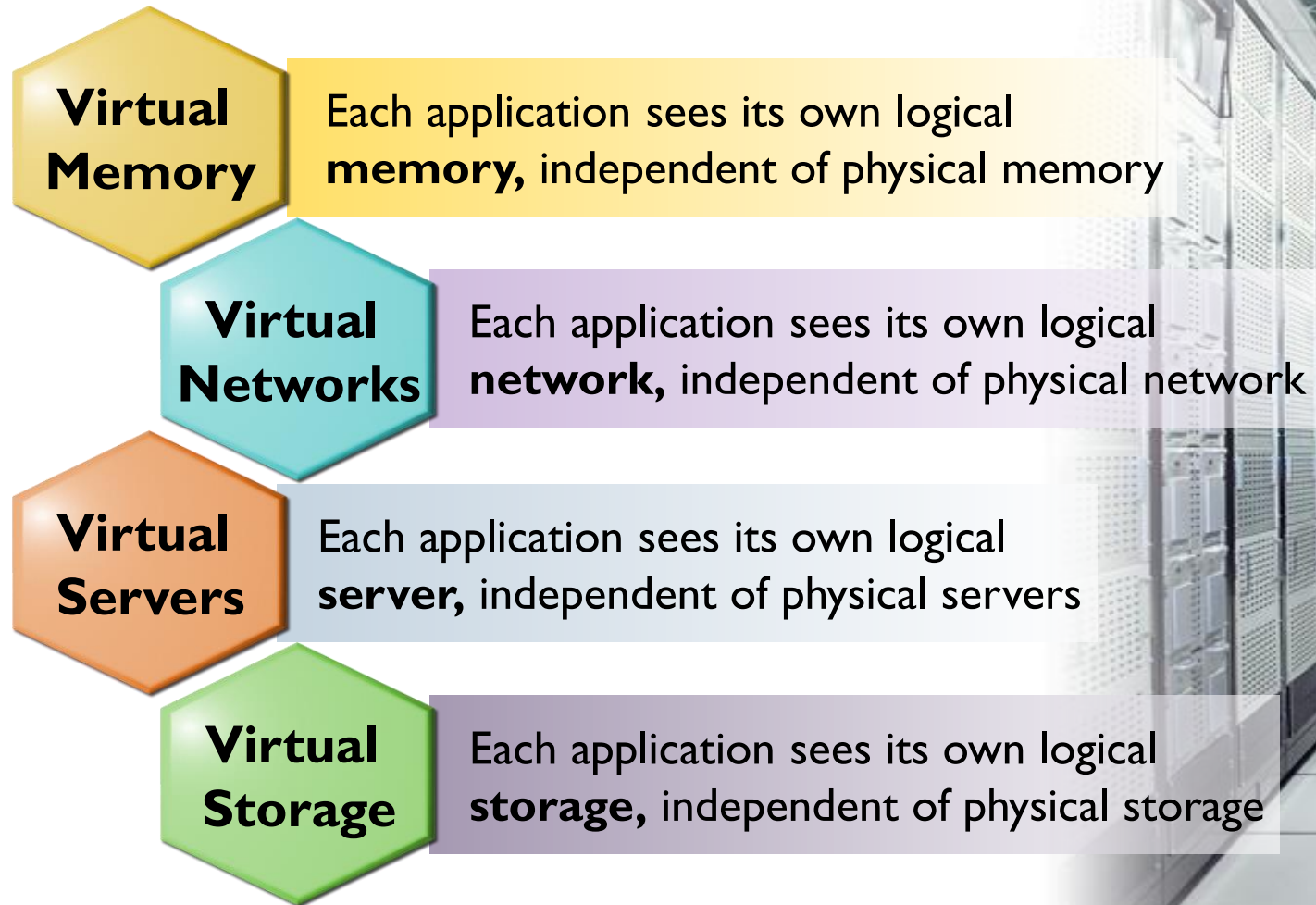
What Makes Virtualization Interesting

Potential Benefits:

- Higher rates of usage
- Simplified management
- Platform independence
- More flexibility
- Better availability

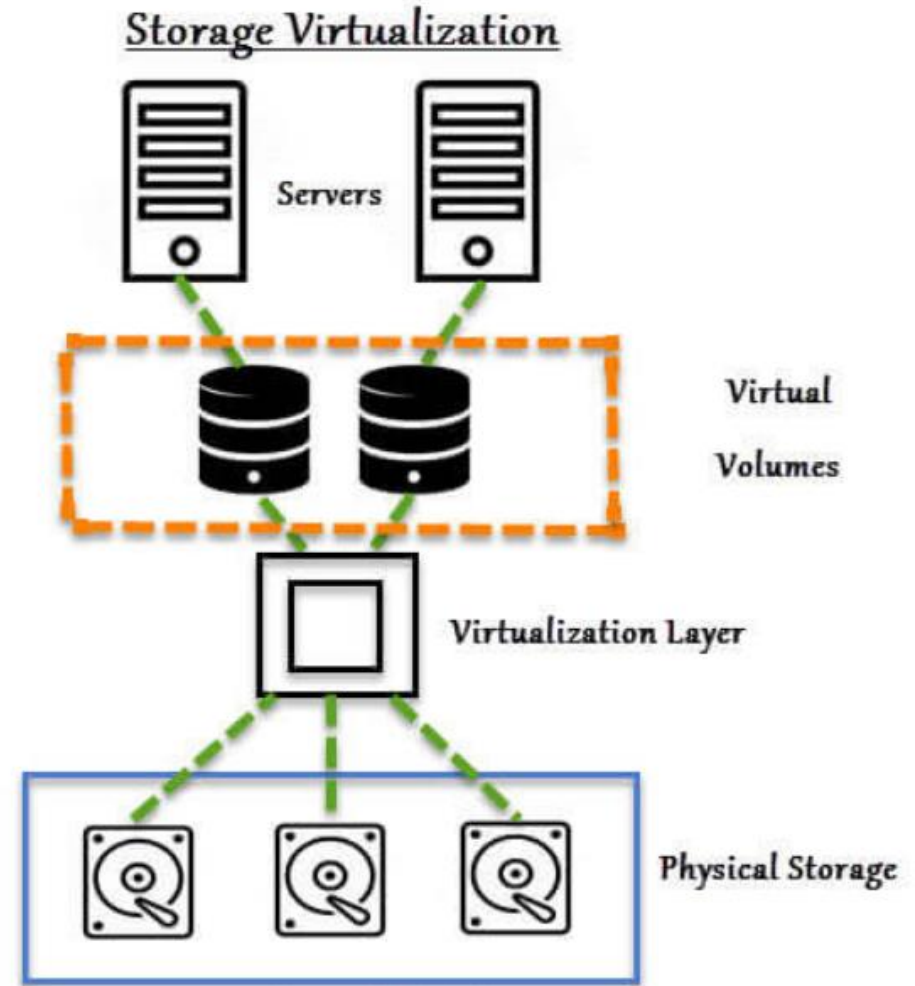


Virtualization Comes in Many Forms



Storage Virtualization

- ▶ Storage virtualization is the pooling of physical storage from multiple storage devices into what appears to be a single storage device -- or pool of available storage capacity -- that is managed from a central console.



Storage Functionality Today

Intelligence Lives Primarily on Servers and Storage Arrays

Server

- Path management
- Volume management
- Replication



Storage network

- Connectivity



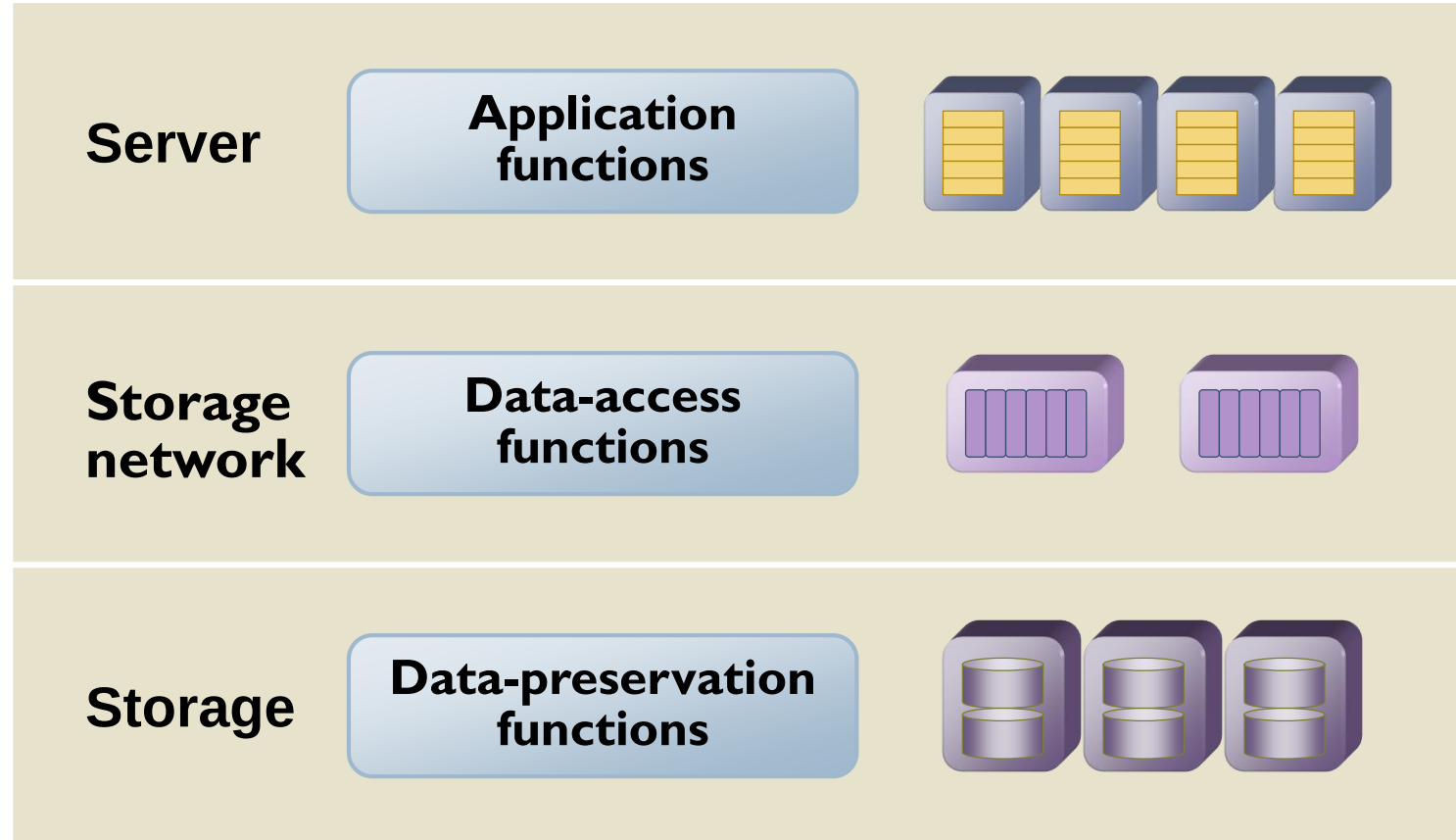
Storage

- Volume management
 - LUNs
- Access control
- Replication
- RAID
- Cache protection



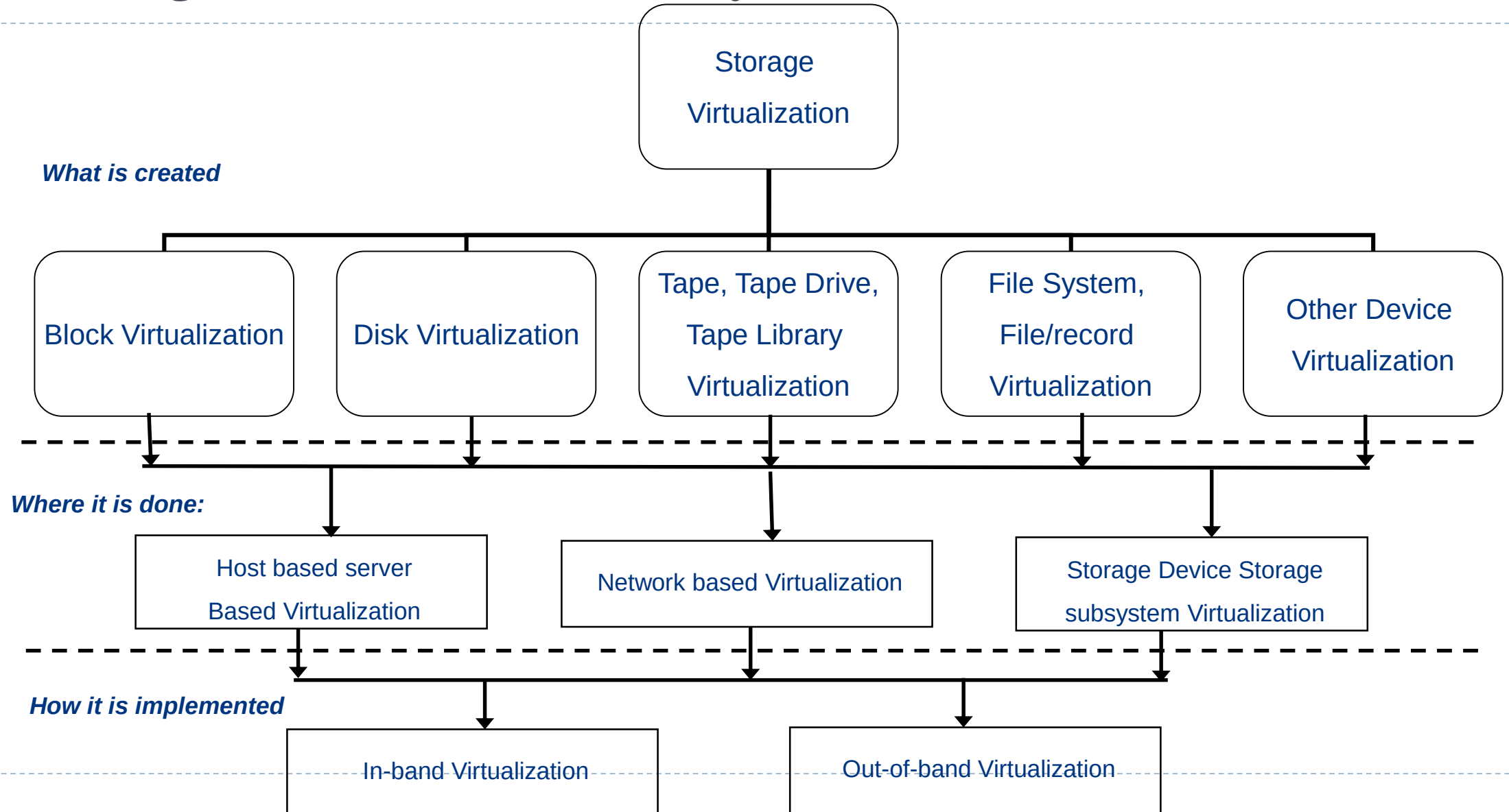
Storage Virtualization Requires a Multi-Level Approach

Intelligence Should Be Placed Closest to What it Controls



Distributed intelligence / centralized management

SNIA Storage Virtualization Taxonomy



Virtualization Solution Comparisons

Host Based Virtualization

► Advantages:

- Independent of storage platform
- Independent of SAN transport
- Software solution

Disadvantages:

- High CPU overhead
- Licensed and managed per host
- Requires a software driver

Array Based Virtualization

Advantages:

- Independent of host platform or OS
- Closer to disk drives
- High performance and scalable

Disadvantages:

- Vendor proprietary
- Management cost
- Can be complex

Network Based Virtualization

Advantages:

- Independent of host platform or OS
- Independent of storage platform
- High performance and scalable

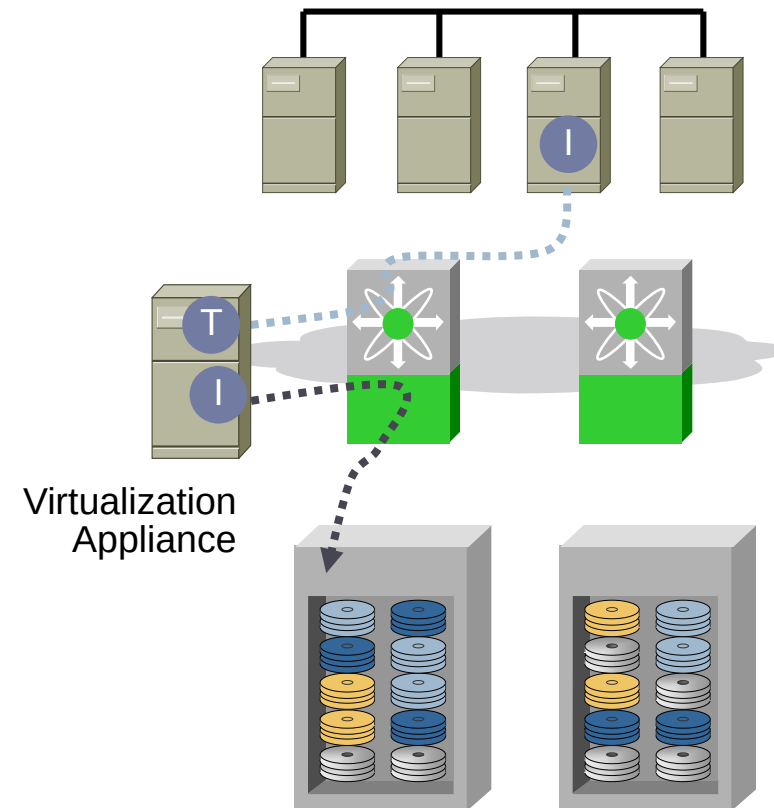
Disadvantages:

- Possible performance issues
- Can have scalability issues
- Vendor proprietary



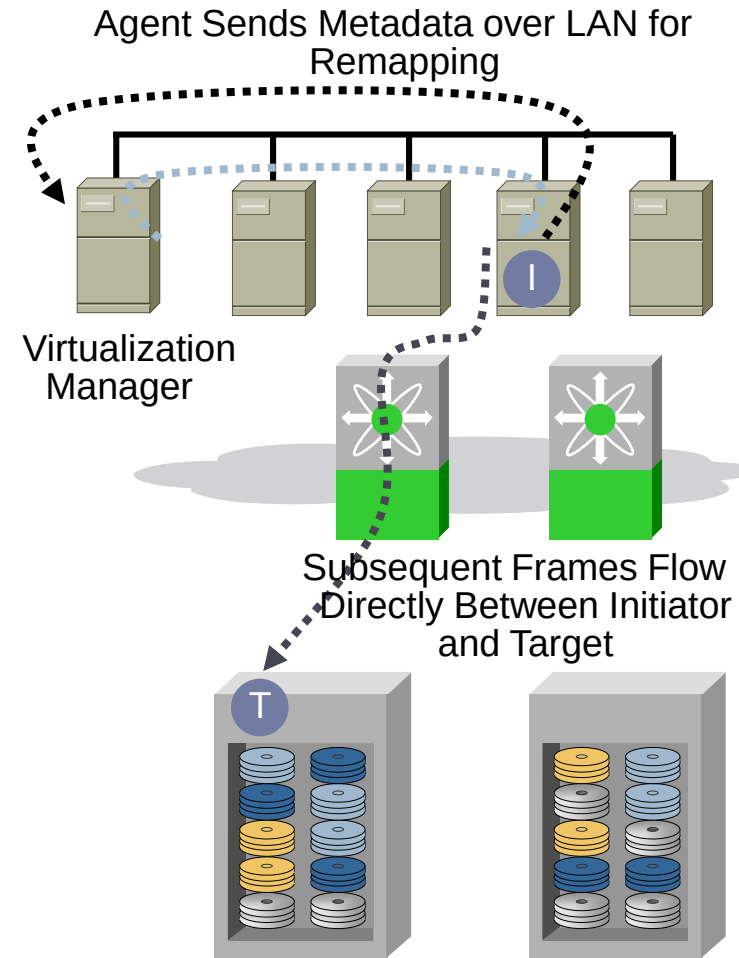
Symmetric Virtualization (Inband)

- ▶ All server ports are zoned with the virtualization appliance, Virtual Target port (T)
- ▶ All storage ports are zoned with the virtualization appliance, Virtual Initiator port (I)
- ▶ All control and data frames are sent to the virtual target and terminated.
- ▶ The CDB and LUN are remapped and a new frame is sent to the real target.
- ▶ Advantages:
 - ▶ Reduced complexity with single point of management
 - ▶ No host agents needed
- ▶ Disadvantages:
 - ▶ All frames are terminated and remapped by the appliance and resent to their destination and adds latency per frame
 - ▶ All traffic passes through the appliance; potential single point of failure and requires clustered solution and potential performance issue and requires huge cache



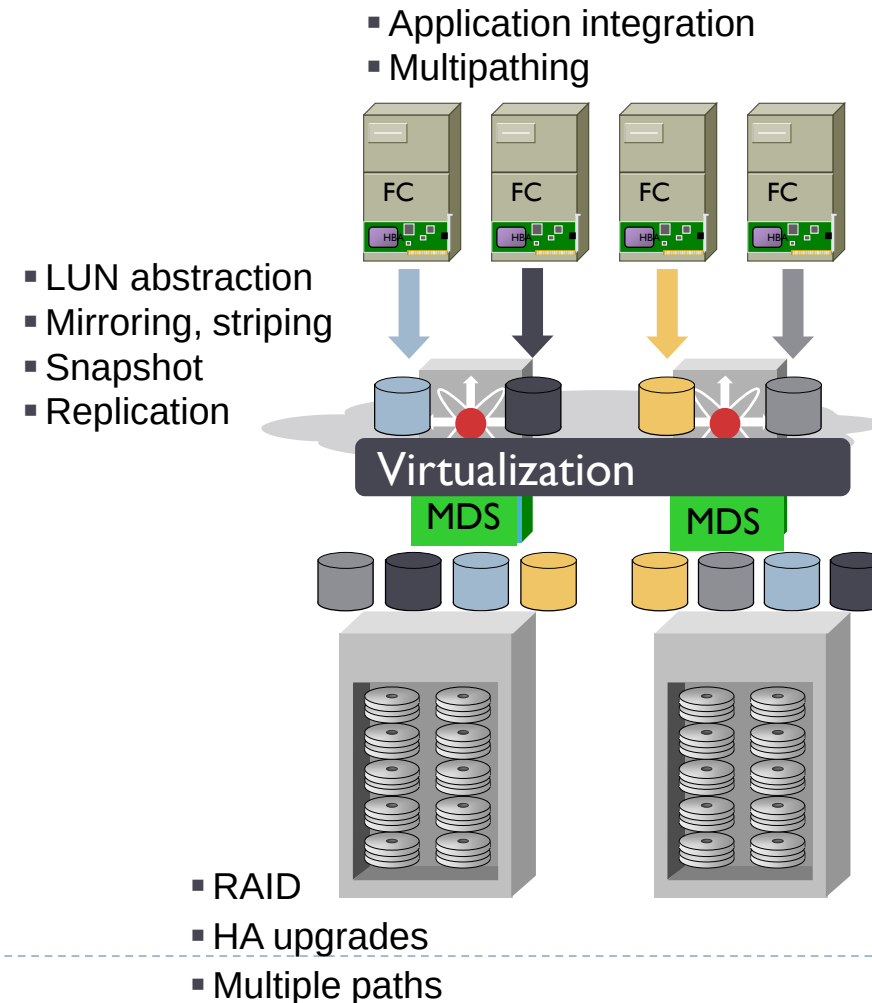
Asymmetric Virtualization (Out of Band)

- ▶ Each server runs a virtualization agent:
 - ▶ Intercepts block I/O requests
 - ▶ Sends the meta-data (CDB and LUN) to a virtualization manager on the LAN
- ▶ The virtualization manager remaps the CDB and LUN and returns it to the server.
- ▶ The server sends the modified control frame to the storage target port.
- ▶ All subsequent data and response frames flow directly between initiator and target.
- ▶ Advantages:
 - ▶ Low latency
- ▶ Disadvantages:
 - ▶ Requires agent in host to intercept control frame
 - ▶ Remapping of CDB and LUN adds latency to first frame in exchange
 - ▶ Virtualization manager could be single point of failure



Network Based Storage Virtualization

- ▶ Provides a single point of management
- ▶ Insulates servers from storage changes:
 - ▶ Data migration
 - ▶ Highly resilient storage upgrades
- ▶ Provides capacity on demand:
 - ▶ Increased utilization
- ▶ Enables consolidation:
 - ▶ Legacy investment protection
 - ▶ Heterogeneous storage network
- ▶ Simplifies data protection:
 - ▶ Snapshots
 - ▶ Replication
- ▶ Different classes of storage for different applications

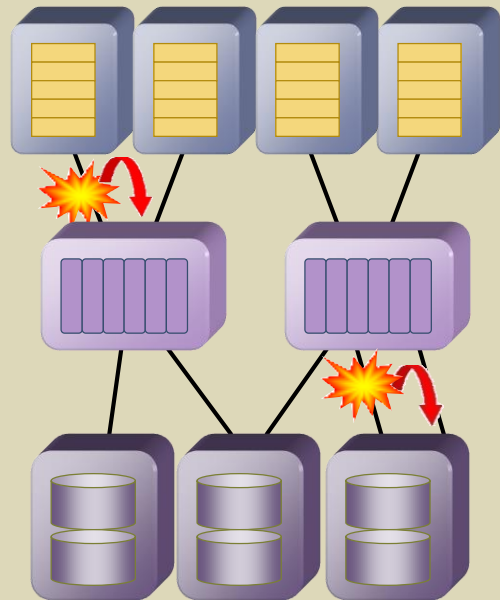


Current Storage Virtualization Examples

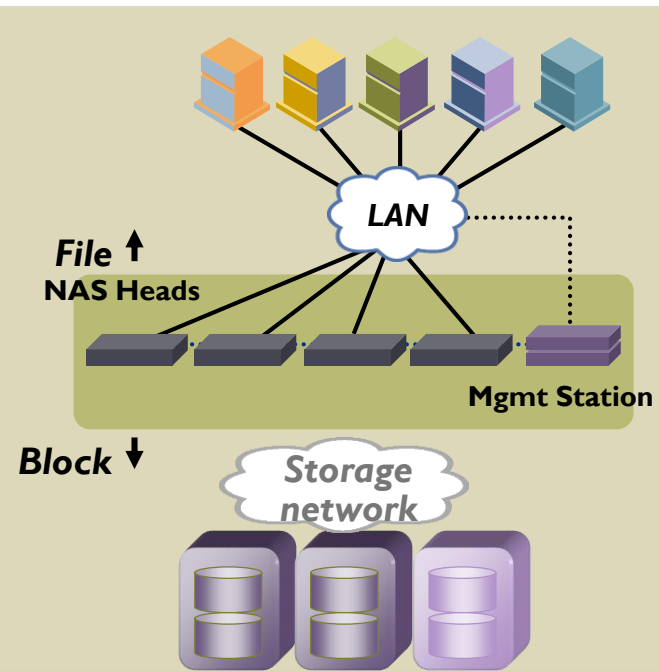
Storage-Virtualization Examples

Problem:

**I/O path performance
and availability**



**Consolidated
file-based storage**

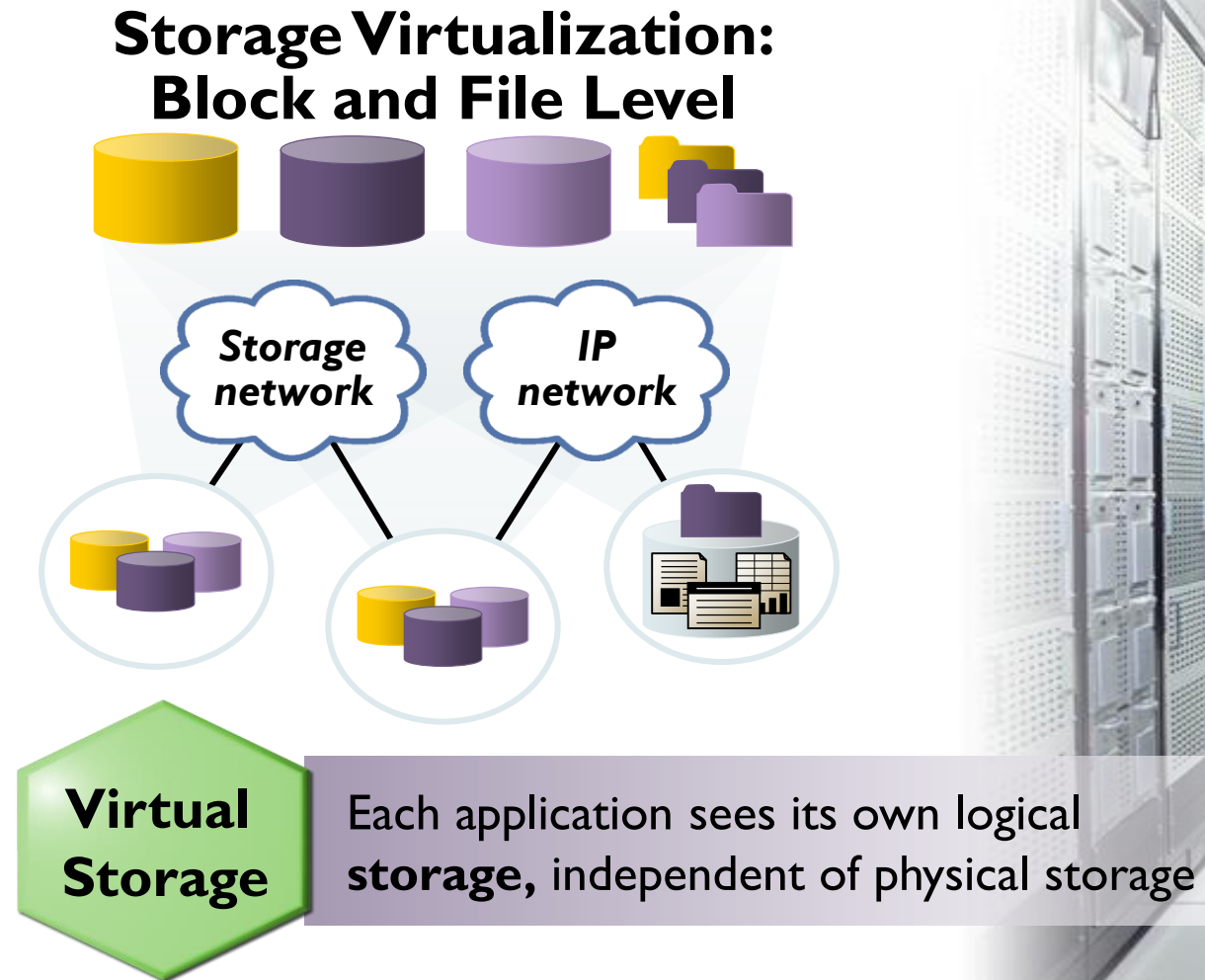


Solution:

MultiPathing Software

NAS Gateway

Virtual Storage

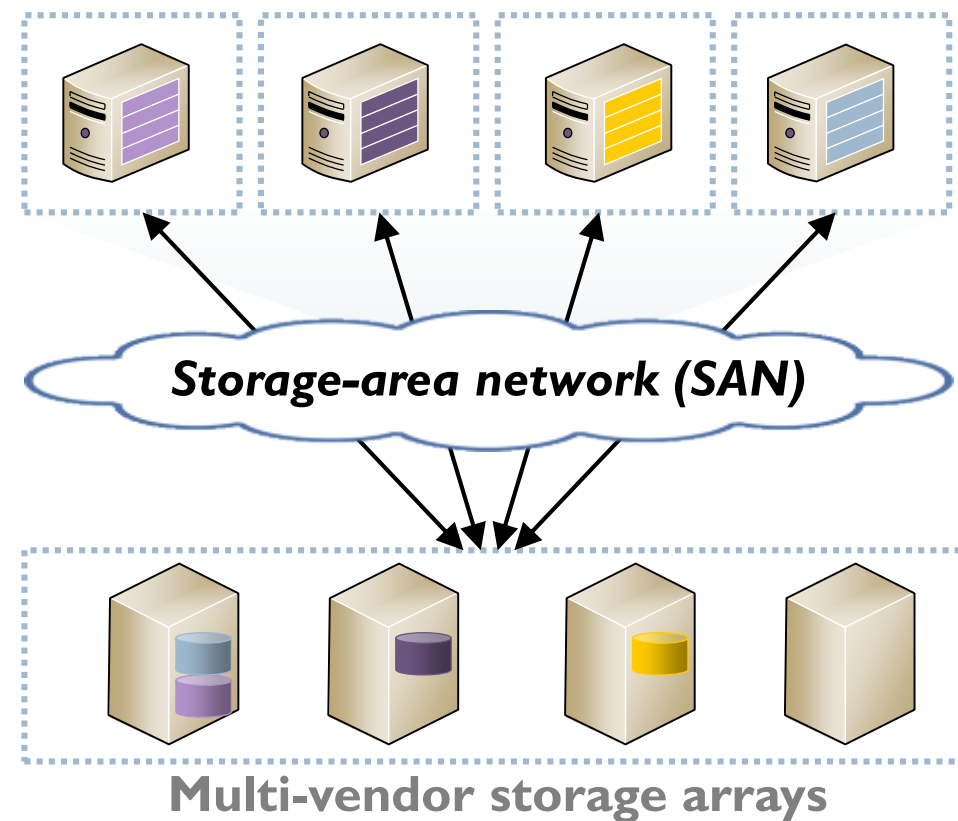


Benefits of Virtual Storage

- Nondisruptive data migrations
- Access files while migrating
- Increase storage utilization

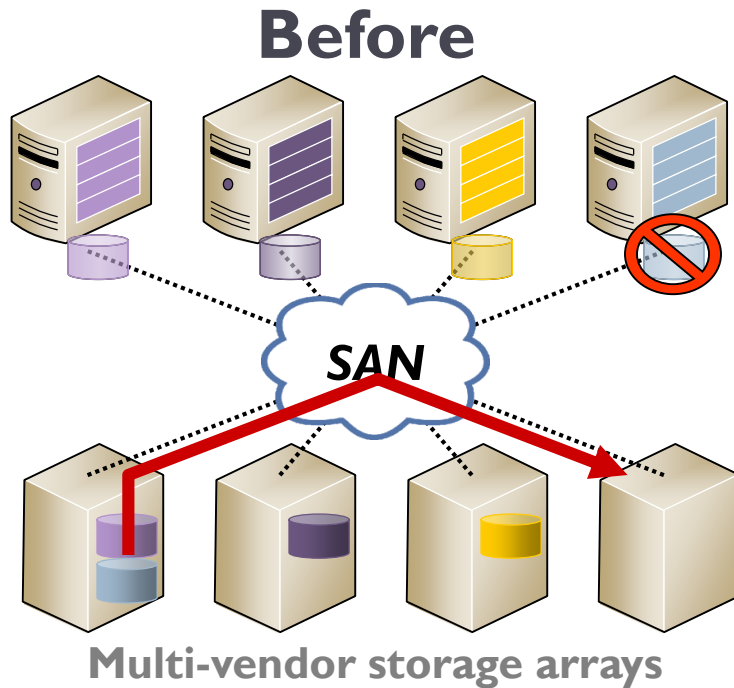
Block-Level Storage Virtualization Basics

- ▶ Ties together multiple independent storage arrays
 - ▶ Presented to host as a single storage device
 - ▶ Mapping used to redirect I/O on this device to underlying physical arrays
- ▶ Deployed in a SAN environment
- ▶ Nondisruptive data mobility and data migration

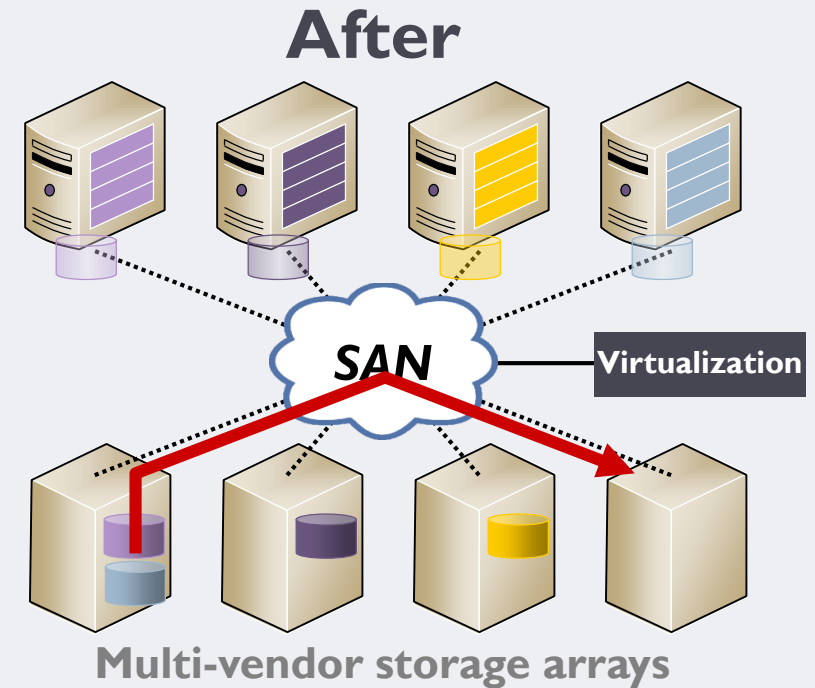


Block-Level Storage Virtualization – Data mobility

Optimizes Resources and Improves Flexibility



All applications have direct knowledge of storage location



- ✓ Simplify volume access
- ✓ Nondisruptive mobility
- ✓ Optimize resources



Monitoring/Reporting/Capacity Management



Senthilkumar Muthusamy

What is Storage Infrastructure Management?



Definition: Storage Infrastructure Management

All the storage infrastructure-related functions that are necessary for the management of the infrastructure components and services, and for the maintenance of data throughout its lifecycle.

- Aligns storage operations and services to an organization's strategic business goal and service level requirements
- Ensures that the storage infrastructure is operated optimally by using as few resources as needed
- Ensures better utilization of existing infrastructure components



Monitoring



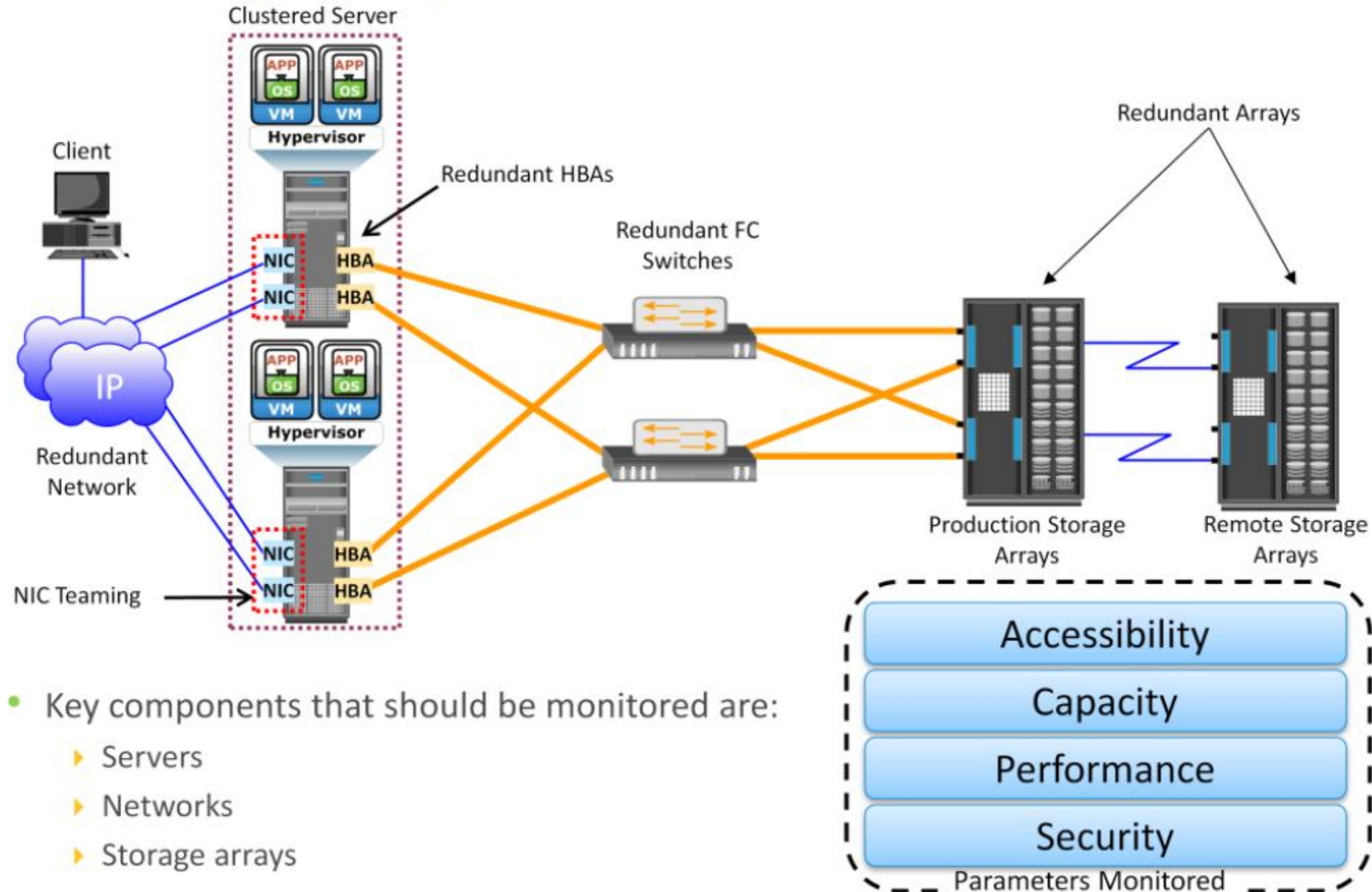
Senthilkumar Muthusamy

Introduction to Monitoring

- Monitoring provides visibility into the storage infrastructure and forms the basis for performing management operations
- It helps to
 - Track the performance and availability status of components and services
 - Measure the utilization and consumption of resources by services
 - Track events impacting availability and performance of components and services
 - Generate reports and triggering alerts
 - Track environment parameters (HVAC)



Monitoring Storage Infrastructure



Monitoring Parameters

- Accessibility
 - ▶ To identify failure of any component that may lead to service unavailability or degraded performance
- Capacity
 - ▶ To ensure availability of adequate amount of resources and prevent service unavailability or degraded performance
- Performance
 - ▶ To evaluate efficiency and utilization of components and identify bottlenecks
- Security
 - ▶ To ensure confidentiality, integrity, and availability of storage infrastructure



Components Monitored – Host

- Accessibility
 - ▶ Hardware components such as HBAs, NICs, and internal disks
 - ▶ Status of various processes/applications
- Capacity
 - ▶ File system utilization
 - ▶ Database: table space/log space utilization
 - ▶ User quota
- Performance
 - ▶ CPU and memory utilization
 - ▶ Transaction response time
- Security
 - ▶ Authentication and authorization
 - ▶ Physical security (data center access)



Components Monitored – Network

- Accessibility
 - ▶ Physical components such as switches and ports
 - ▶ Logical components such as zones
- Capacity
 - ▶ Interswitch links and port utilization
- Performance
 - ▶ Assess individual component performance and help to identify network bottlenecks
 - ▶▶ Monitoring port performance and link utilization
- Security
 - ▶ Unauthorized changes to the fabric
 - ▶ Login failures



Components Monitored – Storage

- Accessibility
 - ▶ Hardware components such as controllers and disks
 - ▶ Software processes such as replication processes
- Capacity
 - ▶ Capacity utilization and consumption trend
- Performance
 - ▶ Utilization rates of storage array components
 - ▶ I/O response time, cache utilization
- Security
 - ▶ Access control and physical security (data center access)



Alerts

- It is an integral part of monitoring
- It keeps administrators informed on the status of components and processes
- Monitoring tools enable administrators to assign different severity levels for different events
 - ▶ Classified as information, warning, and fatal alerts

Type of alerts	Description	Example
Information	• Provide useful information • Does not require administrator intervention	• Creation of zone or LUN
Warning	• Require administrative attention	• FS becoming full • Soft media errors
Fatal	• Require immediate attention	• Component failure • Power failure

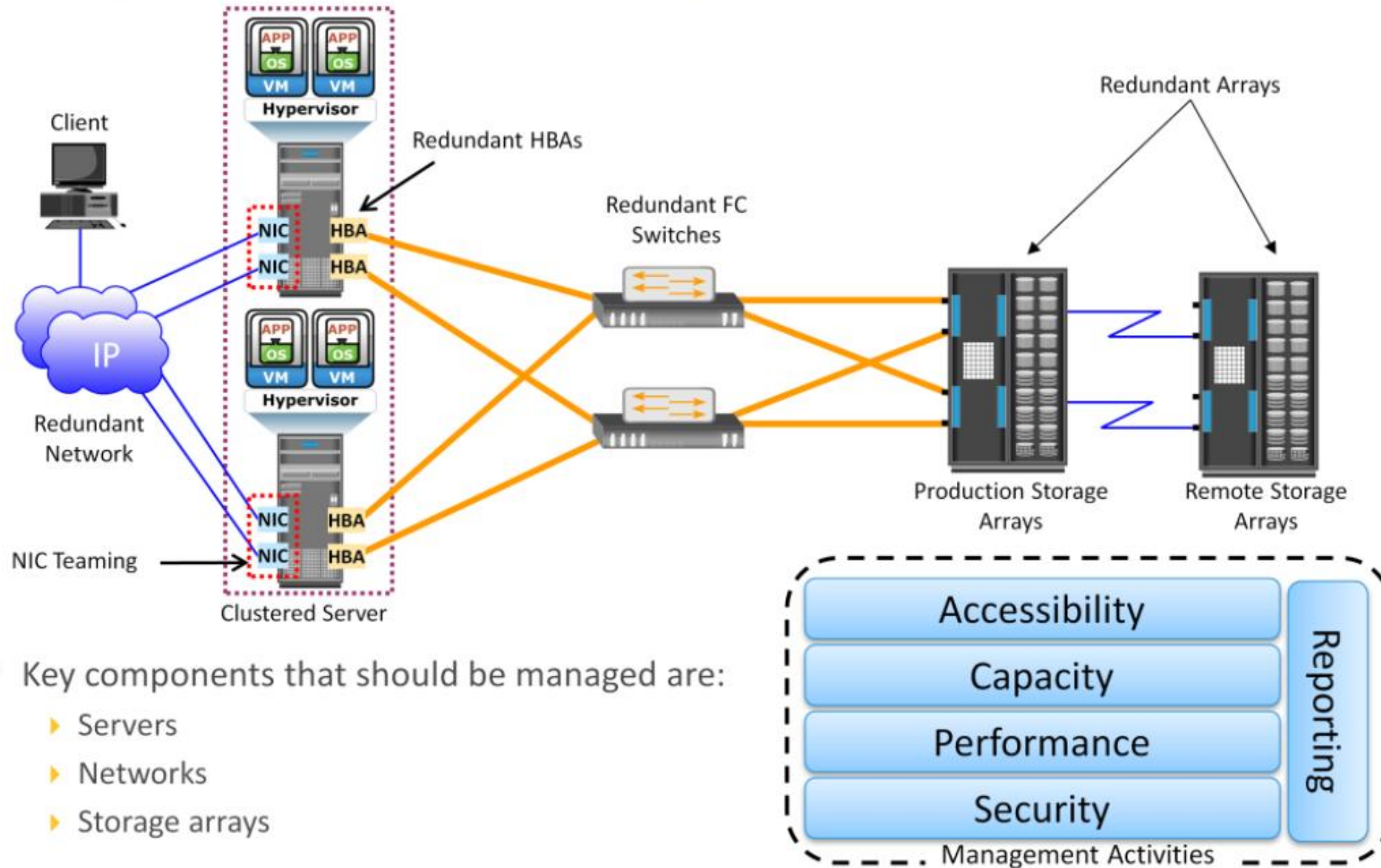




Reporting



Storage Infrastructure Management Activities



Availability Management

Availability Management

The goal of availability management is to ensure that the availability requirements of all the components and services are constantly met.

- Key activity is to provision redundancy at all levels
- Example of availability management tasks are:
 - ▶ Installing two or more HBAs per server
 - ▶ Installing multipathing software
 - ▶ Deploying clustered server
 - ▶ Configuring RAID
 - ▶ Deploying redundant fabric
 - ▶ Configuring data backup and replication



Capacity Management

Capacity Management

The goal of capacity management is to ensure adequate availability of resources, based on their service level requirements.

- Example of capacity management activities are:
 - ▶ Storage provisioning
 - ▶ Enforcing capacity quota for users
 - ▶ Capacity consumption trend analysis
- Technologies such as data deduplication, compression, and virtual provisioning help to manage storage capacity efficiently



Performance Management

Performance Management

The goal of performance management is to ensure the optimal operational efficiency of all components.

- Key activities are:
 - ▶ Fine tuning for performance enhancement
 - ▶ Identifying performance bottlenecks
- Example of performance management activities are:
 - ▶ Configuring multiple paths
 - ▶ Choosing appropriate RAID type and cache configuration



Security Management

Security Management

The goal of security management is to ensure confidentiality, integrity, and availability of information.

- It prevents unauthorized access and configuration of storage infrastructure components
- Examples of security management tasks are:
 - ▶ Managing user accounts
 - ▶ Configuring zoning and LUN masking
 - ▶ Configuring encryption services
 - ▶ Installing antivirus and firewalls
 - ▶ Auditing of event logs



Reporting

- Involves gathering information from various components or processes and generating reports
- Commonly used reports are:
 - ▶ Capacity planning report
 - ▶ Configuration and asset management reports
 - ▶ Performance report
 - ▶ Chargeback report

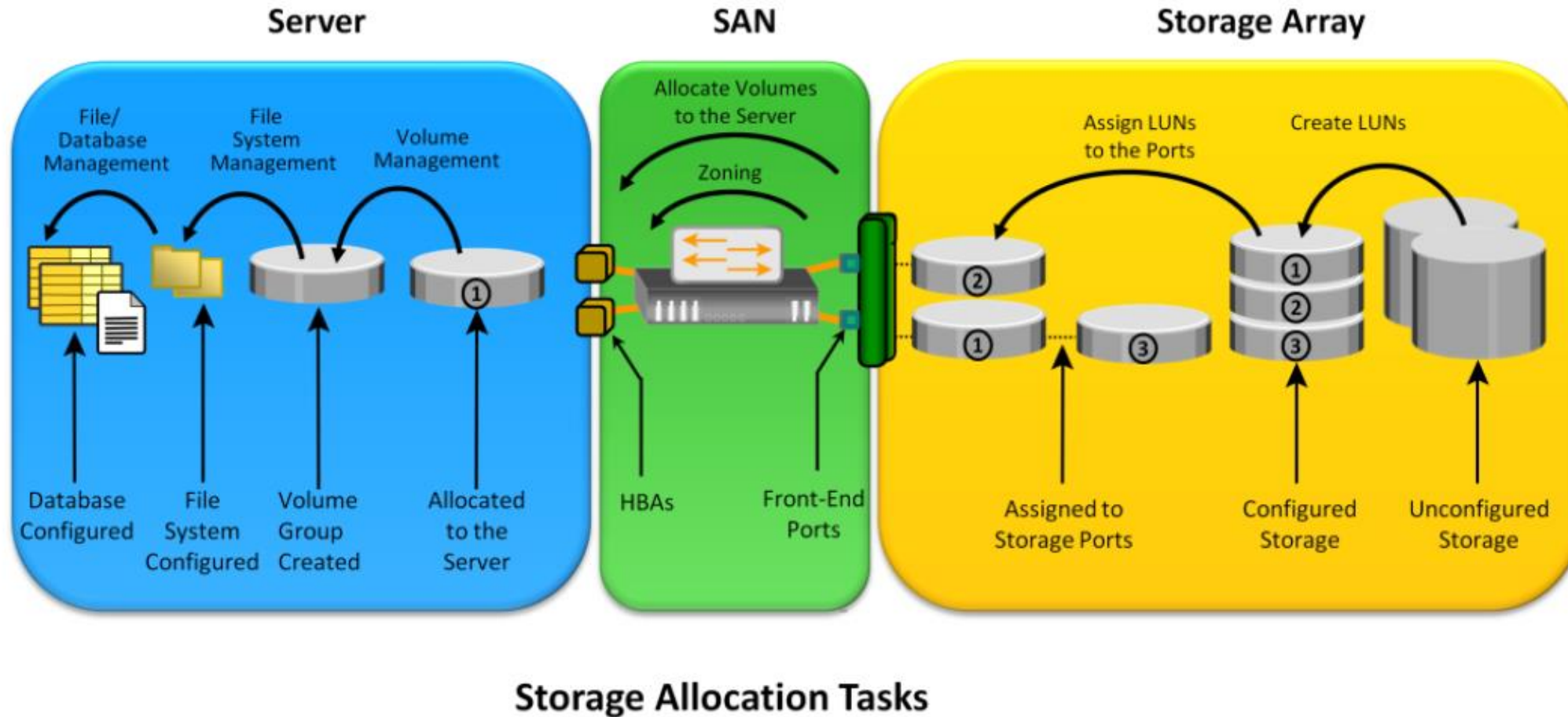


Storage Multitenancy

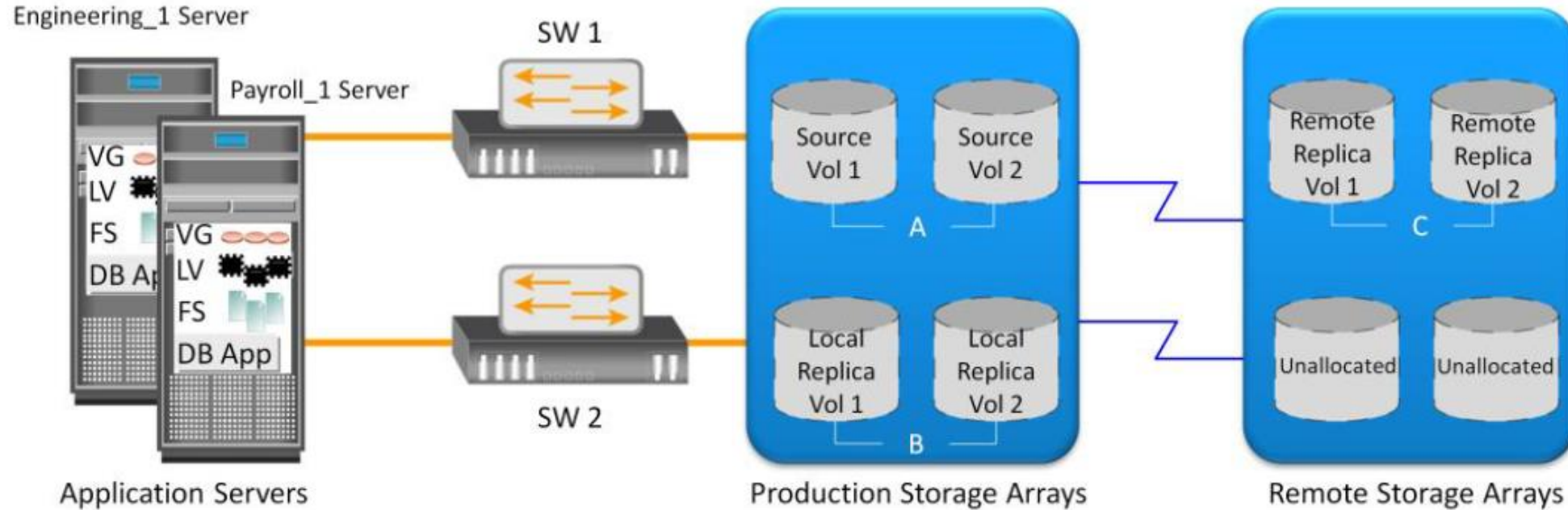
- Enables multiple tenants to share the same storage resources provided by a single landlord (resource provider)
- Security and service level assurance are key concerns in a multitenant storage environment
- Secure storage multitenant environment should follow the four pillars of multitenancy
 - ▶ Secure separation
 - ▶ Service assurance
 - ▶ Availability
 - ▶ Management



Storage Management Example 1 – Storage Allocation to a New Server

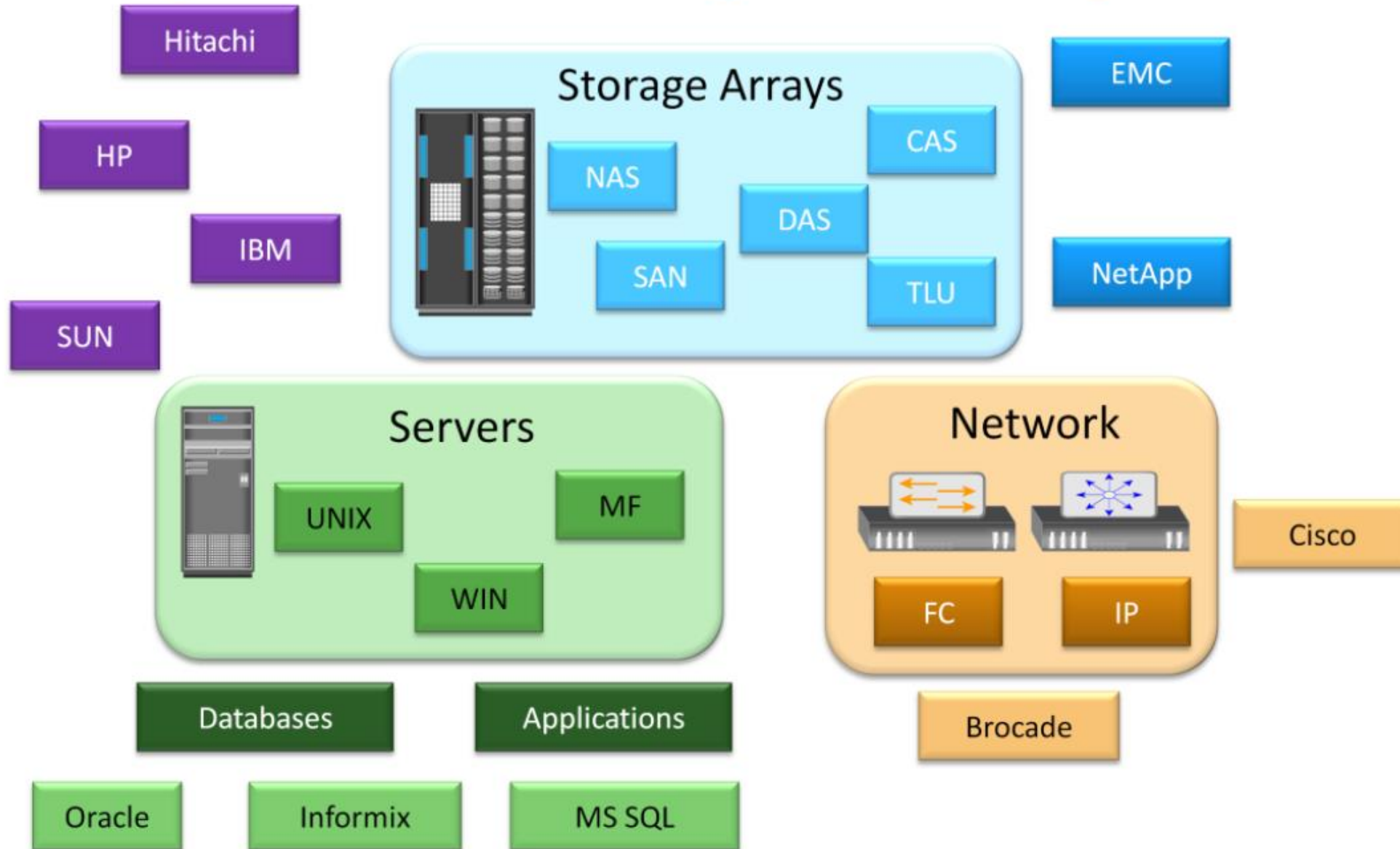


Storage Management Example 2 – Chargeback Report

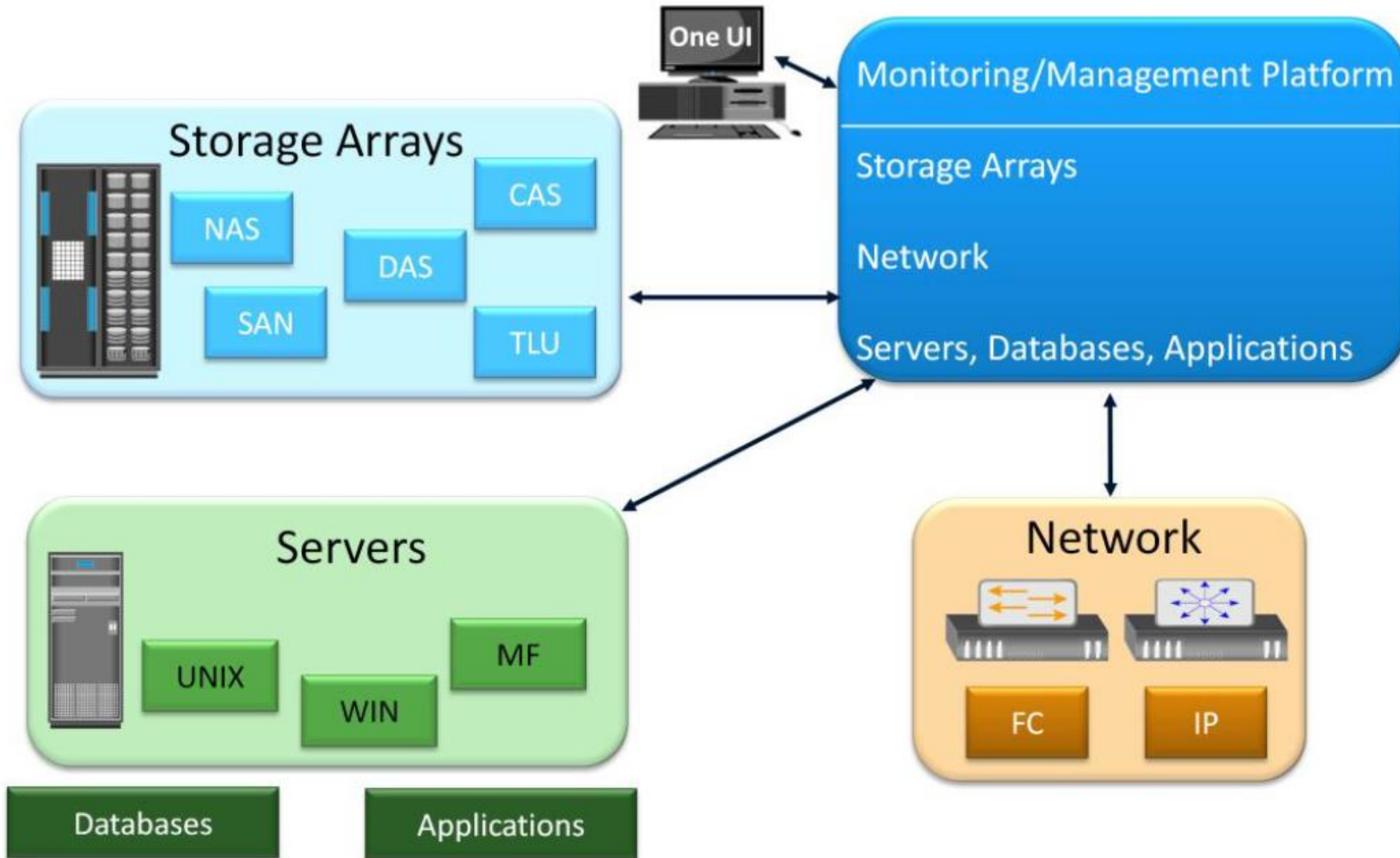


Application	Storage (GB)	Production Storage Raw (GB)	Local Replica Storage Raw (GB)	Remote Replica Storage Raw (GB)	Total Storage RAW (GB)	Chargeback Cost \$5/Raw (GB)
Payroll_1	100	200	100	125	425	2125
Engineering_1	200	250	200	250	700	3500

Storage Infrastructure Management Challenges



Developing an Ideal Solution

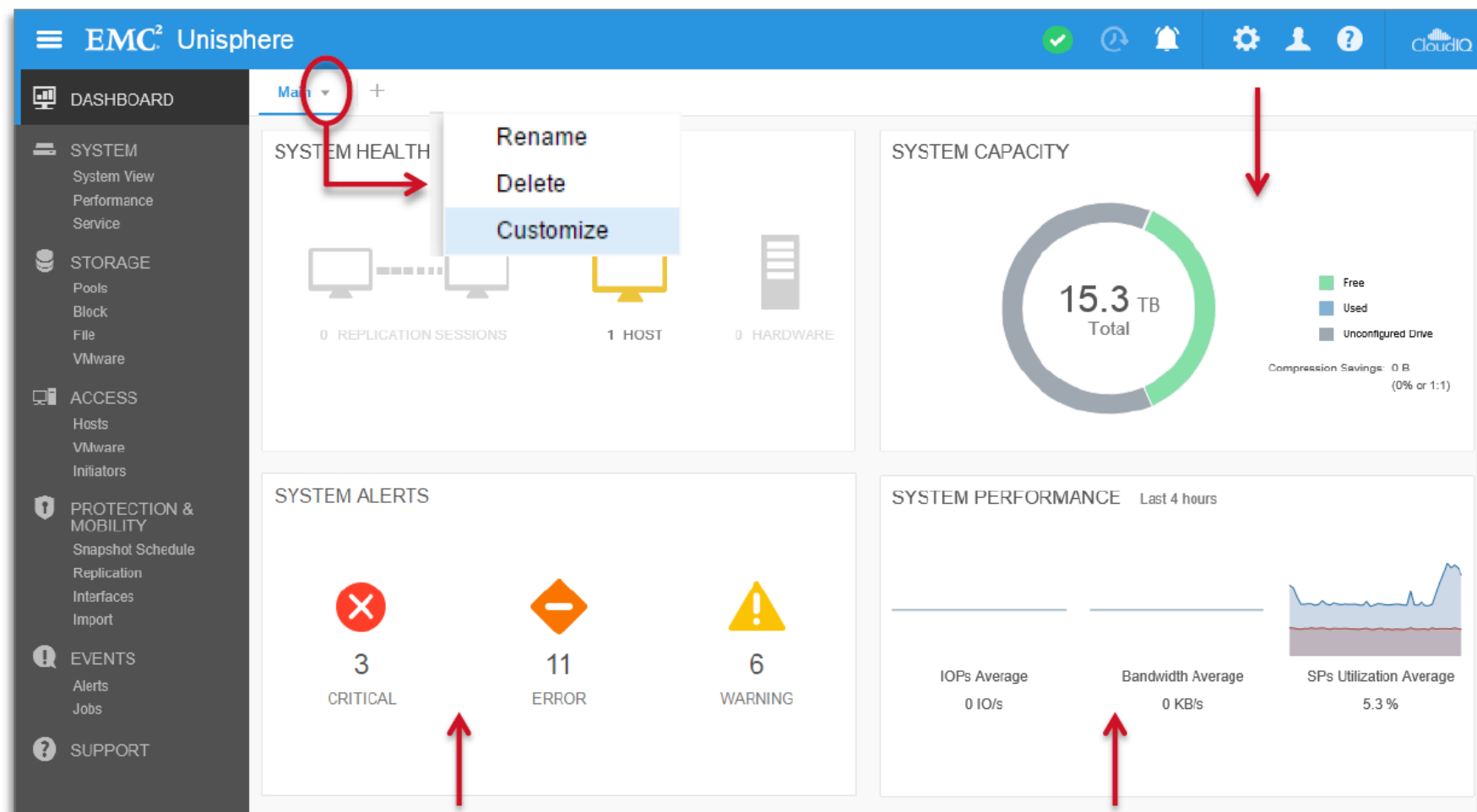


Enterprise Management Platform

- Suite of applications for managing and monitoring storage infrastructure
 - ▶ Provides end-to-end management of physical and virtual resources
 - ▶ Generates alerts to inform the status of components and processes
 - ▶ Schedules operations such as provisioning and configuration management



Unisphere GUI – Dashboard



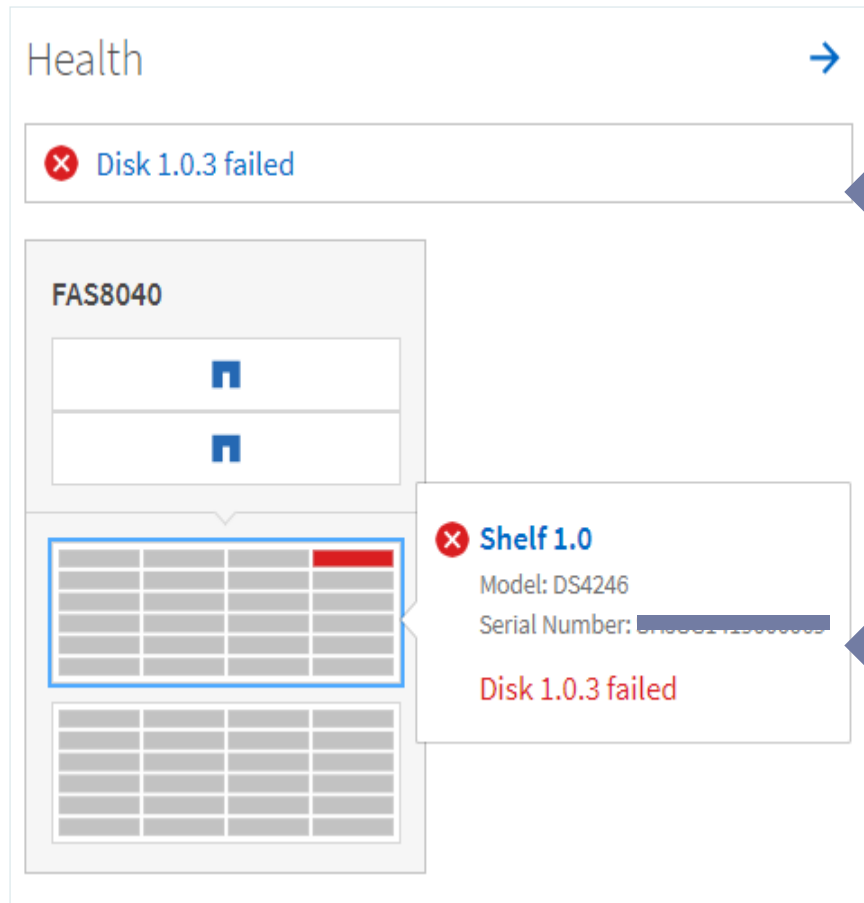
Proactive health score categories

Category		Types of Health Checks
	Configuration	Hosts - non-HA, drive issues: faults, weighted by use (Hot Spare, RAID 6, RAID 5)
	System Health	Components with issues, OE/Firmware compliance issues
	Performance	CPU Utilization, SP Balance
	Capacity	Pools reaching full capacity
	Data Protection	RPOs not being met, last snap not taken



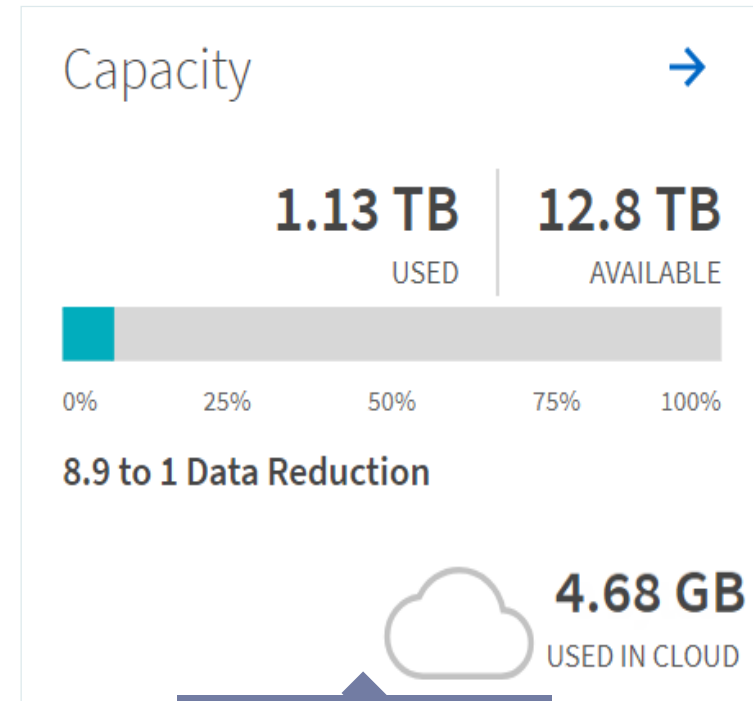
Dashboard

Health and Capacity Portlet



Hardware health status of cluster

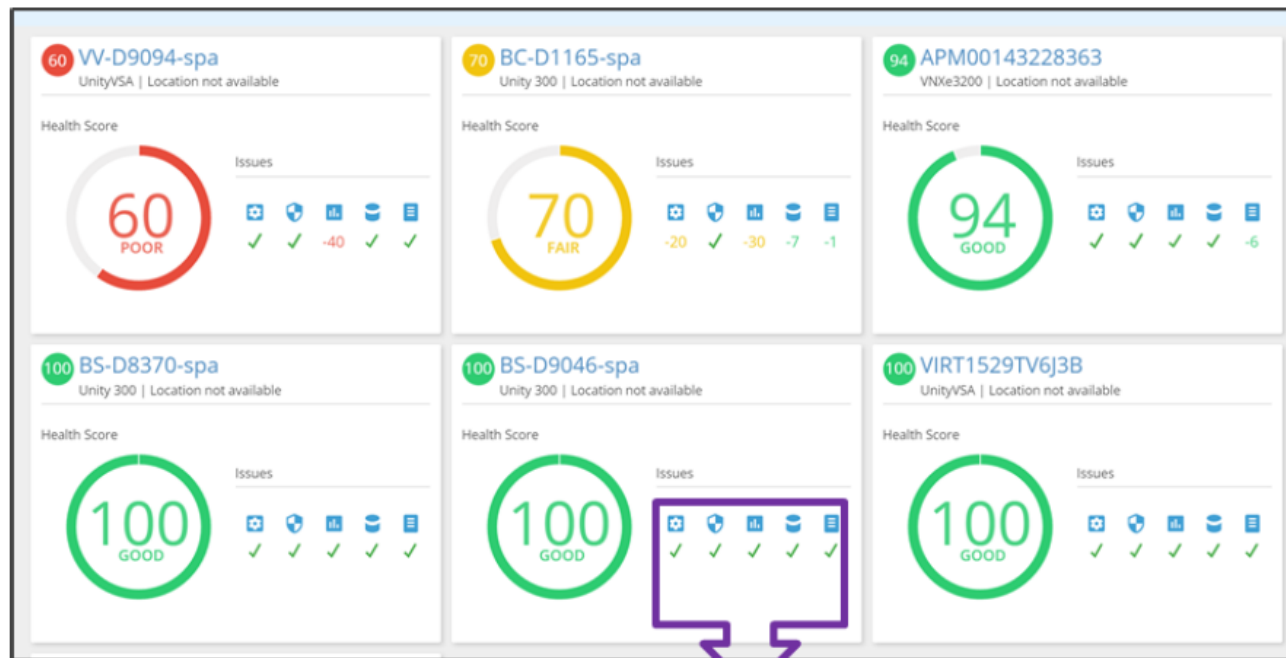
Disk, fan, and power supplies failure



On-premises and in-cloud capacity reporting



Proactive health score view

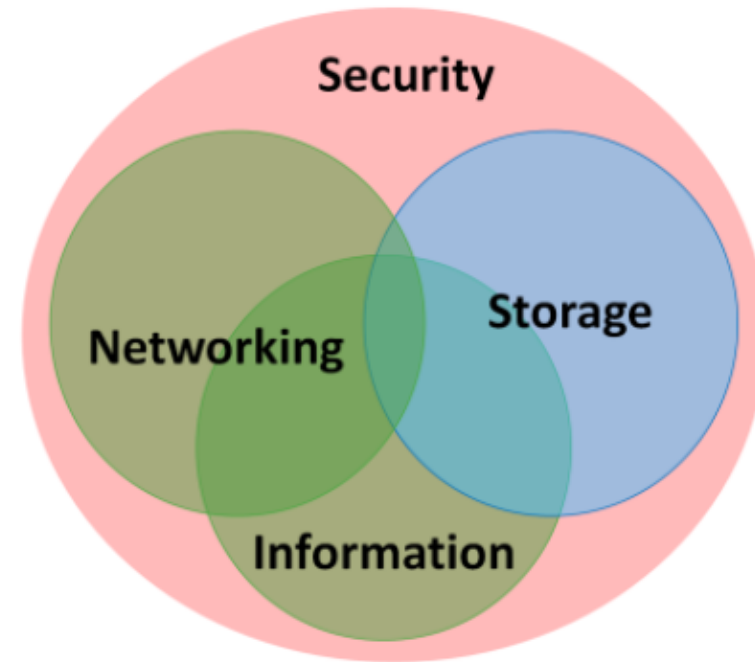


Range	Color	Status
90 – 100	Green	Good
70 – 89	Yellow	Fair
0 – 69	Red	Poor



Storage Security

- Process of applying information security principles and practices within the domain of storage networking technologies
- Storage security focuses on securing access to information by implementing safeguards or controls
- Storage security begins with building 'information security framework'



Information Security Framework

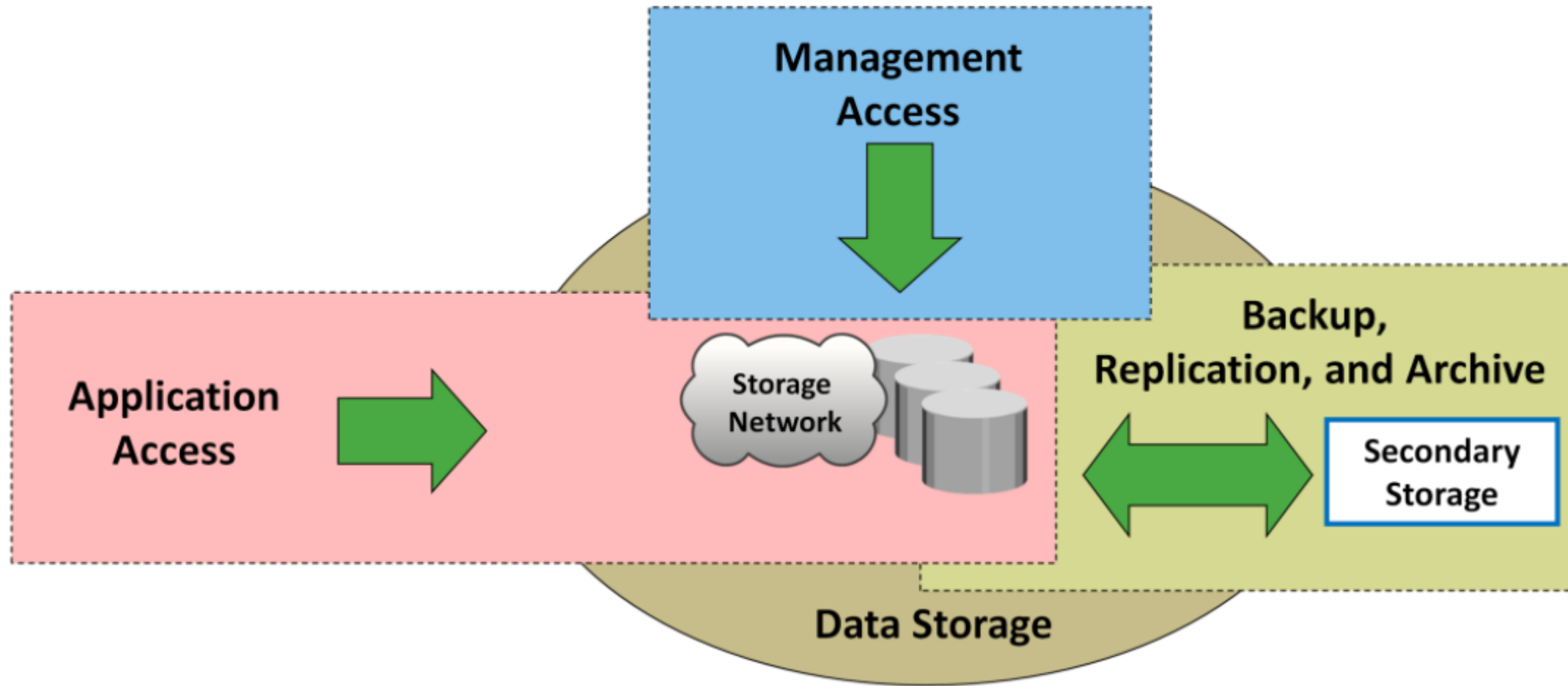
- A systematic way of defining security requirements
- Framework should incorporate:
 - ▶ Anticipated security attacks
 - ▶▶ Actions that compromise the security of information
 - ▶ Security measures
 - ▶▶ Control designed to protect from these security attacks
- Security framework is built to achieve four security goals:
 - ▶ Confidentiality
 - ▶ Integrity
 - ▶ Availability
 - ▶ Accountability
- Securing infrastructure begins with understanding the risk

Security Controls

- Reduces the impact of vulnerabilities
- Any control measure should involve all the three aspects of infrastructure
 - ▶ People, process, and technology
- Controls can be technical or non-technical
 - ▶ Technical: antivirus, firewalls, and intrusion detection system
 - ▶ Non-technical: administrative policies and physical controls
- Controls are categorized as:
 - ▶ Preventive
 - ▶ Corrective
 - ▶ Detective



Storage Security Domains



Security Implementation in SAN

- Common SAN security mechanisms are:
 - ▶ LUN masking and zoning
 - ▶ Securing FC switch ports
 - ▶ Switch-wide and fabric-wide access control
 - ▶ Logical partitioning of a fabric: VSAN



Security Implementation in NAS

- Permissions and ACLs
 - ▶ Protection to NAS resources by restricting access
- Other authentication and authorization mechanisms
 - ▶ Kerberos and Directory services
 - ▶▶ Implemented to verify the identity of network users and define their privileges
 - ▶ Firewalls
 - ▶▶ To protect the storage infrastructure from unauthorized access and malicious attacks



QUESTIONS ?????

THANK YOU !!!!!

